

Physics Principles of Inertial Confinement Fusion and U.S. Program Overview: LLNL-JRNL-819317

O.A. Hurricane*

*Lawrence Livermore National Laboratory,
P.O. Box 808, L-472,
Livermore, California 94550,
USA*

P.K. Patel¹

*Focused Energy Inc.,
11525-B Stonehollow Drive, Suite 200,
Austin TX 78758 USA*

R. Betti, D.H. Froula, and S.P. Regan

*Laboratory for Laser Energetics and Department of Mechanical Engineering and Physics and Astronomy,
University of Rochester,
250 East River Road, Rochester,
New York 14623-1299,
USA*

S.A. Slutz, M.R. Gomez, and M.A. Sweeney

*Sandia National Laboratories,
Albuquerque, New Mexico 87185,
USA*

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Inertial confinement fusion (ICF) has existed as a field of study since the 1970s, but the field was born out of the Cold War. In the decades since the 1960s, pioneering research developing the principles and technologies of ICF culminated in the creation of three major Department of Energy facilities that still exist today, the National Ignition Facility (NIF) at Lawrence Livermore National Lab, the OMEGA laser at the Laboratory for Laser Energetics, and the Z pulsed power facility at Sandia National Laboratories. While the technology of ICF facilities themselves is fascinating, this review concentrates upon the physics principles of the targets fielded on U.S. ICF facilities and upon results from the last decade of research. While there have been periods of frustration on the road to ICF ignition, recent research has demonstrated great leaps in understanding what aspects of the implosions need more control. Tangible progress in ICF is evident as burning plasmas and ignited plasmas have recently been generated, repeatedly, on the NIF stemming from decades of science and engineering understanding generated from work on the above three facilities and international community.

CONTENTS

I. Fusion and related thermodynamic processes	2	II. How to achieve a high energy density ICF plasma	8
A. Fusion fuels	4	A. Hot spot model, adiabatic implosion, analytic solution for alpha-heating	8
1. Reaction rates as a function of temperature, T	4	B. Principal physics parameters controlling ICF implosion performance	9
B. Pressure and internal energy in a DT plasma	4	C. Implosion acceleration and hydrodynamic instability	12
1. "Ignition Temperature"	4	III. X-ray driven implosions	13
2. Alpha-particle stopping range and areal density required to stop alpha-particles	5	A. Brief History	13
3. Lawson's Criterion	7	B. Indirect drive with lasers	14
C. Self-heating without a magnetic field	7	1. Hohlraum Physics	14
		C. Z-pinch driven hohlraums	15
		D. Ablator Physics	16
		E. Present status of results	17
		1. Low-Foot (LF) design: National Ignition Campaign (NIC): 2010-2012	18
		2. High Foot (HF) design: 2013-2015	19
		3. Adiabatic Shaped (AS) design: 2014-2015	22

¹ Formerly at Lawrence Livermore National Laboratory, Livermore, California 94550, USA

* hurricane1@llnl.gov

4. Low Gas-Fill (LGF) Hohlraum Designs:	22
2016-present	
5. Experimental demonstration of burning plasmas and ignition	25
6. Alternate indirect-drive concepts	26
IV. Laser driven implosions	27
A. Brief history of direct drive	27
B. Key differences between direct and indirect drive ICF	29
1. Direct-drive scaling advantages	31
2. Hydrodynamic instabilities in direct-drive implosions	32
3. Initial nonuniformity seeding of hydrodynamic instabilities in direct drive implosions	38
4. Hot electron preheat in direct drive implosions	42
5. Cross Beam Energy Transfer in direct drive implosions	45
C. Present status of direct-drive implosion performance and proximity to ignition	47
1. Hydrodynamic Scaling from OMEGA to megajoule laser energies	47
2. Current OMEGA implosion performance and proximity to hydroscaled ignition	48
3. Polar Direct Drive	49
V. Magnetic Drive Implosions	52
A. History of Pulsed Power Fusion	52
1. Development of Pulsed Power	52
2. Electron beam driven fusion	53
3. Ion beam driven fusion	53
4. Z pinch-driven indirect drive	54
B. Direct magnetic drive using the MagLIF concept	56
1. Basic Principles	56
2. Experimental demonstration of MagLIF concepts	59
3. The effect of mix	61
4. Implosions Instabilities	61
5. High yield and gain from MagLIF on future pulsed power machines	63
VI. Summary and outlook	65
Acknowledgments	65
References	66

I. FUSION AND RELATED THERMODYNAMIC PROCESSES

For many decades the running joke in fusion research has been that “fusion is twenty years away and always will be.” Yet, in 2023 we find ourselves in a position where we can talk about the milestones of burning plasmas, fusion ignition, and target energy gain greater than unity (i.e. ‘Scientific Breakeven’) in the past tense – a situation that is remarkable! In this review paper, we describe the three main inertial confinement fusion (ICF) paths that the U.S. chose to pursue (i.e. laser indirect-drive, laser direct-drive, and magnetic drive) and we tell some of the story of the applied physics and engineering challenges that needed to be overcome by the U.S. ICF program in order to achieve these milestones. To help understand the story, several key physics principles of inertial fusion are presented, and as the reader will see, much of the key

ICF target physics can be understood from the standpoint of classical mechanics and thermodynamics. Moreover, classical mechanics and thermodynamic statements are what underpin what the scientific definitions of the milestones of a burning plasma, an igniting plasma, and gain are in the context of ICF research.

In particular, a ‘burning plasma’ is one where a fusion plasma’s self-heating exceeds external sources of heating, which is a necessary but not sufficient condition for ‘ignition’ which occurs when a fusion plasma’s self-heating exceeds the sum of all the cooling mechanisms present in the plasma, resulting in a thermodynamic instability that sustains and rapidly increases fusion *power* output, for some interval of time. Fusion power integrated over sufficiently long times can result in energy gain. Since ICF systems are energy density concentrators that have elements of successively smaller components of decreasing size nested inside each other, an ‘energy gain’ can be defined for each layer of the system. The central most element of physical interest in an ICF system is the fusion fuel, thus one can define a ‘fuel gain’ (G_{fuel}), which is the ratio of fusion energy produced over the net energy that was externally delivered into the fusion fuel. The fusion fuel in an ICF system is carried inside a shell of material, the capsule, thus ‘capsule gain’ ($G_{capsule}$) defines the ratio of fusion energy produced over the net energy absorbed by the capsule. In the case of x-ray driven ICF designs, a metallic outer structure, a “hohlraum”, surrounds the capsule, completing the ICF target, and thus one can define a ‘target gain’ (G_{target} , or sometimes simply G) which is the ratio fusion energy produced as compared to the energy delivered into the hohlraum. For ICF schemes not involving a hohlraum, capsule gain and target gain are the same thing. None of these gain definitions account for the energy expended by the facility (usually orders of magnitude greater than the energy that is actually delivered to an ICF target). Thus fuel gain, capsule gain, or target gain greater than unity do not imply net energy production. ‘Engineering gain’ ($G_{engineering}$) is usually defined to include the energy used by the facility, thus an engineering gain > 1 , would imply net energy gain in the practical sense of interest to the general public. No manmade fusion system in existence has yet achieved $G_{engineering} > 1$.

$G_{target} \sim 1.5$ was recently achieved (LLNL, 2022), where 3.15 MJ of fusion energy was produced by a target on the NIF with 2.05 MJ of energy delivered by the facility to the target (details will be forthcoming in future publications), while the NIF facility itself consumed several hundred MJ of energy. As a practical reference point, note that a 1 *kWh* (kilowatt-hour) = 3.6 *MJ* and that the average U.S. household energy use is 30 *kWh* $\approx$ 108 *MJ* per day. Target gain and ignition are terms that are often conflated, even though the concepts and physics behind the terms are different. Unfortunately, this conflation of terms was reinforced by a 1997 National

Academy of Sciences (NAS) review committee (Koonin *et al.*, 1997), because at the time of 1997 NAS review there was no point of consensus in the ICF community of what ignition was (and magnetic fusion community definitions were oddly ignored). To paraphrase, the NAS committee adopted an operative definition of ignition as $G_{target} = 1$ for the purpose of their report based upon a, now known to be optimistic, calculated curve of fusion yield vs. NIF laser energy (Fig. 1 of (Koonin *et al.*, 1997)). Nevertheless, the U.S. Department of Energy adopted the 1997 NAS definition and continues to use it to this day.

Since many decades of work have gone into putting ignition on a firm scientific foundation, this review article treats target gain and ignition as separate concepts as the reader will see in the following sections, particularly Section II.A, III, and IV. Interestingly, as the reader will see in the physics development of Section I.B., G_{fuel} , which is more directly tied to thermonuclear fusion processes, and ignition are more directly connected.

Like many aspects of nuclear energy, large strides in understanding and harnessing thermonuclear fusion processes were made during the Manhattan Project era and the subsequent Cold War. A key member of the Manhattan Project driving fusion research was Edward Teller in pursuit of his idea of a “Super” bomb based upon thermonuclear fusion using deuterium. Cold War research and development resulted in the practical application of thermonuclear fusion and research on controlled fusion for peaceful purposes was initiated during the same period.

Based upon the work of G. Gamow (D’e. Atkinson and Houtermans, 1929; Gamow, 1928), it had been known from quantum mechanics that fusion reactions occur when particles tunnel across the Coulomb barrier bringing the nuclei into close proximity. Moreover, that under conditions approaching thermodynamic equilibrium the particles will have enormously greater penetrating-power. So, it was already known decades before the Cold War that significant thermonuclear energy cannot be obtained from particle beam to target configurations since the incident energy is rapidly dissipated by ionization and x-ray emission losses. What is required instead is a completely ionized light element plasma that is either large volume or high density in order to have a high number of fusion reactions. Large volume fusion is naturally adapted to magnetic fusion approaches, while high density fusion is naturally adapted to implosion based concepts that today form the field of ICF.

Laboratory-scale ICF originated in the early 1960s (Nuckolls, 2006; Tarter, 2018) with the realization that “micro-fusion” explosions could be generated if sufficient compression could be achieved. However, like fusion itself the principles of the implosion technology that are key to ICF originated during the Manhattan Project. According to the recollections of key people involved in the

Manhattan Project, the conceptual idea behind implosions has been credited to Richard C. Tolman and/or to Seth Neddermeyer (Hoddeson *et al.*, 1993) in the 1942-1943 period, albeit the interest then was in fission-based implosion systems.

Basically, implosion systems entail surrounding a capsule material on all sides with a source of high pressure that exceeds the yield-strength of the material and any internal pressure. The pressure difference will cause the capsule to ‘implode’ inwardly upon itself thereby increasing the density of the capsule shell and any internal fill (e.g. fusion fuel) as it is squeezed into a smaller volume. It is easier to describe in words what such an implosion is than to do it successfully, since the difficulty of using the tactic increases exponentially with the desired degree of compression. In order to better appreciate the difficulty in making fusion implosions work, we estimate the extreme conditions needed, which is the topic of the sections immediately following this introduction.

The organization of this review article is as follows. Section I.A and I.B describe basic fusion fuel considerations illustrating why deuterium-tritium fusion is by far the most efficient reaction and the importance of compression in limiting the energy cost of igniting a fusion fuel. Section I.C makes the connection between G_{fuel} and the “Lawson triple product” and introduces the key role of alpha-particle self-heating. Section II.A. describes the highly non-linear thermodynamics of fusion fuel inside an implosion explaining why high implosion velocities are so critical for ICF. Section II.B provides a simple classical mechanics analog of an implosion that is quite effective for understanding the principle parameters that must be controlled in an ICF implosion, including the effect of asymmetry on an implosion. Section II.C describes the most essential hydrodynamic instability considerations for ICF that often conflict with the needs of high velocity and high compression in an ICF implosion. Section III then focuses upon the indirect-drive ICF scheme, starting with a brief history, but mostly focusing upon the key works that have been pursued on the NIF over the past decade (Section III.E). Key topics for indirect-drive are those of hohlraum physics and ablator physics, each discussed with some important scaling relationships in Sections III.B and III.D respectively. Since indirect-drive can be generated using Z-pinchs in addition to lasers, Section III.C addresses this alternate way of making x-rays. As part of the ablator physics discussion in Section III.D, the very critically important, yet obscure, physics of ‘coast time’ is emphasized. Laser direct-drive is the focus of Section IV, starting with a brief history in Section IV.A with key advantages and disadvantages, as compared to indirect-drive, described in Section IV.B. Section IV.C summarizes the present state of direct-drive theory and experiment. Section V.A describes a variety of Z-pinch concepts that have been examined, then Section V.B focuses in on the most re-

cent studies using magnetized liner inertial fusion and associated physics and electrical engineering considerations. Sections III, IV, and V are nearly independent, so a reader may skip any of these three sections in favor of jumping directly to a the section of interest to the reader, without much confusion. Section VI then concludes this article with a brief outlook.

A. Fusion fuels

Interestingly, understanding of what are the most practical fusion reactions has not really changed since early work on measuring the fusion reaction rates done by the F-3 group at Los Alamos in 1944 (Hoddeson *et al.*, 1993). It has long been recognize that the most favored reaction is deuterium(D) tritium(T) fusion $D+T \rightarrow n$ (14.1 MeV) + ${}^4\text{He}$ (3.5 MeV) (Post, 1956) because it is orders magnitude more reactive for a given inter-particle collisional energy than any other fusion reaction for $T < 100$ keV. Simply, the fewer protons in the reactants, the less electric repulsion needs to be overcome in order to bring the nuclei into close proximity for fusion. However, other reactions such as $D+D \rightarrow n+{}^3\text{He}$, $D+D \rightarrow p+T$, $D+{}^3\text{He} \rightarrow {}^4\text{He}+p$, and $p+{}^{11}\text{B} \rightarrow 3\alpha$ are often used for experimentation in order to avoid directly handling radioactive tritium.

1. Reaction rates as a function of temperature, T

Accurate analytic expressions for fusion reaction-rates over a broad range of temperatures (see Fig. 1) can be cumbersome, so it is often useful and usually sufficient to use power-law fits over a limited range of temperature. Power-law expressions of the form $\langle\sigma v\rangle_{DT} \sim T^{(2-5)}$ fit the DT reaction-rate well, with the lower power of T decreasing as the temperature range of interest increases. A useful approximation to the DT reaction-rate in regime of present practical interest to ICF implosions is

$$\langle\sigma v\rangle_{DT} \approx 4.2 \times 10^{-20} T^{3.6}, \quad 3.5 < T < 6.5, \quad (1)$$

with T in units of keV and $\langle\sigma v\rangle$ in units of cm^3/s .

B. Pressure and internal energy in a DT plasma

Here we will estimate the thermodynamic conditions that are expected for an igniting inertially confined fusion plasma using basic thermal considerations. Assuming nearly equal ion and electron temperatures (T_{ion} and T_e , respectively), the plasma pressure (p) is

$$p = \frac{\bar{Z} + 1}{\bar{A}} \frac{\rho}{m_p} k_B T \quad (2)$$

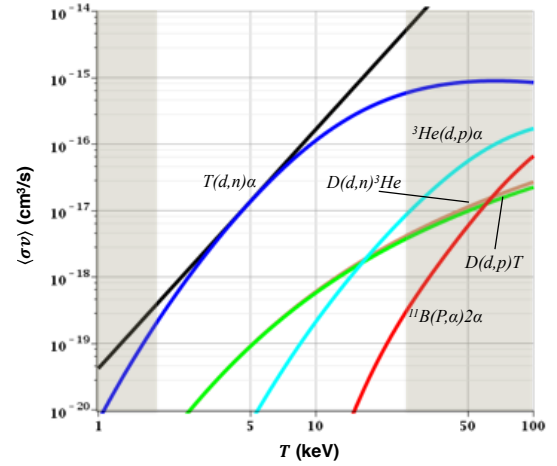


FIG. 1 The reaction-rates of select reactions are plotted based upon formulas given by (Bosch and Hale, 1992) and (Nevins and Swain, 2000) in the case of p-boron. The temperature region, $2 \text{ keV} < T < 20 \text{ keV}$ of practical interest to ICF is highlighted. The top curve is the DT reaction rate, while the tangent line is the power-law approximation, Eq. 1, to the DT reaction rate.

where $\bar{Z}$ is the effective average ionization, $\bar{A}$ is the average atomic mass, ρ is the mass density, k_B is the Boltzmann constant, and T is the thermal temperature. So, for a DT plasma of equal parts D and T, $\bar{Z} = 1$ and $\bar{A} = 2.5$. Thus

$$p = 0.77 \rho T \quad (3)$$

in Gbar, g/cm^3 , keV units. The heat capacity of a DT plasma is then $c_{DT} = (3p)/(2\rho T) = 115 \text{ MJ}/(\text{g}\cdot\text{keV})$, which is key quantity for understanding fusion energy requirements for a given mass and temperature of heated DT. By comparison, with $\bar{Z} = 3$ and $\bar{A} = 6$ the heat capacity of fully ionized pB_{11} fuel is $c_{pB_{11}} = 96 \text{ MJ}/(\text{g}\cdot\text{keV})$. Note that in ICF, and fusion in general, the fusion plasma is not in thermal equilibrium with the radiation field since the photon mean-free path is very long compared to the size of the plasma, so the blackbody energy ($4\sigma T^4/c$, where σ is the Stefan-Boltzmann constant and c is the speed of light) is not included in the heat-capacity.

1. "Ignition Temperature"

Any plasma heated to keV temperatures is a strong x-ray radiator and that radiation is a key cooling mechanism that must be overcome. For a mixture of atoms in local thermodynamic equilibrium, the volumetric bremsstrahlung x-ray (Heitler, 1954) emission power per unit mass (in g) is [e.g., (Spitzer Jr., 2006; Zel'dovich and Raizer, 2002)]

$$Q_{Brems} = \frac{64}{3\sqrt{2}\pi} \frac{e^6 \rho}{m_e \hbar c^2} \sqrt{\frac{k_B T_e}{m_e c^2}} \frac{\bar{Z} \bar{Z}^2}{(\bar{A} m_p)^2} \quad (4)$$

where e is the electron charge (in stat-coulombs), m_e is the electron mass (g), $\hbar$ is the conventional Planck's constant over 2π (erg-s), and c is the speed of light (cm/s). For a pure mixture of 50% D and 50% T, the bremsstrahlung emission per unit mass, Eq. 4, reduces to

$$Q_{B,DT} = 3.1 \times 10^7 \rho \sqrt{T_e} \quad (5)$$

in GJ/(g · s) units with T_e is keV.

The “ignition temperature” (T_{ign}) is defined as the temperature where the x-ray power lost due to bremsstrahlung emission just balances fusion power (Post, 1956). The fusion power per unit mass in α -particles for DT fusion is

$$Q_\alpha = 8.2 \times 10^{24} \rho \langle \sigma v \rangle, \quad (6)$$

in GJ/(g · s). The total fusion yield production-rate for a mass of fuel, m , would then be $dY/dt = 5mQ_\alpha$ where the factor of 5 is due to the fact that only 1/5 of DT fusion yield comes in the form of α 's. Equating, Eq. 5 and Eq. 6 with $T_e = T_{ion} = T_{ign}$, using Eq. 1, and solving gives $T_{ign} = 4.3$ keV. For fusion fuels other than DT, T_{ign} can be substantially higher than 4.3 keV and T_{ign} would also be higher for DT in the presence of impurities that rapidly increase x-ray emission as reflected in the $\bar{Z}$ and $\bar{Z}^2$ terms of Eq. 4.

2. Alpha-particle stopping range and areal density required to stop alpha-particles

In order to self-heat a mass of fusing plasma, some of the fusion by-products must be stopped by collisions in that plasma thereby adding energy to the burn, which in turn increases the temperature and reaction rate. This argument has a fission chain-reaction analogy from the Manhattan Project days as it was used to make the first estimates of the critical mass of fission material needed to stop fission neutrons. Generally, stopping 14 MeV fusion neutrons is not practical in ICF systems because of the high areal density required, so it is only the stopping of the α particles that is practical.

In order to estimate the size and mass of fuel that will self-heat, the α -particle stopping range (λ_{DT}) is needed. A modern fit (Zylstra and Hurricane, 2019) to the Maynard-Deutsch (Maynard and Deutsch, 1985) calculation of ion stopping in a degenerate electron fluid is

$$\lambda_{DT} = \frac{1}{\rho} \frac{0.0171T^{1.071}}{1 + 0.0113T^{1.071}} \left[1 + 1.579 \left(\frac{\rho}{100} \right)^{0.199} \right] \quad (7)$$

where λ_{DT} is in cm, ρ is in g/cm³, and T is in keV. So, at the ignition temperature $T = 4.3$ keV the areal density that must be assembled is

$$(\rho\lambda_{DT})_{ign} = 0.077 \left[1 + 1.58 \left(\frac{\rho}{100} \right)^{0.2} \right] \quad (8)$$

in g/cm² units. While a number of α -particle stopping models exist (Brown *et al.*, 2005; Li and Petrasso, 2015; Singleton, 2008), for usual ICF conditions the impact of the different models is generally small (Zylstra and Hurricane, 2019). In addition to stopping α particles assembling a high areal density DT plasma has a beneficial reduction in electron condition heat losses from the burning region as will be seen later following Eq. 14.

Assuming spherical geometry the mass of DT needed to stop α 's inside the burning volume is $m_{DT} = (4\pi/3)[\mathcal{F}(\rho\lambda_{DT})]^3/\rho^2$, where $\mathcal{F} > 1$ is a ‘safety’ factor that would guarantee the α 's are stopped inside the volume, rather than just at the edge. The mass can be related to energy needed to be delivered to the DT to ignite ($E_{hot,DT}$) and ignition temperature using the heat capacity found above and $m_{DT} = E/(c_{DT}T_{ign})$ giving an implicit expression for the ignition density by using Eq. 8

$$\rho_{ign} = \frac{0.97\mathcal{F}^{3/2}}{\sqrt{E_{hot,DT}}} \left[1 + 1.58 \left(\frac{\rho_{ign}}{100} \right)^{0.2} \right]^{3/2} \quad (9)$$

with density in units of g/cm³ and energy in units of MJ. Note the $1/\sqrt{E_{hot,DT}}$ dependence in Eq. 9, which also implies the associated ignition pressure, $p_{ign} = 0.77\rho_{ign}T_{ign}$, also has the same energy dependence. For a given value of $E_{hot,DT}$, Eq. 9 can be solved numerically and the result can be used to calculate the pressure, mass, and give a fusion yield estimate as shown in Table I below. For $E_{hot,DT}$ less than several hundred MJ, Eq. 9 results in DT fuel densities that are higher (usually by orders of magnitude for laboratory levels of energy) than the solid cryogenic density of DT (0.25 g/cm³), implying that some sort of density compression scheme is needed.

TABLE I For a given energy delivery to the DT, the minimum density and minimum pressure required to bring a mass of DT of two ($\mathcal{F} = 2$) α -particle stopping radii to 4.3 keV with the associated DT mass and total fusion yield (including charge particles and neutrons) estimated from $Y \approx \phi[m_{DT}/(2Am_p)] \times (17.6 \text{ MeV/reaction})$ with characteristic burn efficiency $\phi \sim 0.2$ assumed.

$E_{hot,DT}$ (MJ)	ρ_{ign} (g/cm ³)	p_{ign} (Gbar)	m_{DT} (μ g)	Y (MJ)
1.0	7.4	25	2000	136
0.1	29	96	200	13.6
0.01	117	387	20	1.36
0.001	490	1622	2	0.136

While based upon simple physics arguments, the numbers shown in Table I are very representative of what is obtained from sophisticated modern multi-physics simulations of detailed ICF implosion designs. Moreover, the third row of Table I is representative of the near term

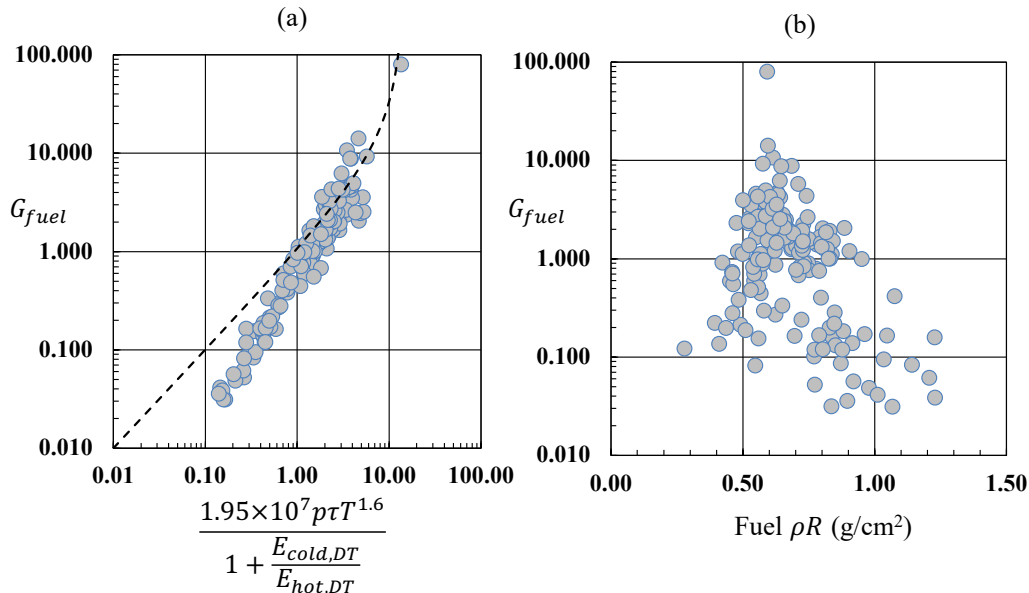


FIG. 2 (a) NIF DT implosion data, as of September 2021, is plotted against Eq. 11 (dashed curve). Here G_{fuel} is calculated from the ratio of fusion yield over peak fuel kinetic energy (an approximation that neglects energy losses in the conversion of kinetic energy into internal energy at stagnation due to x-rays or asymmetry). The low performing experiments that deviate from the theory expression are largely those from the National Ignition Campaign (see Section III.E.1) that experienced significant hydrodynamic instability and mixing – the enhanced bremsstrahlung emission that break the assumptions of Eq. 11. (b) G_{fuel} is plotted vs. measured burn average DT fuel areal density (inferred via the neutron downscatter ratio of 10-12 MeV neutrons over 13-15 MeV neutrons). As can be seen the figure, fusion performance, rapidly falls off at low areal density (as expected since 1D inertial confinement is reduced), but surprisingly also at higher areal density (contrary to conventional wisdom, see Section III.E.1-3) and an apparent optimum exists for fuel $\rho R \sim 0.6 - 0.7$ g/cm².

goal and performance expectation for indirect drive ignition on the National Ignition Facility [NIF, (Nicola *et al.*, 2018)] and the energy, density, and pressure are characteristic of the goals for the central ‘hot-spot’ of the assembled implosions. It should be clear from Table I that heating a large mass DT fusion fuel is energetically costly. Therefore, for facility-based ICF where only a small fraction of the driver energy can be coupled to the fuel, much smaller amounts of DT fuel are targeted for heating but then require achieving extraordinarily high pressures and densities (which implies that electron-ion collisions will be high, justifying the $T_e \approx T_{ion}$ assumption made earlier).

The ‘fuel energy gain’ G_{fuel} (ratio of total fusion yield to the total energy in the DT) (Atzeni and Meyer-Ter-Vehn, 2004) factor of $\sim 136\times$ implied by Table I appears attractive, but it can also be misleading if one does not appreciate that the energy delivered to the fusion fuel is very much less (by one or two orders of magnitude) than the total energy used to get the fusion fuel to those conditions because of the natural inefficiency of ICF implosion systems. To make up for the total energy expended to start ignition, propagation of the burn into another mass of relatively cold fusion fuel is required. For example, $G_{fuel} > 1$ was first achieved in 2014 (Hurricane *et al.*, 2014a), but the *total* energy gain (Atzeni and Meyer-Ter-

Vehn, 2004; Rosen, 1999), G , as measured against the laser energy input was only $\sim 1\%$.

It is interesting to estimate the fuel energy gain for a variety of fusion fuels (see Table II) for burning a mass of fuel heated to the ignition temperature for that particular fuel. The fusion yield can be estimated as from a product of burn-efficiency, $0 < \phi < 1$, the number of possible fusions, $m/(2\bar{A}m_p)$, and the energy released per fusion, traditionally denoted as Q , while the fuel energy, E_{fuel} is the product of heat capacity, mass, and temperature (again assuming ion-electron equilibrium) so,

$$G_{fuel} \equiv \frac{Y}{E_{fuel}} \rightarrow \frac{\phi Q}{3(\bar{Z} + 1)k_B T_{ign}} \quad (10)$$

when $T = T_{ign}$. Since the somewhat misnamed ignition temperature is the *minimum* temperature needed for ignition, it is clear from Eq. 10 that low T_{ign} and low $\bar{Z}$ in combination with high Q is highly desirable for a fusion fuel. As can be seen in Table II, DT fusion is orders of magnitude more attractive as a fusion fuel for any hope of having fusion output greatly exceed the energy consumed to get to fusion conditions because of the relatively low heat capacity and ignition temperature. It is important to recognize a subtlety in Eq. 10 in that E_{fuel} is the sum of energy from external drivers and any energy that comes from self-heating, so the fuel gain as measured

against an external source of energy can be higher than what Eq. 10 implies.

TABLE II Fusion fuel gain for a given mass of fuel brought to the ignition temperature when $T_e \approx T_{ion}$. The result for p-boron fusion includes the recent re-evaluation of the reaction cross-section (Sikora and Weller, 2016) and ignition temperature (Putvinski *et al.*, 2019) that are tens of percent more optimistic than past values.

Reaction	$k_B T_{ign}$ (keV)	Q (MeV/reaction)	G_{fuel}/ϕ
D+T $\rightarrow$ n+ ^4He	4.3	17.6	1364
D+D $\rightarrow$ n+ ^3He	35	3.25	31
D+D $\rightarrow$ p+T	25	4.03	54
D+ ^3He $\rightarrow$ ^4He +p	28	18.3	175
p + ^{11}B $\rightarrow$ 3 $\times$ ^4He	300	8.7	5

3. Lawson's Criterion

An alternate expression for G_{fuel} , that accounts for the energy deposition of self-heating back into the fusion fuel, shows a direct and important connection to ignited fusion requirements and fuel gain. Since, from Eq. 6 the total fusion yield produced by a mass of DT, over a characteristic confinement-time, τ , is $Y \sim 5mQ_\alpha\tau$ and since the internal energy in that DT is $E_{hot,DT} = c_{DT}mT$ the fuel gain can be written as

$$G_{fuel} = \frac{Y}{E_{pdV}} \approx \frac{\frac{Y}{E_{hot,DT}}}{1 + \frac{E_{cold,DT}}{E_{hot,DT}} - \frac{q}{10} \frac{Y}{E_{hot,DT}}} \quad (11)$$

$$\text{with } \frac{Y}{E_{hot,DT}} \approx 4.6 \times 10^{26} p \frac{\langle \sigma v \rangle}{T^2} \tau \quad (12)$$

where p is in Gbars, T in keV, and τ in s units. Note that $E_{cold,DT}$ is the energy in the cold fuel which is a fraction, that depends upon fuel entropy, of the maximum kinetic energy acquired by the fuel during the implosion. In Eq. 11, the energy delivered (by external means) is determined from the hot and cold fuel (if present) energy at stagnation, $E_{hot,DT}$ and $E_{cold,DT}$ respectively, minus a correction for additional energy retained by self-heating of the fuel and not lost as bremsstrahlung as measured by a “quality” factor, q . So, $E_{pdV} \approx E_{hot,DT} + E_{cold,DT} - qY/10$, $0 \leq q \leq 1$, where the $1/10$ is from $1/5$ of the fusion energy being in α -particles over $1/2$ the burn duration.

Albeit generally arrived at in a different fashion than above, the product $p(\langle \sigma v \rangle / T^2) \tau$ is a Lawson-like (Lawson, 1957) product for ignition of an ICF implosion, which has been studied and re-stated in many alternate forms for decades (Betti *et al.*, 2010; Tipton, 2015; Zhou and Betti, 2009). Eq. 11 makes it clear that fuel energy gain and generalized Lawson product are directly

related. Note that Lawson’s original analysis of a pulsed power producing thermonuclear reactor determined that a function of $\rho\tau$ (the “Lawson parameter”) and T was fundamental to ignition, later combined into a “Lawson triple product,” $\rho\tau T$, but some modification of these expressions is needed for ICF because of the dynamic nature of implosions, as will be discussed below.

Using Eq. 1 in Eq. 12 gives the simple expression

$$\frac{Y}{E_{hot,DT}} \sim 1.95 \times 10^7 p(\text{Gbar}) T^{1.6}(\text{keV}) \tau(\text{s}), \quad (13)$$

which makes it clear, with Eq. 11, that for high G_{fuel} , the product of pressure, temperature, and confinement-time should be maximized by whatever means. A plot of Eq. 11 is given in Fig. 2 for DT implosion experiments on the NIF.

C. Self-heating without a magnetic field

The temperature development of the DT fusion plasma in an ICF implosion is determined by a thermodynamic balance of energy sources and sinks (see Fig. 3). The source of fusion energy gain, Q_α , Eq. 6, and one source of energy loss, Q_B , Eq. 5, have been discussed above. The complete power balance in an ICF plasma includes electron conduction losses and mechanical ‘ pdV ’ work

$$c_{DT} \frac{dT}{dt} = f_\alpha Q_\alpha - f_B Q_{B,DT} - Q_e - \frac{1}{m} p \frac{dV}{dt} - Q_{other} \quad (14)$$

where $Q_e = 5.4 \times 10^3 T^{3.2} / (\rho^{0.8} R^2)$ or $Q_e = 5.9 \times 10^3 T^{7/2} / (\rho R^2)$ is the electron conduction loss in GJ/(g · s) units [assuming a SESAME (Lyon and Johnson, 1995) conductivity form or an Spitzer (Spitzer and Härm, 1953) form, respectively], and R is the radius (assuming spherical geometry). The term f_B is the fraction of x-rays lost to the hot plasma, which is < 1 if the optical depth of the hot region is high enough to re-absorb x-rays or > 1 if the presence of high-Z material enhances x-ray loss beyond that of what pure DT would radiate. The quantity $f_\alpha < 1$ is the fraction of alpha-particles stopped in the hot DT plasma and is related to λ_{DT} , e.g., with Eq. 7 the appropriate expression is (Zylstra and Hurricane, 2019)

$$f_\alpha = 1.0228 - 0.4254 \left(\frac{\rho\lambda}{\rho R} \right) + 0.07101 \left(\frac{\rho\lambda}{\rho R} \right)^2 - 0.004 \left(\frac{\rho\lambda}{\rho R} \right)^3, \quad (15)$$

In the presence of magnetic fields (see Section V), the f_α and electron conduction terms are modified. The pdV work term, Eq. 14, is a source of mechanical power injection into the plasma on implosion ($dV/dt < 0$) or a loss term upon explosion ($dV/dt > 0$) as the volume V surrounding the hot plasma changes. The loss term Q_{other} in Eq. 14 is associated with energy sinks that *may* exist

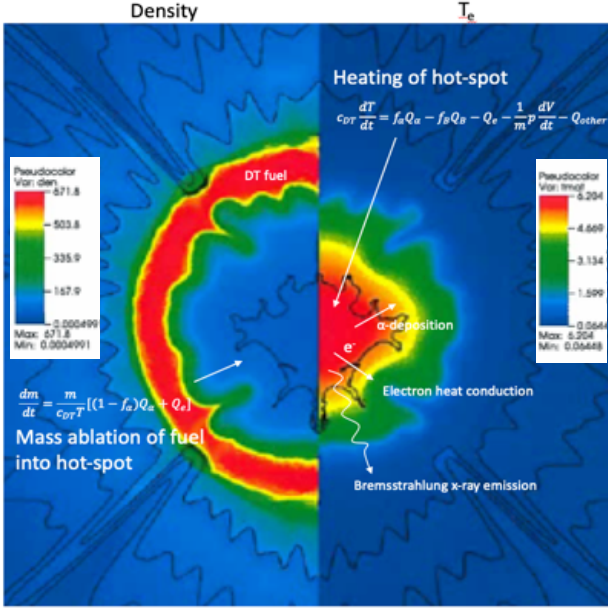


FIG. 3 A simulation image of a characteristic hot-spot ignition design implosion at peak compression illustrating what a configuration may look like. Black outlines illustrate material boundaries and the breaking of perfect symmetry results from the input of realistic surface finishes and asymmetry breaking engineering features into the simulation – while initially very small these perturbation inevitably grow as the tendency of an implosion is to act as an amplifier. The left portion of the figure shows the density (g/cm^3). The right portion of the figure shows electron temperature (keV). A few key processes are illustrated.

in a real implosion that are not captured in the conventional thermodynamic power balance.

A tipping-point in DT plasma self-heating can be obtained if the $f_\alpha Q_\alpha$ term in Eq. 14 dominates over the other terms on the right-hand-side of the power balance for sufficient time. Once this tipping-point, “ignition,” is reached thermal instability follows and T increases in a finite-time singular (explosive) fashion over the “bootstrapping” time-scale, dimensionally determined per Eq. 14 by the ratio of heat capacity and alpha heating rate at stagnation (subscript s) namely $dt/d\ln T \sim c_{DT}T/(f_\alpha Q_\alpha) \sim 2.9 \times 10^{-6} c_{DT}/(f_\alpha \rho_s T_s^{2.6}) \sim 10$'s of ps. Since ICF implosions are dynamic, thermodynamic instability is eventually terminated by expansion as the system blows itself apart.

II. HOW TO ACHIEVE A HIGH ENERGY DENSITY ICF PLASMA

Eq. 14 shows that ICF implosions are hydrodynamic systems that do mechanical work on fusion fuel to compress it to reach a high energy-density state in order to trigger thermonuclear reactions. In ICF a shell surrounding the fusion fuel is accelerated to high velocity inwards

upon itself (the aforementioned ‘implosion’) until there is nowhere to go (‘stagnation’). At stagnation the built-up kinetic energy is turned into internal energy increasing temperature and pressure in the DT. At the boundary of self-heating ($f_\alpha Q_\alpha - f_B Q_B - Q_e \approx 0$), where x-ray and conduction losses balance alpha-heating, the minimum characteristic velocity needed can be estimated from the balance between internal energy and kinetic energy as seen in Eq. 14, assuming spherical symmetry and $Q_{other} = 0$,

$$c_{DT}mT \leq \frac{1}{2}m_{shell}v_{imp}^2. \quad (16)$$

Taking $T \approx T_{ign}$ in Eq. 16 gives a velocity estimate [noting that $c_{DT} = 115 \times 10^3 (\text{km/s})^2/\text{keV}$]

$$v_{imp}(\text{km/s}) \geq 960 \sqrt{\frac{m}{m_{shell}}}. \quad (17)$$

For hot spot ignition, where only a small fraction of the DT fuel is heated $m/m_{shell} \sim 1/10$, so Eq. 17 gives $v_{imp} \geq 320 \text{ km/s}$. Designing the conditions needed to achieve these velocities (and higher) while controlling the tendency towards instability (a consequence of the tremendous accelerations/decelerations and material density gradients involved in the implosion) and asymmetry, requires great finesse and engineering control.

A. Hot spot model, adiabatic implosion, analytic solution for alpha-heating

Various implosion design strategies have been explored but hot-spot ignition, where most of the DT fuel is compressed but not significantly heated (Lindl, 1995; Lindl *et al.*, 2004; Nuckolls *et al.*, 1972), is still the leading strategy using either indirect x-ray drive or direct laser drive. In hot-spot ignition as the implosion heats, heat-conduction electrons and any alpha-particles leaving the hot-spot are stopped in the inner-most DT fuel and the heating ablates the fuel (Gus'kov *et al.*, 1976), from the inside-out, into the hot spot increasing the hot spot mass according to the rate equation

$$\frac{dm}{dt} = \frac{m}{c_{DT}T} [Q_\alpha(1 - f_\alpha) + Q_e]. \quad (18)$$

Eq. 18 underestimates the mass ablation of the fuel into the hotspot due to a boundary layer effect in the electron conduction solution (Albright *et al.*, 2021), but corrections to the electron conduction solution become less significant in the limit of $Q_\alpha \gg Q_e$. Since the energy in the hot-spot is $E_{hs} = c_{DT}mT = (3/2)pV$, one can differentiate the expression for the hot-spot internal energy, dE_{hs}/dt , and then combine the result with Eq. 18 and Eq. 14 to obtain an expression for hot-spot pressure, recovering a form popularized by Betti (Betti *et al.*, 2002)

and after a little algebra, one finds

$$\frac{dp}{dt} + \frac{5}{3} \frac{p}{V} \frac{dV}{dt} = \frac{2}{3} \rho (Q_\alpha - f_B Q_B - Q_{other}). \quad (19)$$

Since α -particles that leave the hotspot are stopped in the DT fuel, ablating it, thus returning α -particle energy to the hotspot, f_α is not explicitly present in Eq. 19 and thus the expression is insensitive to α -stopping model uncertainties. Dropping Q_B and Q_{other} , it was noted by Betti, et al., (Betti *et al.*, 2002) that, assuming $\langle \sigma v \rangle \sim T^2$ near stagnation ($dV/dt \sim 0$), Eq. 19 takes the form of a Riccati equation, which has a finite-time singular solution. In particular, one finds the pressure becomes singular when a normalized form of the Lawson criteria, $\chi_{no-\alpha} \sim p\tau/S(T)$ (‘generalized Lawson criteria,’ GLC) exceeds unity, $S(T)$ being the hot-spot volume average of $T^2/\langle \sigma v \rangle$ at stagnation, and a number of other theory works have built upon this idea (Betti *et al.*, 2010, 2015; Chang *et al.*, 2010; Cheng *et al.*, 2013; Christopherson *et al.*, 2018a; Spears *et al.*, 2012; Springer *et al.*, 2018; Zhou and Betti, 2009).

Noting that $\rho = 3p/(2c_{DT}T)$ and that $\dot{p} + 5/3(p/V)\dot{V} = p[d\ln(pV^{5/3})/dt]$ (the over-dot denoting a time derivative) allows Eq. 19 to be integrated simply, obtaining (Hurricane *et al.*, 2019a)

$$\frac{pV^{5/3}}{p_{pv}V_{pv}^{5/3}} = \exp \left[\int_0^t \frac{Q_\alpha - f_B Q_B - Q_{other}}{c_{DT}T} dt \right] \quad (20)$$

where $p_{pv}V_{pv}^{5/3}$ is an integration constant and the subscript refers to the conditions at peak velocity (pv). The assumptions leading to Eqs. 19 and 20 are only good if the 3D asymmetries are limited to low-modes, Legendre modes $\ell < 6$, since for higher modes the hot-spot ‘bubbles’ could be cold and the isobaric assumption becomes worse as flows in the hot-spot become more significant.

A diagram version of Eq. 20 is $pV^\gamma/(p_{pv}V_{pv}^\gamma) \sim \exp(E_\alpha/E_{hs})\exp(-E_B/E_{hs})$ where E_α is the α -heating energy and E_B is the energy lost to bremsstrahlung x-rays. Since E_α/E_{hs} has been identified as a key quantity for the ignition of 1D implosions (Christopherson *et al.*, 2018b; Lindl *et al.*, 2018) it is expected that a rapid and nearly discrete jump in pV^γ is strongly related to fusion yield amplification and ignition as well. Eq. 20 does not solve the pressure equation because the integral depends upon p through the density dependence of the gain and loss terms, but Eq. 20 puts the equation into a form where, for high- T (> 4.3 keV), an accurate but approximate solution can be easily found by the method of Steepest Descent (Hurricane *et al.*, 2019a,b) without making any simplifying assumptions about the x-ray loss term or reaction-rate and the resulting solution shows that many properties of the hot-spot increase exponentially with another variation of a Lawson-like parameter, e.g., $pV^{5/3}/(p_{pv}V_{pv}^{5/3}) \approx \exp[pH(T)\tau_{BW}]$, where $\tau_{BW} = \sqrt{T/\ddot{T}}$ and $H(T)$ is a complicated function of

temperature (Hurricane *et al.*, 2021), with the property that $H(T) > 0$ for $T > T_{ign}$ and $H(T_{ign}) = 0$. Namely, (see Fig. 4)

$$H(T) = \frac{\sqrt{\pi}F(T)}{\sqrt{\frac{2\gamma}{\gamma-1} + \frac{2T}{F(T)}\frac{\partial F}{\partial T}}}. \quad (21)$$

where $\gamma = 5/3$ is the polytropic index and the function $F(T)$ is a measure of the competition between alpha-heating and bremsstrahlung cooling evaluated at peak hotspot pressure,

$$F(T) = \frac{3}{2} \frac{Q_\alpha - f_B Q_B}{\rho(c_{DT}T)^2}. \quad (22)$$

The parameter f_B reflects the DT bremsstrahlung enhancement at peak hotspot pressure. Note that $F = 0$ at the ignition temperature $T_{ign} = 4.3f_B^{0.3}$ in keV. A maximum in $F(T)$ exists near 14 keV nearly (within ~ 0.5 keV) independent of the value of radiation loss parameter, f_B (see Fig. 5). The T dependence of $F(T)$ and $H(T)$ illustrate why $T \approx T_{ign}$ is insufficient for ignition and that a more sensible range of thermal temperatures to target in the design of an ICF implosion is where $F(T)$ is nearly flat ($\partial \ln F / \partial \ln T \ll 1$), maximal, and insensitive to radiation loss, namely $T \sim 8 - 14$ keV. Ignition based upon the Lawson-like formulation, $pH(T)\tau_{BW} > 1$, appears consistent across a variety of implosion designs (MacLaren *et al.*, 2021).

From Eq. 19 or Eq. 20 it can be seen that when $(Q_\alpha - f_B Q_B - Q_{other})\tau \ll c_{DT}T$ that

$$pV^{5/3} \sim p_{pv}V_{pv}^{5/3}, \quad (23)$$

meaning that the pressure and volume in an ICF implosion will behave in an adiabatic fashion as the shell of the implosion decelerates from peak velocity to zero (‘stagnation’) – this relation reflects the pressure amplification nature of an ICF implosion. This adiabatic approximation is sufficient for obtaining basic understanding of how important properties of an implosion scale with velocity, areal density, and measures of asymmetry.

B. Principal physics parameters controlling ICF implosion performance

With Eq. 23 and a simple abstracted physical picture of a shell decelerating on hot DT (see Fig. 6), a useful model can be created (Hurricane *et al.*, 2020) for understanding the key parameters controlling an implosion. The asymmetric pistons, which represent segments of the implosion shell, are characterized by a minimum and maximum areal density ($\rho\delta R_{min}$ and $\rho\delta R_{max}$) and initially have the same implosion velocity. The pistons are incompressible and constant mass, which clearly neglects some of the physics details of a real ICF implosion

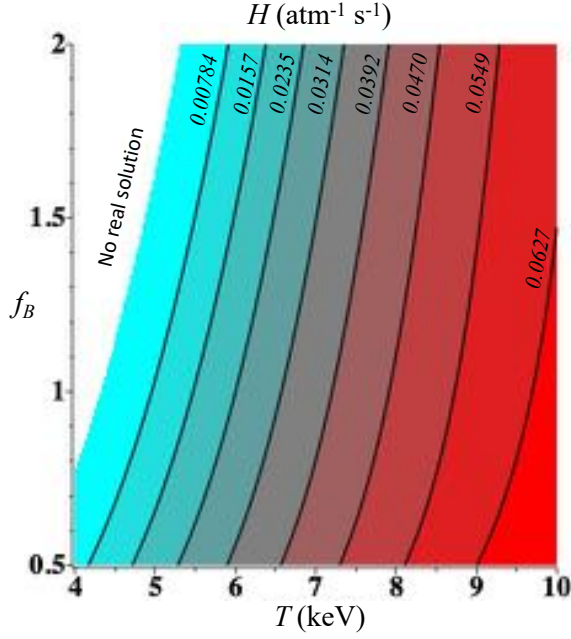


FIG. 4 The contour plot of H shows the dependence upon f_B and T . At fixed T , higher f_B lowers H essentially reflecting the fact that enhanced bremsstrahlung cooling makes the ignition temperature higher than it would be for pure DT and this makes ignition more difficult to achieve. If the implosion can be engineered to have a high enough optical depth to reabsorb bremsstrahlung x-rays ($f_B < 1$), the ignition temperature is reduced.

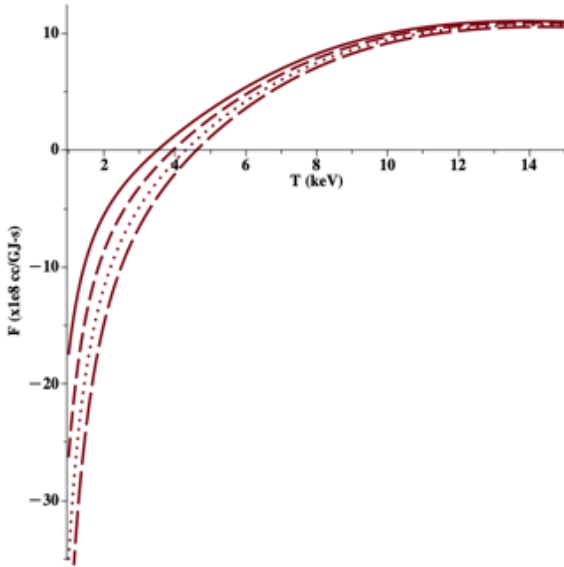


FIG. 5 A plot of $F(T)$ for four different values of f_B (solid, 0.5; dash, 0.75; dotted, 1.0; long-dash, 1.25). A maximum in $F(T)$ exists, nearly independent of f_B around ~ 14 keV and above 14 keV temperature $F(T)$ slowly declines due to the reaction-rate dependence. $F(T)$ is only positive when alpha-heating exceeds radiative loss and for $T \gg T_{ign}$ the sensitivity of F to radiative loss is of diminishing significance.

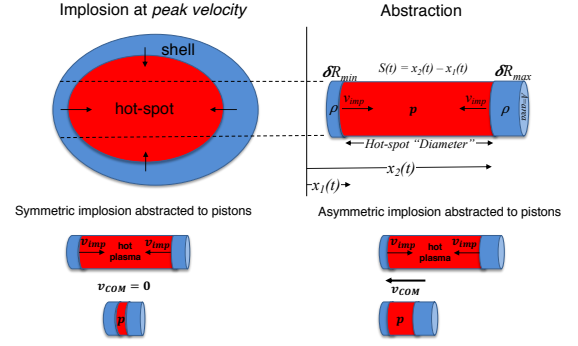


FIG. 6 An implosion with asymmetry at peak velocity is abstracted into an implosion of 2 asymmetric pistons. An analytic model constructed around this simple physical picture captures the principal impact of asymmetry on an implosion. Along the direction defined by the extremes of shell thickness variation (upper left), the implosion is abstracted into an “apple core” of thin and thick pistons at opposing ends of the hot spot (upper right). A symmetric configuration results in no center of mass motion (lower left) while an asymmetric configuration has a net center of mass motion that carries kinetic energy. Figure adapted with permission from (Hurricane *et al.*, 2020).

shell as it stagnates on the hot-spot, but as will be seen below this classical mechanics model appears to be adequate for predicting several hot-spot properties.

The asymmetric pistons are separated by a “hotspot” plasma that obeys Eq. 23. Combining the equations of motion for each piston together gives an autonomous non-linear differential equation for the piston separation, S ,

$$\ddot{S} = \frac{p_{pv}}{\rho \delta R_{ave}} \left(\frac{2}{1 - f^2} \right) \left(\frac{S_{pv}}{S} \right)^{5/3} \quad (24)$$

where f , the shell asymmetry fraction, is defined as a measure of areal density variation from the mean shell areal density

$$f = \frac{\rho \delta R_{max} - \rho \delta R_{min}}{2 \rho \delta R_{ave}} \quad (25)$$

p_{pv} is the pressure at peak velocity (pv) and S_{pv} is the piston separation at peak velocity. Data (Rinderknecht *et al.*, 2020) has corroborated that Eq. 25 is a relevant measure of asymmetry and that f is directly connected to alternate measures of asymmetry such as hot-spot drift velocity (v_{hs} , calculated from the time-integrated neutron spectra (Ballabio *et al.*, 1998) along multiple lines-of-sight that allow one to determine the bulk burn-averaged fluid velocity in the fusing plasma (Munro, 2016)). The

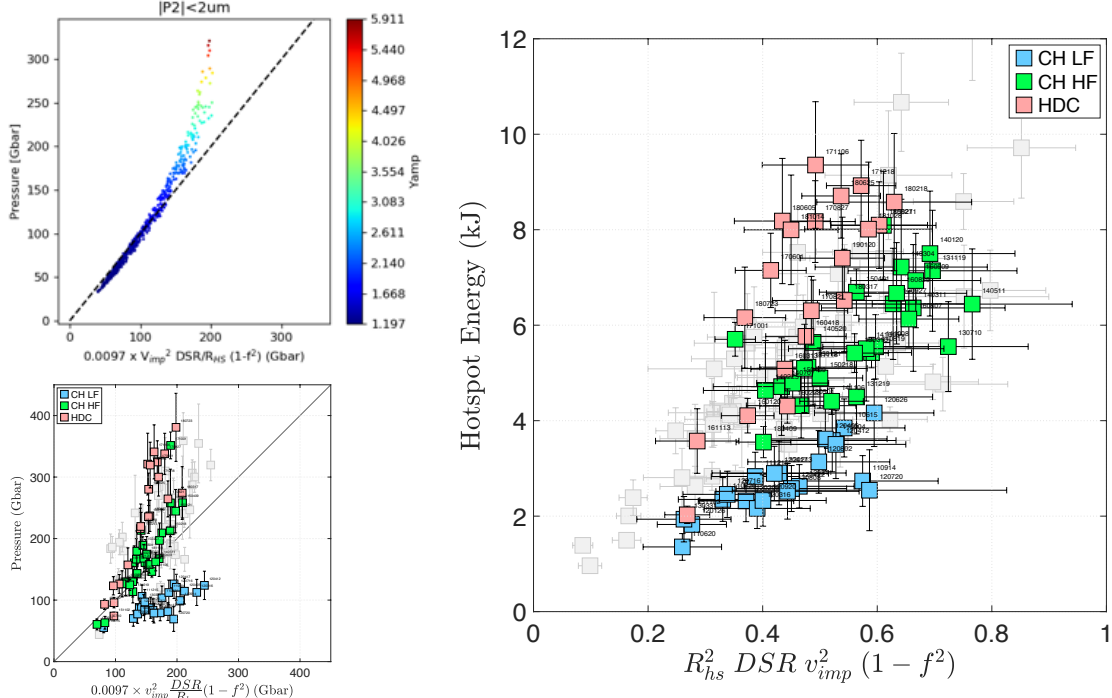


FIG. 7 Upper left: Stagnation pressures from a suite of ensemble model simulations are plotted against the piston pressure formula, $P_{stag}(\text{Gbar}) = 0.0097 DSR(\%) v_{imp}^2 (\text{km/s}) / R_{hs} (\mu\text{m})$, which is a rewriting of Eq. 26 in terms of more directly measured variables. The ratio of 10-12 MeV fusion neutrons to 13-15 MeV fusion neutrons defines DSR , which is directly related to shell ρR . Lower left: Stagnation pressures from indirect drive DT experiments on NIF are plotted against the piston pressure formula. Right: Hot-spot energy inferred from indirect drive DT experiments on NIF are plotted against the piston formula for hot-spot energy. Key implosion campaigns are highlighted in color. The data inference of stagnation pressure and hot-spot energy follow from a procedure first described in (Cerjan *et al.*, 2013; Springer, P.T. *et al.*, 2013). Figure taken with permission from (Hurricane *et al.*, 2020).

parameter f is related to the concept of residual kinetic energy (RKE , e.g. (Kritcher *et al.*, 2014)) via the relationship $f = v_{hs}/v_{imp} \approx \sqrt{RKE/(c(\alpha_{if}) \cdot KE)}$, where $KE = \frac{1}{2} m_{shell} v_{imp}^2$ and $c(\alpha_{if})$ is a fuel adiabat (α_{if}) compressibility factor, $c(\alpha_{if}) = 1/(1 + 1.53/\alpha_{if}^{1.35}) \sim 0.66$ on average and typically ranges between 0.5-1 (Hurricane *et al.*, 2022). It is useful to note that in an implosion shell KE is split into a portion that compresses the hotspot and a portion that compresses the shell itself and that higher α_{if} shifts more of the shell KE into the hotspot, i.e. $E_{cold,DT}/E_{hot,DT} = 1/c(\alpha_{if}) - 1$.

Eq. 24 is analytically solvable and from the solution expressions for hot-spot diameter, the stagnation pressure, hot-spot energy at stagnation, inertial confinement-time, Lawson product ($p\tau$), hot-spot temperature, and fusion yield can be obtained when the levels of α -particle self-heating are limited to levels of yield amplification $Y_{amp} < 2$. A subset of key scaling expressions are (Hurricane *et al.*, 2020)

$$P_{stag} \sim \frac{v_{imp}^5}{p_{pv}^{3/2}} \left[\frac{2c(\alpha_{if})\rho\delta R_{ave}}{S_{pv}} (1 - f^2) \right]^{5/2} \quad (26)$$

$$E_{hs} \sim A c(\alpha_{if}) \rho \delta R_{ave} v_{imp}^2 (1 - f^2) \quad (27)$$

$$P_{stag} \tau \sim c(\alpha_{if}) \rho R_{ave} (1 - f^2) v_{imp}. \quad (28)$$

where $A = m_{shell}/(2\rho\delta R_{ave})$ is the piston cross-sectional area. The ratio of Eq. 28 to Eq. 26 gives the inertial confinement-time. As already implied by Eq. 17, Eqs. 26-28 reflect the importance of velocity. In addition to high v_{imp} , high ρR_{ave} , low f , low p_{pv} , and low S_{pv} are seen to be desirable properties. A three dimensional extension of the piston model indicates that the impact of mode 2 asymmetry can be captured by the above relations by substituting for $1 - f^2$ the ratio of hotspot area weighted harmonic mean (WHM) of ρR to the area weighted average ρR (Hurricane *et al.*, 2022). Namely,

$$\frac{\rho\delta R_{WHM}}{\rho\delta R_{ave}} = \frac{(\int dA)^2}{\left(\int \frac{dA}{\rho\delta R}\right) (\int \rho\delta R dA)} \quad (29)$$

$$\frac{RKE}{c(\alpha_{if})KE} = 1 - \frac{\rho\delta R_{WHM}}{\rho\delta R_{ave}} \quad (30)$$

where the integrals are taken over hotspot surface. Assuming a power-law fusion reaction-rate dependence, $\langle\sigma v\rangle \sim T^a$ then gives a fusion yield degradation (in the low alpha-heating limit) due to asymmetry of $(1 - f^2)^a$ or $(\rho\delta R_{WHM}/\rho\delta R_{ave})^a$, respectively for mode-1 or general

asymmetry. The asymmetry induced yield degradation with significant alpha-heating is more complicated, but a formula is available (Hurricane *et al.*, 2022). Other models have also pointed to the importance of the areal density harmonic mean (Woo and Betti, 2021). The key parameters for reaching ignition suggested by the piston model are the Lawson product, Eq. 28, reflects the importance of high shell (piston) areal density, high peak implosion velocity, and minimal asymmetry for a generic implosion.

Asymmetries can originate from capsule fabrication (Casey *et al.*, 2021) and/or from drive imbalances (MacGowan *et al.*, 2021). The fact that good symmetry is important for converting the implosion kinetic energy into useful thermal energy at stagnation is not surprising, as noted by many researchers (Bodner, 1995; Bose *et al.*, 2018, 2017; Cheng *et al.*, 2018; Chittenden *et al.*, 2016; Gatu Johnson *et al.*, 2016; Gu *et al.*, 2014; Kritcher *et al.*, 2014; Scott *et al.*, 2013; Spears *et al.*, 2014; Springer *et al.*, 2018; Woo *et al.*, 2018). However, there has been debate over what aspect of asymmetry (e.g. hotspot, shell areal density, time dependent of either vs. at stagnation) is key. This piston model works well when compared against simulations and data (see Fig. 7) and it provides a concise statement of what aspect of asymmetry, namely shell areal density asymmetry at stagnation, is most important. The essential physics of asymmetry illuminated by the piston model is that in the lab frame of reference, some elements of the shell are still moving at the time of peak compression, thus the kinetic energy of those moving elements is not converted into internal energy at peak compression.

C. Implosion acceleration and hydrodynamic instability

Asymmetry is not the only potential degradation that impacts implosions. Inwardly accelerating shells of material to high velocity and then having them decelerate as their central pressure builds puts the implosion at risk of unstable growth wherever the gradient in pressure, $\vec{\nabla}p$, is oppositely directed of the density gradient, $\vec{\nabla}\rho$, i.e., wherever $\vec{\nabla}p \cdot \vec{\nabla}\rho < 0$. Any small imperfection can be amplified and, if the growth is severe enough, will defeat the ability of the implosion to effectively heat and compress DT because the shell that provides the inertial confinement gets shredded by the instability and will (in some cases) inject cold shell material in the central hot-plasma region cooling it by enhancing the bremsstrahlung emission due to the $\bar{Z}$ dependence of Eq. 4.

Rayleigh-Taylor (RT) instability (Rayleigh, 1883; Taylor, 1950), under continuously varying acceleration, and Richtmyer-Meshov (RM) instability (Meshkov, 1969; Richtmyer, 1960), under impulsive accelerations are the primary modes of concern. Magnetically driven implosions have additional instabilities to be concerned about

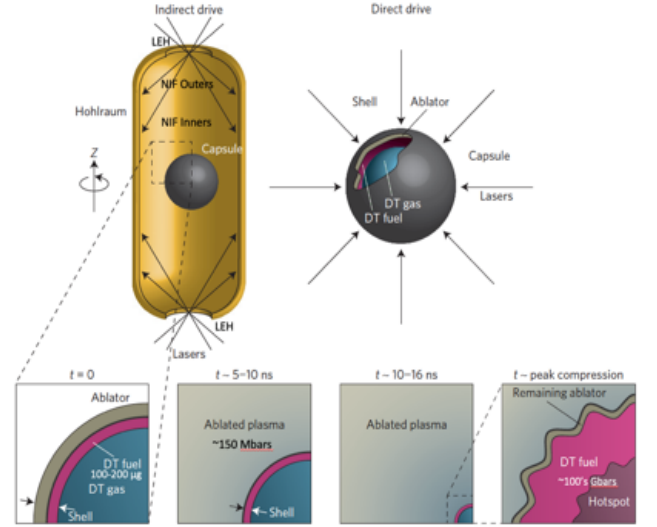


FIG. 8 Typical targets used in laser-driven ICF are indirectly driven (upper left, where the laser energy is incident on a hohlraum generating a bath of x-rays) or directly driven (upper right, where the laser beams are directly incident on the capsule). In either case, a spherical capsule is prepared at $t = 0$ with a layer of DT fuel on its inside surface. As the capsule surface absorbs energy and ablates, pressure accelerates the shell of remaining ablator and DT fuel inwards, an ‘implosion.’ By the time the shell is at approximately one-fifth of its initial radius it is imploding at a speed of many hundreds of kilometers per second as dictated by Eq. 17. By the time the implosion reaches minimum radius, a hotspot of DT has formed, surrounded by colder and denser DT fuel. The time-scales shown in the row of frames at the bottom are characteristic of plastic ablaters. The time scales for higher density ablaters, like high density carbon (HDC), are generally shorter owing to their reduced thickness. Figure taken with permission from (Betti and Hurricane, 2016).

such as electro-thermal and kink modes as well as the magnetohydrodynamic equivalent of RT (MRT) and RM modes (see Section V.B.4).

For linear instability (which assumes perturbation amplitude $<$ perturbation wavelength), the growth of the RT instability at the ablation-front in an ICF implosion is a product of the effect of convergence [essentially mass conservation in the form known as the Bell-Plesset effect (Bell, 1951; Plesset, 1954)] and exponential growth.

The effect of convergence on the amplitude of perturbation growth is estimated as (Goncharov *et al.*, 2000a)

$$\eta_{BP} = C_R^{-1/4} \exp \left[\sqrt{2\ell} \left(\arcsin \sqrt{1 - 1/C_R} - \sqrt{1 - 1/C_R} \right) \right] \quad (31)$$

where $C_R > 1$ is the convergence ratio which is the ratio of initial to final radius of the unstable region and ℓ is the Legendre-mode number.

The exponential growth-rate is approximately given

by (Betti *et al.*, 1998a)

$$\gamma_{A-RT} = \alpha_2(Fr, \nu) \sqrt{\frac{kg}{1 + kL_\rho}} - \beta_2(Fr, \nu)kv_a \quad (32)$$

where $k = 2\pi\ell/R$ is the perturbation wavenumber, g is the shell acceleration, $L_\rho = \rho/|\vec{\nabla}\rho|$ is the density gradient scale-length of the ablation front, v_a is the velocity of ablation [a stabilizing term first identified by Bodner (Bodner, 1974) and described as “fire-polishing”], while α_2 and β_2 are parameters of order unity whose exact values depend upon a heat-conduction scale-length parameter, ν , and the Froude number, $Fr = v_a^2/(gL_\rho)$. Combining Eq. 31 and 32 then gives the total mode dependent growth-factor, $G(\ell)$ (ratio of amplitude of final perturbation to initial amplitude) $G(\ell) \sim \eta_{BP} \exp[\int \gamma_{A-RT}(t)dt]$. As a consequence of convergence and exponential growth, the degree of unstable perturbation growth is potentially many orders of magnitude (see Fig. 9).

In the low-mode limit $kL_\rho \ll 1$, a simple analysis of Eq. 32 suggests an important design parameter for controlling instability growth. Namely, taking $\partial\gamma_{A-RT}/\partial k = 0$ gives the maximum growing mode, $k_{max} \approx \alpha_2^2 g/(\beta_2^2 v_a^2)$, and maximum growth-rate, $\gamma_{max} \approx \alpha_2^2 g/(4\beta_2 v_a)$. Noting that $R \sim gt^2/2$ and $2g\delta R \sim v_a^2$, one finds that $\int \gamma_{max} t \sim [\alpha_2^2/(4\beta_2)]\sqrt{IFAR}$, where $IFAR = R/\delta R$ is the “in-flight aspect ratio”, highlighting the instability danger of accelerating a thin shell.

The particular form of Eq. 32 is from (Betti *et al.*, 1998a), but numerous variations of Eq. 32 exist in the extensive literature on this topic and equations of that form are often referred to as the Takabe formula (Takabe *et al.*, 1985, 1983). Eq. 32 is enough to see that, given an initial condition of perturbation seeds, there are three principle ways to mitigate RT instability in an ICF implosion: increase the ablation velocity, increase the density gradient scale-length, or reduce convergence.

Because the shell of an ICF implosion has a finite thickness, perturbation growth on the outside of the shell feeds though to the inside of the shell (and vice versa) as a consequence of shock-imprinting, continuity (mass conservation), and the circulating flows inside the shell that bouyancy-driven instabilities generate. Post-shock, for a given thickness of shell, δR , any perturbation one side of the shell develops on the other, but attenuated by a factor of $\exp(-k\delta R)$ (Mikaelian, 1995), which can lead to a complex nonlinear interaction between the inside and outside of the shell.

Interfaces of the shell, such as the fuel-ablator interface, can be subject to very high mode growth, with growth-rate $\gamma \sim \sqrt{k}A_t g$, if the density jump, as measured by the Atwood number (A_t , being the density difference across an interface over the sum of densities across that same interface), is unfavorable. Nonlinearly (for modes with perturbation amplitude $>$ perturbation

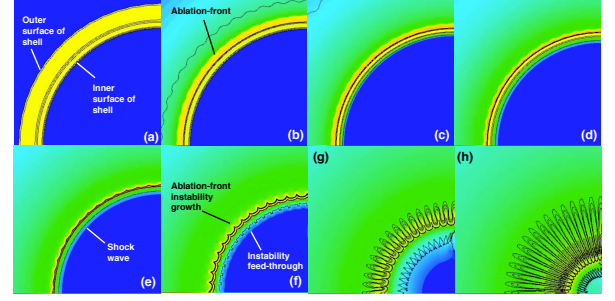


FIG. 9 A simulation of instability growth on an indirectly-drive ICF gas capsule implosion is shown in a variable logarithmic density color-scale from low (blue) to high (red). The particular implosion design is that of the “low-foot” (see Section III.E.1 of this article). A small amplitude mode-60 perturbation is imposed on the outside of the capsule at $t = 0$ (a). (b) By $t = 10$ ns the outer layer of the capsule has ablated away and the shell has compressed up. (c) By $t = 15$ ns the shell begins to rapidly accelerate inwards. (d) By $t = 16$ ns perturbation growth on the outside of the capsule becomes visible. (e) By $t = 18$ ns the shockwave that transited the ablator and entered the gas races inward ahead of the shell. (f) By $t = 19$ ns instability growth on the inside of the shell develops. (g) By $t = 20$ ns the amplitude of instability growth exceeds the thickness of the shell. (h) By $t = 21$ ns the shell of the implosion has been shredded by instability

wavelength), the amplitude growth is expected to asymptote to a free-fall scaling, $\sim gt^2$ [e.g., (Dimonte *et al.*, 2004)]. While the instability behavior of ICF implosions can be understood analytically in special situations, usually computer simulations are used to assess the potential for instability growth and mixing (e.g., Fig. 9).

Like all areas of fusion research, controlling hydrodynamic instability growth is a major challenge. None of the three major ICF thrusts (laser indirect drive, laser direct drive, and magnetic direct drive) are immune from instability concerns. However, the idea of “starting over,” advocated by J. Nuckolls in the 1970’s, with a smooth field of x-rays driving a capsule was thought to be the best chance of minimizing instability problems and drove early work at Livermore Lab.

III. X-RAY DRIVEN IMPLOSIONS

A. Brief History

At Lawrence Livermore National Laboratory (LLNL), the first laser fusion program, “Q group” led by Ray Kidder, spanned the period of 1962-1972 (Nuckolls, 2006) (and it later became apparent the Soviets also had an ICF program led by Nikolai Basov starting during this period). During that period the key target design simulation tool LASNEX (Zimmerman and Kruer, 1975) was developed and that was the principle tool used to study

ICF at LLNL for many decades. In 1971, elements of ICF began to be declassified and made public (but not the concept of x-ray drive), e.g., the famous Nuckolls *et al.* paper (Nuckolls *et al.*, 1972). Some aspects of ICF remain classified even today.

From 1972-1992 a second laser fusion program existed, centered around the 10 kJ class SHIVA laser system from 1977-1982, and it was with this laser system that the first indirectly-driven ICF targets were demonstrated. A second higher energy and shorter wavelength laser system, NOVA (operating from late 1984-1999) later replaced SHIVA as it became apparent the laser wavelength was a key issue for mitigating deleterious laser-plasma interactions.

In 1993 plans were laid for the mega-Joule class NIF laser system with the physics basis described in a review paper by John Lindl (Lindl, 1995). Construction of NIF began in 1997 with the laser fully being completed about a decade later. NIF is presently the primary indirect-drive experiment facility in the U.S. In principal, indirect-drive implosions should have an advantage at small spatial-scales because the x-ray drive generated has a smoothing effect which mitigates high-mode asymmetry and fluid-instabilities. The sacrifice made for the smoothing effect of x-rays is in terms of energy coupled to the implosion, because a large fraction of energy delivered to an indirect-drive target is absorbed by the walls of the x-ray generating cavity known as the “hohlraum,” because the capsule surface area is small compared to the hohlraum surface area.

B. Indirect drive with lasers

A principal mechanism to drive an implosion inwards upon itself and achieve high implosion speeds, that estimates suggest are needed, (e.g., Eq. 17), is to expose the outer surface of a shell of material (an “ablator”), to an intense bath of x-rays (see Fig. 8). Absorption of x-ray energy by a thin layer of ablator material will ionize it, generate high pressures (of order hundreds of Mbar) producing an inward-directed acceleration that literally rockets the remaining ablator and DT fuel to high velocity. The generation of ablation pressure is accompanied by the generation of inwardly directed shock waves (Guderley, 1942) the control of which are a key aspect of any modern design (Haan *et al.*, 2011; Landen *et al.*, 2010, 2011; Lindl, 1995; Robey *et al.*, 2012b). The implosion effectively amplifies the ablation pressure, e.g., Eq. 23, as the volume shrinks to the levels required, e.g., Table I, but a majority of the x-ray energy is consumed in the process as the majority of the ablator (typically 90-95%) is turned from a solid state into a plasma.

1. Hohlraum Physics

As laser energy is deposited into the hohlraum, power balance determines the radiation temperature (T_r) of any high energy density physics (HEDP) hohlraum,

$$P_{laser} = \sigma T_r^4 (1 - a_h) A_H, \quad (33)$$

where P_{laser} is the input power, σ is the Stefan-Boltzman constant (1.03×10^{11} MJ/cm²·s·keV⁴), a_h is the hohlraum wall albedo which scales as (Lindl, 1995)

$$1 - a_h \sim T_r^{-2/3} t^{-1/2}, \quad (34)$$

where A_H is the inner hohlraum wall area. Eq. 33 reasonably assumes that losses are dominated by wall absorption rather than loss through apertures such as the laser entrance holes (LEH's) or to a capsule, assumptions that are not correct when the LEH area or capsule area are a significant fraction of A_H . Eq. 34 reflects the fact that energy diffuses into the hohlraum wall in the form of Marshak (Marshak, 1958) wave – the most comprehensive modern treatment being found in (Hammer and Rosen, 2003).

Integrating Eq. 33 with Eq. 34 gives a useful scaling of T_r with laser energy (E_{laser}), namely

$$E_{laser} \sim T_r^{3.3} A_H \sqrt{t_{laser}}, \quad (35)$$

where t_{laser} is the duration over which laser power is applied. With a prescribed temporal history of laser power and with consideration of the details of capsule heat capacity and LEH losses, a more precise model of $T_r(t)$ based upon Eq. 33 can be obtained that compares favorably to data and that is useful for scoping studies (Callahan *et al.*, 2020) because it has a much shorter time-to-solution than massive computer simulations [e.g., (Marinak *et al.*, 2001; Suter *et al.*, 1996; Zimmerman and Kruer, 1975)] that are a primary tool for the field – the status of common baseline simulations for laser indirect-drive can be found in (Clark *et al.*, 2018; Kritcher *et al.*, 2018a).

The need to get laser energy into the hohlraum while also minimizing x-ray energy losses back out the way the lasers came-in leads to hohlraum geometries (e.g., see Figure 8) and individual laser beam pointing's that greatly complicate the symmetry control of the implosion. The hohlraum can drive asymmetry because the sky of x-ray flux, as seen by the capsule, is non-uniform and changing in time. Much like hydro-instability, an implosion is particularly sensitive to x-ray drive asymmetry during the periods of peak acceleration (Masse *et al.*, 2019). Asymmetry is typically expressed by amplitudes of Legendre polynomials (e.g., P_0 , P_2 , P_4 standing for the amplitude of Legendre mode 0, 2, and 4, respectively). Principal techniques of managing P_2 and P_4 symmetry control of hohlraum driven implosions involve

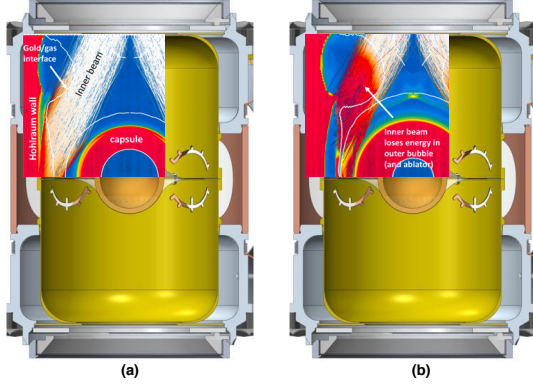


FIG. 10 Coupled hohlraum-capsule simulation images are inset on-top of a typical indirect-drive target geometry. (a) At time early in peak laser power, in the shallow polar angle ‘inner’ beams propagate to their intended location on the inner hohlraum wall unimpeded. (b) At later time, a plasma bubble generated by the incidence of the large polar angle ‘outer’ beams (not shown) interferes with the propagation of the inner beam, upsetting the intended x-ray symmetry inside the hohlraum. Inset figures taken with permission from (Callahan *et al.*, 2018).

adjustments to beam pointing, rebalancing power between sets of beams [‘cone fraction’, e.g., (Glenzer *et al.*, 2010; Town *et al.*, 2011)], hohlraum gas-fill (usually helium gas), and designing around plasma features inside the hohlraum (see Fig. 10) that evolve to obstruct the desired laser energy deposition (Callahan *et al.*, 2018; Ralph *et al.*, 2018).

Depositing laser energy into a hohlraum is often somewhat frustrated by LPI (MacGowan *et al.*, 1996), which occurs when the laser wave beats resonantly with an outgoing scattered light wave and one or more plasma waves. When the laser wavevector, $\vec{k}_L$, and laser frequency, ω_L , satisfy the phase-matching conditions, $\vec{k}_L = \vec{k}_s + \vec{k}_p$ and $\omega_L = \omega_s + \omega_p$ (where s and p stand for scattered wave and plasma wave respectively), unstable amplification of the scattered wave and plasma wave will occur (Kruer, 2003). The locations where the local electron plasma frequency, $\omega_{pe} = \sqrt{e^2 n_e / \epsilon_0 m_e}$ (n_e being electron density and ϵ_0 being the vacuum permittivity), matches ω_L defines the “critical surface” and critical electron density, n_c – resonant absorption of laser energy occurs in the plasma with number density at or below n_c . One quarter critical density, $n_c/4$, is the limiting density for electron plasma wave Stimulated Raman Scattering (SRS) and two-plasmon decay instabilities. While acoustic plasma wave Stimulated Brillouin Scattering (SBS), parametric decay, and filamentation instabilities can occur at densities up to and including n_c . The principal tactic for dealing with LPI in ICF experiments is to keep n_e low (Hinkel *et al.*, 1998) and T_e (electron temperature) high in regions where laser needs to propagate which reduces the chance of LPI occurring. In some cases LPI, in the form of cross-beam energy transfer (CBET), can be used to

intentionally direct energy from one set of beams to others changing the effective cone-fraction (Kritcher *et al.*, 2018b; Michel *et al.*, 2011, 2009; Moody *et al.*, 2012), which can be a useful symmetry control tactic.

C. Z-pinch driven hohlraums

Lasers are not unique in their ability to generate x-rays. Magnetic z-pinchs are also ubiquitous x-ray sources and, moreover, they avoid LPI. For example, the Z facility at Sandia National Laboratories is a pulsed power accelerator with 6 MJ of stored electrical energy that discharges in 120 ns. On Z, the peak currents delivered to the load approach 27 MA. Wire-array loads (see Section V.A.4) have been the main approach to producing x-rays on Z for use in high energy density physics and inertial confinement fusion experiments. In particular, vacuum hohlraums (Hammer *et al.*, 1999), which are suitable for driving capsules, have achieved peak radiation temperatures (T_r) of 90 eV. This temperature is achieved in a secondary hohlraum driven by a primary hohlraum containing a wire array pinch. ‘Hybrid hohlraum’ experiments have heated a small secondary hohlraum to (T_r) of about 145 eV (Olson *et al.*, 2001), but lack the control of illumination symmetry afforded by the vacuum hohlraum. In contrast, hohlraums at the Omega laser in Rochester regularly achieve 200 eV peak radiation temperatures while, in the NIF, 300 eV is typical. Part of the problem of obtaining higher radiation temperatures using wire-arrays is that they are not compact compared to laser driven targets.

In wire-array loads, explosive resistive heating of the fine wires quickly generates a sheath of metal plasma as the Z machine discharges. This cylindrical sheath of conducting plasma is driven radially inward by the Lorentz force, $\mathbf{J} \times \mathbf{B}$ ($\mathbf{J}$ being current density and $\mathbf{B}$ being magnetic field, i.e., the z-pinch effect). As the plasma implodes on the z-axis, x-rays are produced, with approximately 10% variation from shot to shot. The radius of the plasma sheath, R_p , is related to the peak current, I_p , and the current rise time, t_R through magnetohydrodynamic (MHD) force balance, $\rho_s dR_p/dt = -\nabla(B^2/2\mu_0)$, which yields the dimensional analysis scaling

$$R_p \sim \frac{\sqrt{I_p t_R}}{\rho_s^{1/4}} \quad (36)$$

where ρ_s is the density of the plasma sheath. The energy imparted to the load must then scale as $E \sim R_p^3 B^2 \sim I_p^{5/8} t_R^{1/2} / \rho_s^{1/4}$.

From Eq. 33 and assuming fixed hohlraum area, one finds that the radiation temperature scales as $T_r \sim I_p^{5/8} \rho_s^{-1/16} t_R^{-1/8}$ for constant hohlraum albedo. Since the hohlraum albedo is generally not constant, Eq. 34, a more accurate radiation temperature scaling is found to

be

$$T_r \sim \frac{I_p^{3/4}}{\rho_s^{3/40}} \quad (37)$$

which, interestingly, is independent of current rise time. The sub-linear scaling of T_r with current reflects the difficulty of obtaining high radiation temperatures in magnetically driven hohlraums. However, while any load design must adhere to the dynamical scaling of Eq. (36), higher radiation temperatures can, in principal, be obtained, thereby breaking the scaling of Eq. (37) by forcing the radiating plasma into a smaller primary/secondary geometry (Hammer, 2016).

D. Ablator Physics

The hohlraum generated x-ray energy is deposited into a unit area of ablator surface given by Planck's expression for radiant energy flux, σT_{rad}^4 . If the flow of radiation into the ablator is subsonic, which is generally the case in an x-ray driven implosion, the x-ray energy flux onto the ablator, corrected for the ablator albedo (a_{abl}), is balanced by the rate of work done in expanding the ablator plasma at the ablation pressure ($p_{abl,x}$), $c_s p_{abl,x} \sim \sigma T_r^4 (1 - a_{abl})$, but here the speed is the sound speed of the isothermal rarifying plasma, $c_s \sim \sqrt{p/\rho} \sim \sqrt{((Z+1)/A)(k_B T_e/m_p)}$, with the use of Equation 2. So, assuming $T_e \approx T_r$ in the ablator, the x-ray driven ablation pressure scales as

$$p_{abl,x} \sim \sqrt{\frac{\bar{A}}{Z+1}} (1 - a_{abl}) T_r^{(7/2)}. \quad (38)$$

This pressure generates the force that accelerates the implosion to high velocity as described in Section II. In order to obtain the required implosion velocities, a good x-ray ablator must be a material that is fairly dense, will readily ionize, and not be too opaque to the incident x-rays – a condition which generally excludes consideration of materials much above carbon on the periodic table. For realistic ablator materials, the effect of a albedo is to reduce the T_r scaling in Eq. 38 from $T_r^{3.5}$ to $T_r^{(2.4-2.9)}$ depending upon material (Lindl *et al.*, 2021).

Of course, ablator materials start in the solid state and are then shock compressed and melted as the ablation pressure builds in response to the x-rays incident upon the capsule. Melting the ablator upon initial shock, without subsequent refreezing, is an important aspect of implosion design especially for crystalline materials. In order the guarantee a melted state, the pressure of the initial shock must exceed the melt pressure, p_{melt} , and the post-shock ablator material must subsequently maintain a high temperature. Table III lists the melt pressure for common indirect drive ablator materials based upon references (Benedict *et al.*, 2014; Fratanduono *et al.*, 2016).

An additional, yet not completely understood, complication of crystalline ablators is the effect of grain structure which can generate post-shock-front velocity perturbations (Ali *et al.*, 2018) that seed hydro-instability; perturbations that are minimized with complete transition to melt before the implosion proceeds.

TABLE III Some useful characteristics of typical ablator materials. Initial room temperature density can vary with crystal grain size (e.g. nano-scale or micro-scale) and with the presence of any dopant atoms. Typical 1st shock pressure design target, p_{1st} , varies with desired fuel adiabat, but is usually chosen to minimally exceed the melt pressure, p_{melt} . Due to its amorphous nature glow discharge polymer (GDP) plastic (a.k.a. “CH” due to its atomic composition) allows for a large range of p_{1st} , the numbers below being associated with the low- and high- foot designs of Section III.e.

Ablator	Approx. ρ_0 (g/cc)	p_{1st} (Mbar)	p_{melt} (Mbar)
GDP	1.1	1.7-3.4	1.0
Be	1.9	3.5	2.6
B ₄ C	3.5	3.4	2.2
HDC	3.3 <i>nano</i> - 3.5 <i>micro</i>	12-14	12

Early in the implosion, when the central pressures are small, the trajectory of the implosion is well approximated by a simple rocket model (Saillard, 2006). Namely,

$$m_{shell} \frac{dv_{imp}}{dt} = -4\pi R_{shell}^2 p_{abl,x} \quad (39)$$

$$\frac{dm_{shell}}{dt} = -4\pi R_{shell}^2 \dot{m}_a \quad (40)$$

where R_{shell} is the ablation-front radius, $v_{imp} = \dot{R}_{shell}$ is the mean implosion velocity of the remaining capsule shell, and $\dot{m}_a = k_m p_{abl}/\sqrt{T_r}$ is the mass ablation rate per unit area (k_m being a numerical constant). Note that in Eq. 39 the reduction in x-ray absorption efficiency due to the capsule surface area shrinking in time ($\sim R_{shell}^2$), is compensated for by mass ablation (m_{shell} also declines in time), so acceleration at late time can still be significant even though the capsule radius is small. If T_r , and therefore $p_{abl,x}$, is constant in time over the duration of the implosion then Eqs. 39 and 40 have a simple “rocket-equation” solution, which is $v_{imp} = (\sqrt{T_r}/k_m) \ln[m_{shell}(0)/m_{shell}(t)]$ – that is, high velocities are obtained by ablating most of the shell and/or by having a very high T_r in the hohlraum. The ratio of late-time shell mass, $m_{shell}(t)$, to the initial shell mass, $m_{shell}(0)$, is the fractional “mass-remaining,” which plays a key role in managing the integrity of the implosion as the need for high velocity is traded-off against the need for hydrodynamic stability control and good inertial confinement at stagnation. In the thin-shell approximation, the replacement $p_{abl,x} \rightarrow p_{abl,x} - p$ in Eq. 39, coupled with Eq. 14 and Eq. 18 or Eq. 19, provides

a complete 1D dynamic model of a DT implosion and hotspot with the deceleration and explosion phases.

In typical indirect-drive designs, $p_{abl,x}$ is not constant in time, as assumed in the solution in the paragraph above, but is instead more typically a triangular-shaped function (see Fig. 11), resulting in a more complex solution to Eqs. 39 and 40 than the rocket-equation solution. In dimensionless radius and time terms, the solution of Eqs. 39 and 40 is determined by three dimensionless parameters, $4\pi R_0 p_{max} t_{max}^2 / m_{shell}(0)$, $4\pi R_0^2 k_m p_{max} t_{max}^{6/7} / m_{shell}(0)$, and $p_{max} / (\dot{p}_{cool} t_{max})$, where R_0 is the initial shell radius, p_{max} is the peak ablation pressure, t_{max} is the rise time of peak ablation pressure, and $\dot{p}_{cool}$ is the rate of ablation pressure decline after peak pressure that results from hohlraum cooling. The time duration between when $p_{abl,x}$ peaks and implosion stagnation is termed “coast time,” albeit that definition is not unique. The coast time is an important aspect of an implosion design since experimentally short coast-times strongly correlate with many measures of improved hotspot performance (see Fig. 12) and compression (Zylstra *et al.*, 2014), such as the stagnation pressure (Döppner *et al.*, 2020; Hurricane, 2016; Hurricane *et al.*, 2016a, 2017b) (see Fig. 13) and total areal density (Landen *et al.*, 2020, 2012; Lindl *et al.*, 2014; Robey *et al.*, 2013). Reducing the coast time results in less sensitivity to $\dot{p}_{cool}$, a reduction in the radius at which peak velocity is achieved (Hurricane *et al.*, 2022), e.g., a reduction in the $S_{pv} \sim 2R_{pv}$ term in Eq. 26, and this results in a higher deceleration rate and increased rate of mechanical power transfer as kinetic energy is converted into internal energy (Callahan *et al.*, 2020). For example, a key metric for ignition is the product of hotspot stagnation radius and pressure, $p_{stag} R_{hs}$ (which is equivalent to the product of hotspot ρR and T) and this quantity rapidly increases with reduced R_{pv} (Hurricane *et al.*, 2022)

$$p_{stag} R_{hs} \sim \frac{m_{shell} c_s^2}{4\pi R_{pv}^2} (1 - f^2) c(\alpha_{if}) \quad (41)$$

where c_s is the hotspot plasma sound speed. In the expression above, it has been assumed that there is enough alpha-heating to offset x-ray loss cooling in addition to the plasma heating coming from compression. Eq. 41 is the form of a kinetic energy areal density and we note that $p_{stag}^2 E_{hs} = 2\pi (p_{stag} R_{hs})^3$. For a coast time much less than the cooling time of the hohlraum (τ_{cool}), R_{pv} saturates at a minimal value due to hotspot back pressure, generally of the order of 100-200 μm .

E. Present status of results

The National Ignition Campaign (NIC) (Lindl and Moses, 2011) focused upon demonstrating ignition and thermonuclear burn in the laboratory based upon a

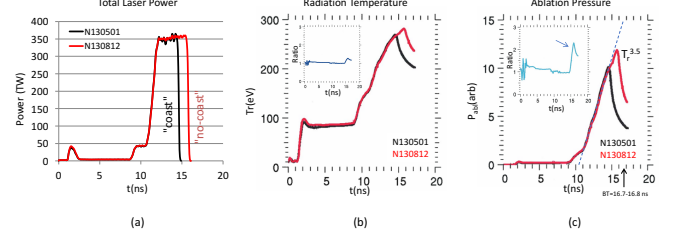


FIG. 11 Example laser pulse-shapes, hohlraum radiation temperatures, and ablation pressures for two indirect-drive experiments on the NIF. (a) Laser power vs. time for ‘coasting’ 1.3 MJ pulse, N130501, and ‘no-coasting’ 1.7 MJ pulse, N130812. The colloquial terms ‘coast’ and ‘no-coast’ are misnomers since both of the pulse-shapes shown result in implosions that coast. It is just that the longer duration pulse results in less coasting. (b) The resultant simulated hohlraum radiation temperatures for shot N130501 and shot N130812 with the ratio of the two shown in the inset. (c) The ablation pressure (arbitrary units) that results from $T_{rad}^{7/2}$ with the ratio of the two shown in the inset. The shape of the ablation pressure in time is nearly triangular with a near-linear rise and fall. Time duration between bang-time and the peak in ablation pressure is the coast-time. Inset figures taken with permission from (Hurricane *et al.*, 2017b).

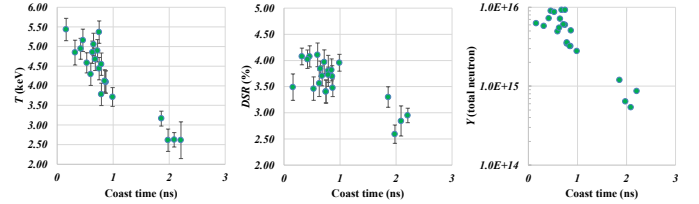


FIG. 12 Data from the high foot series of experiments (2013-2014) show a clear correlation between burn average hotspot ion temperature as measured by neutron time-of-flight detectors (left frame), the neutron down-scatter ratio, DSR , which is the ratio of 10-12 MeV fusion neutrons to 13-15 MeV fusion neutrons and therefore measure of compression (middle frame), and total fusion neutron yield (right frame).

“point-design” that was thought to be capable of ignition and burn based upon a decade of work on the Nova and Omega facilities. Post-NIC, progress on the laser indirect-drive approach to ICF at NIF (Moses *et al.*, 2016) has come in bursts, as new design strategies address limiting physics issues identified as the experimental program on NIF has unfolded (see Table IV). These new design strategies (“basecamp”) are then pushed to their maximum performance limits using the implosion control parameters of velocity, coast time, or adiabat [a measure of DT fuel entropy before stagnation (Haan *et al.*, 2011)] while trying to manage hydrodynamic instabilities, symmetries, and laser backscatter. Exploring the performance limits of these new designs identifies perfor-

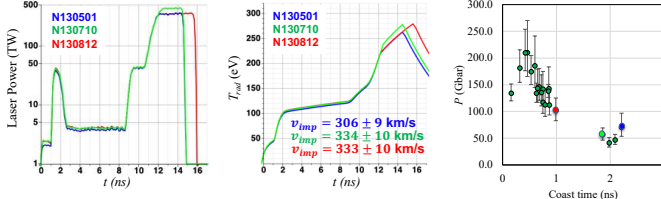


FIG. 13 The very first high foot DT implosion experiment, N130501, formed a baseline to study the impact of increasing implosion velocity with different choices of coast time. (Left frame) Experiment N130710 increased implosion velocity by increasing laser power relative to N130501 while experiment N130812 maintained the same laser power as N130501, but extended the duration of peak laser power, therefore significantly reducing the coast time. (Middle frame) Experiments N130710 and N130812 were designed to have identical peak hohlraum radiation temperature and implosion velocity, but the N130812 experiment has a higher late-time hohlraum radiation temperature. (Right frame) The three experiments N130501, N130710, and N130812 gave the first indications that hotspot pressure would increase with reduced coast time as was later proven by a series of high foot experiments where the coast time was systematically decreased in steps. At very short coast time, the hotspot pressure was observed to drop from its maximum value – a failure mode speculated to be due to a rupture in shell caused by an asymmetry induced perforation at low mass remaining (Hurricane *et al.*, 2016a).

mance limiting factors, which inform the next generation of designs, while focused science experiments delve into the details of the performance limiters and simulations models are updated to capture these factors. Over the past decade of indirect drive ICF experiments on the NIF, the fusion yields have increased by a factor of $1000\times$, recently entering the burning plasma (Kritcher *et al.*, 2022a; Ross *et al.*, 2021; Zylstra *et al.*, 2022a) and ignition (Abu-Shawareb *et al.*, 2022; Kritcher *et al.*, 2022b; Zylstra *et al.*, 2022b) regimes and the following discussion chronicles the steps that built to these achievements.

1. Low-Foot (LF) design: National Ignition Campaign (NIC): 2010-2012

The initial low-foot (LF) design design used high helium gas-fill (~ 1 mg/cm³) hohlraum to hold back the ingress gold hohlraum wall plasma over the long, ~ 20 ns pulse lengths to produce high gain with a high 1D margin via a low adiabat. These targets were sensitive to a number of failure modes (Lindl *et al.*, 2014), primarily due to mix at 330 km/s implosion speeds caused by high ablation-front RT instability growth (Ma *et al.*, 2013; Regan *et al.*, 2013) which result in hotspot performance far from what is required for ignition (see Fig. 14a). In addition to measured T below T_{ign} and fusion yields in the low kJ range instead of MJ range, a key data inference of burn-averaged stagnation pressure using burn-average

measurements and imaging data, essentially via the formula $p \sim \sqrt{T^2 Y / (\langle \sigma v \rangle V \tau)}$, developed during this period (Cerjan *et al.*, 2013; Springer, P.T. *et al.*, 2013), gave pressures $\sim 1/3$ of what was understood to be necessary for ignition.

However, the LF design was used in commissioning platforms (Landen *et al.*, 2011) on NIF for 1) the laser system (Moses *et al.*, 2016); 2) shock timing (Robey *et al.*, 2012a,b); 3) early-time symmetry (Dewald *et al.*, 2011); 4) implosion velocity (Hicks *et al.*, 2010); 5) in-flight symmetry (Rygg *et al.*, 2014; Town *et al.*, 2014); 6) hot-spot symmetry (Kyrala *et al.*, 2010, 2011); and 7) cryogenic layered implosions (Glenzer *et al.*, 2012). The LF campaign also demonstrated: 1) precision pulse shaping and four staged shocks leading to high fuel areal densities, for coast-times < 1 ns [as measured by neutron down-scattered ratio (DSR, the ratio of 10-12 MeV neutrons to 13-15 MeV neutrons) $\sim 6\%$] (Mackinnon *et al.*, 2012); 2) wavelength separation ($\Delta\lambda$) between the inner and outer laser cones was an effective means of controlling $P2$ symmetry (Glenzer *et al.*, 2010; Michel *et al.*, 2009) and between the 23° and 30° cones to control M4 symmetry in a high gas fill (HGF) hohlraum (Michel *et al.*, 2011; Moody *et al.*, 2012); and 3) gold-lined depleted uranium (DU) hohlraums effectively gave a 7% higher x-ray drive (Callahan *et al.*, 2012) as compared to pure Au hohlraums. However, regardless of the hohlraum material, these high gas fill hohlraum designs had problems with strong time-dependent asymmetries, and significant SRS and missing energy (compared to radiation hydrodynamics simulations) (Kline *et al.*, 2013). This motivated the development of the view-factor platform (MacLaren *et al.*, 2014) that confirmed the apparent drive deficit and inaccuracy in simulations of the LEH region of the hohlraum.

It was clear at the end of the NIC, that a number of problems existed in both the low-foot implosions themselves (e.g. hydrodynamic instability, mixing, low-mode asymmetry) and in the coupling of the hohlraum to the implosions. Interestingly, most of the observed problems were anticipated pre-NIC (Lindl, 1995; Lindl *et al.*, 2004), but what was not expected was the inability to manage these problems with the choices made in the “point-design” of the low-foot implosion, the veracity (or lack thereof of computer simulations used in the design), and the real level of control over the laser and target engineering. A different strategy was needed, one that addressed the observed problems, and then addressed anticipated problems that would later reveal themselves, like peeling away the layers of an onion, and thus was born the incremental “basecamp” strategy (Hurricane, 2015). In the basecamp strategy, one steps back from directly trying for ignition and instead targets solving problems in steps, thus an ignition design was created by evolution over time and in this case it took a decade. The first “basecamp” was the high foot implosion.

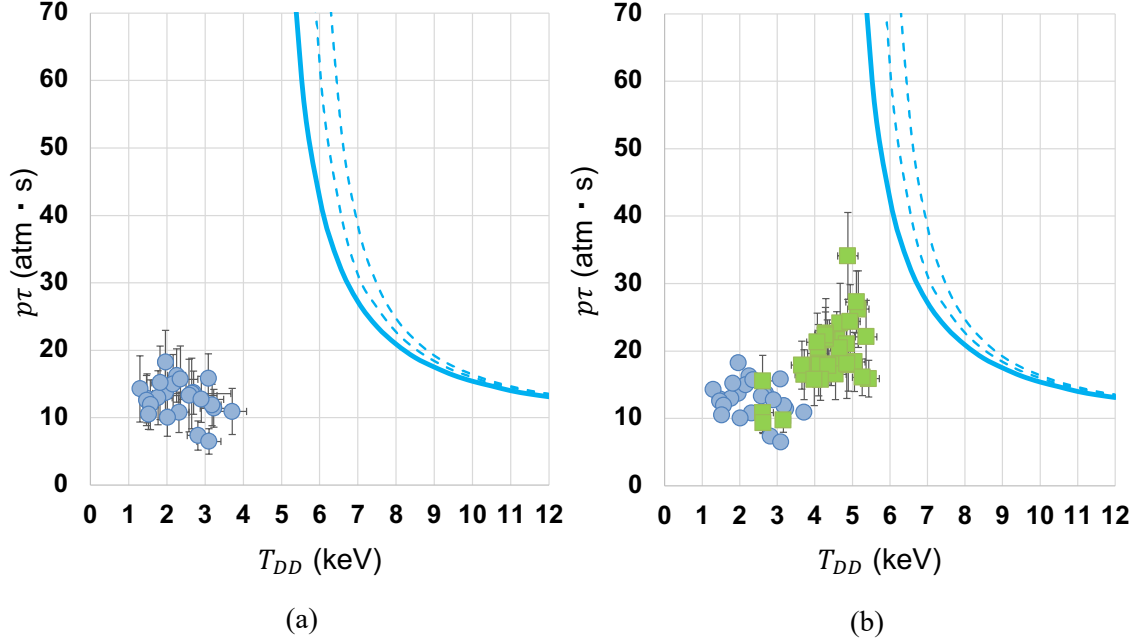


FIG. 14 NIF DT shot data (dots) are plotted in the space of inferred Lawson parameter and hot-spot (thermal) temperature. The curves denote the ignition boundary estimated from $p\tau H(T) \geq 1$ and Eq. 21. The dotted and dashed curves show how the ignition boundary moves to higher temperature under different assumptions of mixing of high-Z material into the DT resulting in enhancement of bremsstrahlung x-ray losses (solid: $f_B = 1$, dotted: a 50% increase in bremsstrahlung, $f_B = 1.5$, and dashed: a 100% increase, $f_B = 2.0$). (a) Data from the Low-Foot series (circles). (b) Data from the High-Foot series (squares). In this analysis, the NToF inferred DD ion temperature, $T_i(DD)$, is used as an estimate of thermal temperature since it is less sensitive to Doppler broadening effects, as previous work (Jarrott *et al.*, 2018) measuring electron temperature (T_e) has shown that $T_e \approx T_i(DD) \leq T_i(DT)$ for these types of implosions.

2. High Foot (HF) design: 2013-2015

The high-foot (HF) design (Dittrich *et al.*, 2014), which mitigated ablation-front RT instability, obtained an order-of-magnitude better fusion yield performance than the low-foot design (Hurricane *et al.*, 2014a,b; Park *et al.*, 2014). Due to being a lower-convergence, higher-adiabat design, with a higher early-time hohlraum T_r , the high-foot design was able to resist ablation-front RT (ART) instability by increasing the two key stabilizing terms in Eq. 32, namely L_ρ and v_a (see Fig. 17 for a simulation demonstration of the improved stability). Better controlling ART instability allowed access to much higher implosion speeds without mixing (Ma *et al.*, 2015) than the low-foot design, and the high-foot experiments showed that the fusion yield rapidly increased with gold-equivalent laser energy, $Y \sim E_{laser}^{5.5-7.9}$ (Callahan *et al.*, 2015), as is expected under the conditions of increasing α -particle self-heating. Stability control with increased implosion speed allowed higher hotspot temperatures, pressures, and Lawson metric (see Fig. 14b). As the implosion speed was pushed to 390 km/s by thinning the initial ablator thickness, the ART instability growth increased (Clark *et al.*, 2016), leading to increased sensitivity to the tent membrane (see Fig. 15) – an engineering feature needed to hold the capsule in the hohlraum that

also seeds localized vorticity and undesirable instability during implosion (Nagel *et al.*, 2015; Ralph *et al.*, 2020; Tommasini *et al.*, 2015) – which ultimately limited any further increase in performance. The tent membrane was inferred to perforate the shell for the highest velocity (i.e., lower mass remaining) high-foot implosions (Clark *et al.*, 2016; Hurricane *et al.*, 2016a; Ralph *et al.*, 2020; Springer *et al.*, 2018).

Focused experiments to measure capsule symmetry during the implosion and to measure ART growth rates using the hydrodynamic growth radiography platform (HGR) (Raman *et al.*, 2014; Smalyuk *et al.*, 2014) were introduced at the same time as the high-foot design in order to test the code predictions of ART growth. The simulation models reproduced the measured perturbation growth for the low- and high-foot platforms (e.g., see Fig. 16) from modes 30-160 (Casey *et al.*, 2014; Peterson *et al.*, 2015b; Smalyuk *et al.*, 2014). During this same period it was recognized that uv-light induced oxygen uptake, over time, in the first few outer microns of the GDP/CH ablators resulted in significant non-uniformity of ablation, effectively increasing the capsule surface roughness (Baxamusa *et al.*, 2015; Haan *et al.*, 2015) – a result which rationalized why low-foot implosions appeared, from simulations, to behave as if the surface roughness was $\sim 4\times$ worse than capsule metrology

TABLE IV Indirect Drive Implosion designs on NIF tested over various time-periods with motivation for design.

Design	Motivation	Approx. Year(s)
Low-foot (LF), aka NIC	Demonstrate ignition and high gain using a 4-shock low-adiabat design with a CH ablator.	2010-13
High-foot (HF)	Using LF targets, trade away any possibility of ignition to obtain a more ART stable implosion, with fewer (3) shocks, with the goal of better simulation-data agreement (i.e., establish a ‘base-camp’). Obtain velocity and energy (e.g., coast time) scaling. Test for performance cliffs.	2013-15
Low-foot Adiabatic Shaped (LFAS)	Demonstrate that high-foot stability and the high-foot Au-lined DU hohlraum can be used on a 4-shock lower-adiabat to obtain higher convergence.	2014
High-foot Adiabatic Shaped (HFAS)	Demonstrate that high-foot stability and the high-foot Au-lined DU hohlraum can be used on a 3-shock lower-adiabat to obtain higher convergence.	2015
HDC 3SH HGF	Test whether an HDC ablator offers any advantage using an otherwise HF-type design	2014-15
HDC 2SH VAC	Test an HDC ablator in a vacuum (VAC) hohlraum with different case-to-capsule ratios. Look for improvements in symmetry control and energy coupling.	2015-2017
Be LGF	Test a beryllium ablator implosion in a low gas fill (LGF) hohlraum with two different picket tactics.	2015-17
CH LGF	Test a high-foot plastic ablator implosion in a LGF hohlraum with two different capsule scales. First tests of using CBET in a LGF hohlraum.	2015-18
HDC LGF	Leverage the short pulse length of HDC to better utilize a LGF hohlraum and test the limits of symmetry and velocity.	2016-19
Bigfoot	Test unconventional tactic of colliding shocks in the DT fuel, trading-off compression for extra stability and minimal pulse-length (for improved symmetry control).	2016-18
2-shock CH	Test a minimal asymmetry and minimal instability (i.e., ‘1D’) implosion by making the shell thick, slow, very high-adiabat and compare results against simulations (MacLaren <i>et al.</i> , 2018).	2016-17
Hybrid Designs (B-E)	Increase energy coupling to implosion via capsule scale while maintaining symmetry, high implosion speed, and low coast time. Recognizing the limited predictive capability of hohlraum simulations for symmetry control, use empirical based model to guide design. Integrate Bigfoot and HDC elements into one design (Hybrid-B), attempt to control CH implosion symmetry in LGF hohlraums using CBET (Hybrid-C), and test Be ablator implosions (Hybrid-D). Efforts converged to HDC ablators with CBET (Hybrid-E).	2018-present
I-raum / Frustraum	Test different hohlraum geometries, literally an i-beam cross-section and frustum shapes, respectively for managing the symmetry control of large-radius capsules.	2018-present

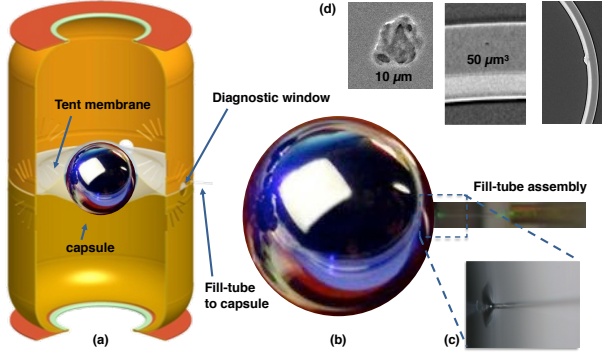


FIG. 15 (a) A schematic indirect-drive target showing the basic engineering elements. The tent membrane, which holds the capsule in the center of the hohlraum is a formvar material of thickness 15 - 100 nm. The most typically used tent thickness being 45 nm which balances fragility with desire to minimize any potential hydrodynamic perturbation. (b) A fill-tube, through which the capsule is fueled, extends out of a hole drilled through the ablator. Fill-tubes have ranged in diameter from 2 μm to 30 μm depending upon capsule material and target type. For plastic and beryllium capsules 10 μm diameter fill-tubes have been standard, while for HDC capsules 5 μm diameters are standard. (c) The fill-tube is epoxied to the capsule where the two components contact each other and the drill-hole around the tube is also grouted with epoxy. (d) Defects typically for HDC capsules are, from left-to-right, particles (or pits) on the outer surface of the ablator, voids in the thru-thickness of the ablator, and residue on the inner surface of the capsule. These defects are all potential seeds for inwardly directed hydrodynamic jets of ablator material.

indicated.

While the high-foot tactic managed hydro-instability of the ablator and reasonable repeatability between like-shots was observed, low-mode asymmetries for many high-foot implosions were observed directly in imaging data (see Fig. 18) and asymmetries were also inferred from time-of-flight nuclear spectra data (Chittenden *et al.*, 2016; Gatu Johnson *et al.*, 2016; Spears *et al.*, 2014). Ultimately, the tent membrane and the low-mode asymmetries, that grew upon implosion, were implicated as the basis for the observed performance-limiting cliff observed in experiments.

Attempts to use longer hohlraums to mitigate the observed *P4* asymmetry resulted in more difficulty with *P2* symmetry control, as a result of trouble in propagating the inner-beam laser to the waist (Pak *et al.*, 2017). Rugby shaped hohlraums were also attempted as a tactic for better symmetry control (Leidinger *et al.*, 2016), but experiments did not show any special advantage. Somewhat better hot-spot symmetry control was obtained with gold-lined DU hohlraums (Döppner *et al.*, 2015). This better control enabled the use of longer duration laser pulses needed for a coast time study.

It was found that lowering coast-time, in addition to velocity, strongly correlated with hot-spot performance

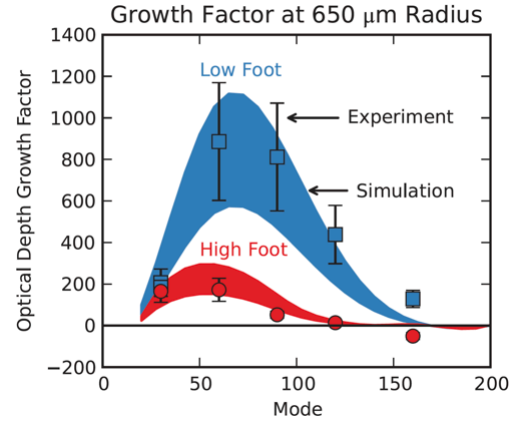


FIG. 16 Observed and simulated growth of an imposed perturbation on the outside surface of the capsule is plotted against the mode number of the perturbation. In this case the capsule has converged from its initial outer radius of 1137 μm to 650 μm . The perturbation amplitude growth is measured in terms of an increase optical depth variation using a face-on radiography technique. As can be observed in the figure, the predicted instability growth is largely in agreement with the data, except for the highest modes, and the high-foot implosion indeed has much less ART growth than the low-foot implosion across all modes, proving the key assertion about the high-foot tactic. Figure taken with permission from (Peterson *et al.*, 2015b).

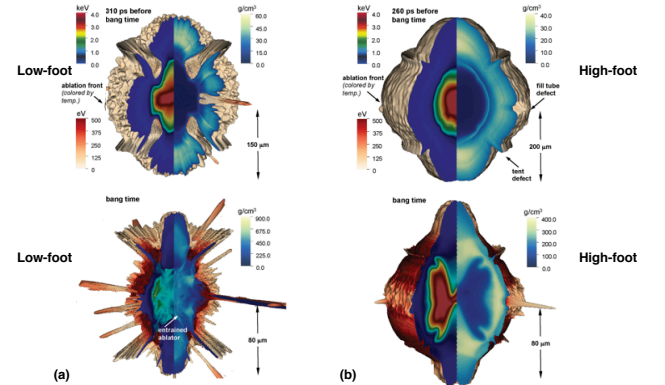


FIG. 17 High resolution 3D simulations of (a) LF implosion N120405 and (b) HF implosion N130927 showing the impact of instability, asymmetry, and the tent perturbation. The reduced sensitivity of the HF to A-RT instability and tent perturbation growth is clear. Both implosions exhibit a mode-2 asymmetry, which manifest itself as a toroidal hot-spot in the HF implosion, albeit the degree of hot-spot asymmetry seen in the 3D simulation is exaggerated as compared to data. Figure adapted, by permission, from (Clark *et al.*, 2016).

(Hurricane *et al.*, 2016a) by reducing premature shell decompression (Zylstra *et al.*, 2014), thereby enhancing the stagnation pressure, scaling as $p_{hs} \sim T_r^{7/5} v_{imp}^3$ (Hurricane *et al.*, 2017b). With a fixed target geometry, a lower coast-time is obtained by increasing the duration of late-time laser power, thus delivering additional laser energy to the target. Longer duration laser pulses keep

the hohlraum hotter at late-time relative to a long coast-time drive, reducing the drop of the late-time ablation pressure, Eq. 38, and reducing the radius at which peak velocity, R_{pv} , is obtained (Hurricane *et al.*, 2022). The reduced R_{pv} reduces the deceleration distance of the implosion, thus increasing the rate at which kinetic energy is converted into internal energy as the implosion stagnates which has very beneficial effects, e.g. Eq.41 (see Section III.D). The effectiveness of lowering coast time was noted experimentally across ablator types and implosion designs (Hurricane, 2016) and thus became a principal design goal for most of the subsequent experiments.

3. Adiabatic Shaped (AS) design: 2014-2015

Adiabatic-shaped designs (Clark *et al.*, 2014; Milovich *et al.*, 2015; Peterson *et al.*, 2015a), in which the laser pulse was tailored to achieve similar resistance to ART instability growth as the high-foot design, but with a lower fuel adiabat to achieve higher compression similar to the low-foot design, were performed in HGF hohlraums. For the 4-shock version, reducing the ablation-front instability growth resulted in a 3-10 times increase in neutron yield with no loss of fuel compression (Casey *et al.*, 2015) compared to the low-foot implosion, but lower yield (by half) compared to the most equivalent high-foot implosion. For the 3-shock version, reducing the fuel adiabat resulted in a 36% increase in fuel compression, but with similar yield (and other hot-spot metrics) to the corresponding high-foot implosions (Baker *et al.*, 2015; Robey *et al.*, 2016; Smalyuk *et al.*, 2016, 2015) and below that expected from analytic 1-D scaling. Low-mode areal density inhomogeneity remained high, confirming that drive asymmetry and tent induced capsule perforation would need to be mitigated for CH designs.

4. Low Gas-Fill (LGF) Hohlraum Designs: 2016-present

Symmetry control and reduced x-ray coupling because of high levels of inner-cone SRS remained a performance limiting problem for HGF hohlraum experiments (Kritcher *et al.*, 2016). Reducing the hohlraum gas fill provided a new avenue for symmetry control and coupling. Research started in 2013 on an indirect drive exploding pusher (IDEP) for neutron diagnostic calibration used hohlraum gas fills that were 50 times less than previous designs. These near-vacuum hohlraums (NVH) efficiently converted laser power to x-rays because of minimal levels of SRS and other LPI (Le Pape *et al.*, 2014; MacKinnon *et al.*, 2014). By using high density carbon (HDC) ablaters, with density three times higher than previous CH ablaters, shorter laser pulses were used that could tolerate more hohlraum wall motion that occurs with lower hohlraum gas fills. Low levels of backscatter

were confirmed in 2- and 3-shock designs (Berzak Hopkins *et al.*, 2015a) and symmetry control was achieved in small capsules (Berzak Hopkins *et al.*, 2015b; Turnbull *et al.*, 2016), but symmetry was harder to achieve at larger scale (Le Pape *et al.*, 2016) and was difficult to predict, possibly due to kinetic and/or plasma inner-penetration effects (Berzak Hopkins *et al.*, 2015b; Higginson *et al.*, 2019) that are absent in radiation-hydrodynamic codes, such as HYDRA (Marinak *et al.*, 2001).

Mode-2 symmetry predictability and control was improved without the onset of LPI by increasing the hohlraum gas fill density from 0.03 mg/cm³ to between 0.3 mg/cm³ and 0.6 mg/cm³ and led to an improvement in stagnation pressure and fusion yield. In particular, a threshold in gas-fill for SRS was observed in a series of experiments that varied gas-fill, demonstrating a significant reduction in SRS with gas-fills below 0.6 mg/cm³ (Hall *et al.*, 2017). Low gas-fill hohlraums favor thinner higher-density ablaters (due to the shorter time-scale needed for shocks to transit the material and thus shorter pulse-lengths). Three designs were pursued to assess low gas fill (LGF) hohlraums: a high-foot CH design (Döppner *et al.*, 2020; Hinkel *et al.*, 2016), a 3-shock HDC design (Divol *et al.*, 2017), and the BigFoot design (Baker *et al.*, 2018). The CH design improved symmetry control, reduced LPI and reduced levels of hot electrons (Hinkel *et al.*, 2016). By further reducing the CH capsule radius and thickness by 10% more complete symmetry control was obtained with CH as well as record stagnation pressure for plastic ablator implosions (Döppner *et al.*, 2020). In general, for LGF designs the discrepancies between simulated and measured radiation production were markedly reduced and, for a fixed laser energy, improved capsule performance was attained as compared to HGF hohlraum implosions.

Using a subscale HDC design (0.844 mm inner radius capsule in a 5.75 mm diameter DU hohlraum), the $P2$ symmetry control over the duration of the laser pulse was demonstrated to be better than $\pm 5\mu\text{m}$. Based on detailed residual kinetic energy (RKE) analysis (Kritcher *et al.*, 2014) for this implosion, it was concluded that intrinsic asymmetry was a minor source of yield degradation (Divol *et al.*, 2017). Scaling this design up by 8% (to a 0.910 mm inner radius capsule in a 6.20 mm diameter hohlraum) achieved a neutron yield of $1.88 \times 10^{16} \pm 5 \times 10^{14}$ whose output fusion energy was estimated to be twice the peak kinetic energy of the imploding shell (Le Pape *et al.*, 2018). The HDC design extended the Lawson $p\tau$ metric (see Fig. 19a) with a significant increase in stagnation pressure as compared to earlier plastic ablator designs, but burn average temperatures remained limited to the vicinity of the ignition temperature. Performance limiting X-ray emission features associated with jetting of ablator material into the hot spot coming from hydrodynamics seeded by the fill

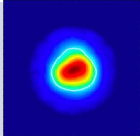
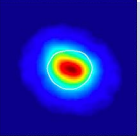
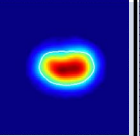
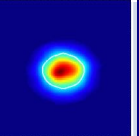
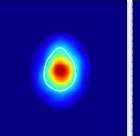
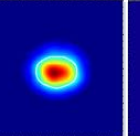
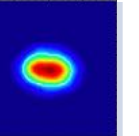
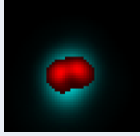
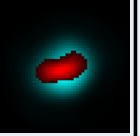
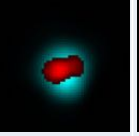
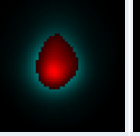
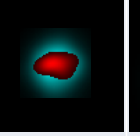
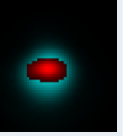
High-foot DT repeatability tests	N131219	N140225 (N131219 repeat)	N141106 (N131219 repeat)	N140520	N141016 (N140520 repeat, bundle misfire)	N150121 (N140520 repeat)	N150409 (N140520 repeat)
X-ray emission at 78-degree view, 100x100 microns							
Neutron emission at 315-degree view (red=13-17 MeV, blue=6-12 MeV)							
E_{laser} (MJ)	1.62	1.57	1.62	1.76	1.69	1.74	1.73
Y_{total} (kJ)	9.83	9.14	9.11	25.4	10.0	20.4	22.9
T_{DT} (keV)	4.91 $\pm$ 0.15	4.51 $\pm$ 0.15	4.44 $\pm$ 0.13	5.54 $\pm$ 0.15	4.07 $\pm$ 0.13	5.21 $\pm$ 0.11	5.5 $\pm$ 0.15
T_{DD} (keV)	4.16 $\pm$ 0.21	3.95 $\pm$ 0.17	4.04 $\pm$ 0.17	4.8 $\pm$ 0.2	3.65 $\pm$ 0.17	4.55 $\pm$ 0.17	4.6 $\pm$ 0.17
DSR(%)	3.8 $\pm$ 0.33	3.7 $\pm$ 0.2	3.82 $\pm$ 0.21	4.08 $\pm$ 0.2	3.73 $\pm$ 0.15	4.02 $\pm$ 0.18	4.21 $\pm$ 0.18
$BT_{\text{x-ray}}$ (ns)	16.03 $\pm$ 0.02	16.28 $\pm$ 0.02	16.25 $\pm$ 0.05	15.71 $\pm$ 0.02	16.17 $\pm$ 0.03	15.86 $\pm$ 0.02	15.93 $\pm$ 0.02
$BW_{\text{x-ray}}$ (ps)	147.49 $\pm$ 21.9	112.89 $\pm$ 33.7	130.16 $\pm$ 7.18	110.93 $\pm$ 25.1	156.0 $\pm$ 40.0	109.96 $\pm$ 19.6	117.16 $\pm$ 8.71

FIG. 18 Two sets of repeat shots were performed using the 175 μm ablator thickness capsule variant of the high-foot design in high gas-fill DU hohlraums (Hurricane *et al.*, 2016b; Ma *et al.*, 2015). Top row, shows the equatorial (i.e. side) view time-integrated x-ray emission at > 6 keV photon energy (Ma *et al.*, 2012). The second row shows equatorial view neutron imaging system data (Volegov *et al.*, 2014). Except for one experiment, N141016 where the NIF laser misfired, the experiments exhibit oblate hotspot symmetry with indications of DT fuel mass accumulation, inferred from the down-scatter neutrons, at the poles of the implosion. Scalar performance metrics are as labeled in the other rows of the figure/table. Fusion yield repeats to within 25%, while most other metrics repeat to better than 10%.

tube (Pak *et al.*, 2020) (see Fig. 15) were also first clearly identified in the HDC campaign.

The BigFoot (BF) approach traded-off high convergence, and therefore areal density, in favor of high implosion velocity, good energy coupling to the hotspot, and reduced hydrodynamic instability growth (Casey *et al.*, 2018). The design pioneered the “cone and quad split” pointing scheme, which smooths the azimuthal intensity variations and reduces the average laser intensity on the hohlraum wall. This design was scaled up by 12% (to a 0.950 mm inner radius capsule in a 6.00 mm diameter hohlraum) and achieved a marginally higher record yield of $1.95 \times 10^{16} \pm 5 \times 10^{14}$ at an implosion speed of 425 km/s. The performance of the Bigfoot design in Lawson parameter space (see Fig. 19a) and yield very closely mirrored that of the contemporary HDC design despite the differences in the designed adiabat ($\alpha_{if} \sim 4$ for BF vs. $\alpha_{if} \sim 2.5$ for HDC). Attempts to lower the adiabat of BF, to increase convergence, and further increase the velocity (by using a DU hohlraum) showed a decrease in

nuclear performance due to suspected ablator-fuel instability during deceleration. In fact, all HDC/BF designs seem to exhibit less convergence for the same adiabat than CH ablator designs and show a lack of compression response to varying α_{if} . (Landen *et al.*, 2021) and understanding the physics origin of this problem is a topic of current research, the leading (but not only) hypothesis being small-scale hydro-dynamic instability of the fuel-ablator interface.

By using the data-base generated from these and other low gas fill (LGF) experiments [such as the 2-shock CH design (Hall *et al.*, 2017) and beryllium ablator designs (Zylstra *et al.*, 2018)] a simple underlying pattern for symmetry control was discovered in cylindrical-geometry hohlraums (Callahan *et al.*, 2018). The data from these experiments indicated that the ingress of hohlraum-wall plasma, generated by the outer beams of NIF during the early-time (picket) part of the laser pulse, blocked the inner beams from reaching the hohlraum wall (Ralph *et al.*, 2018). The key connection between design parameters

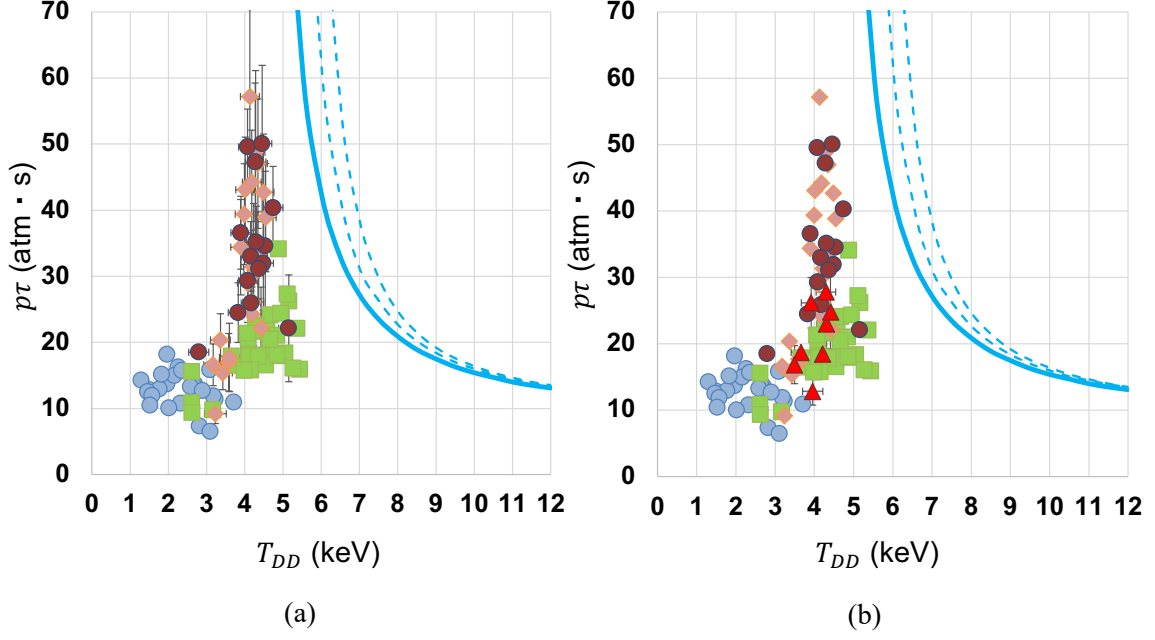


FIG. 19 NIF DT shot data (dots) are plotted in the space of inferred Lawson parameter and hot-spot thermal temperature. (a) Data from the HDC (diamonds) and Bigfoot (dark circles) series of experiments. (b) Data from the Hybrid-E 1100 micron radius capsule series (dark triangles).

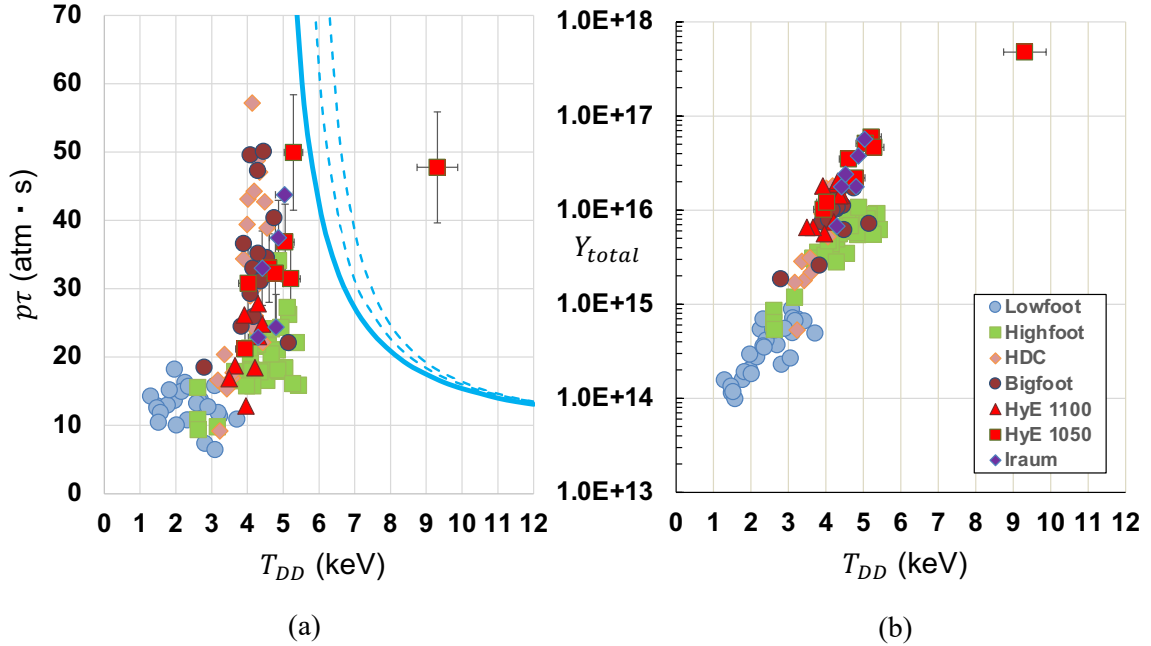


FIG. 20 (a) NIF DT shot data (dots) are plotted in the space of inferred Lawson parameter and hot-spot thermal temperature, here including the I-raum series (dark diamonds) and Hybrid-E 1050 micron capsule (dark squares) series. (b) Total fusion yield vs. temperature across several experimental series with legend. The significant jump in temperature and yield, setting apart the point associated with experiment N210808 (rightmost point), is a qualitative indication of ignition reflecting that thermal instability, indeed, occurred.

and mode-2 symmetry control was reasoned-out to be (Callahan *et al.*, 2018)

$$\Delta P_2 \sim -\sqrt{\frac{E_{picket,outer}}{A_{outer}\rho_{fill}}} \frac{t_{laser}}{R_{hohl}} \frac{R_{cap}}{R_{hohl}}, \quad (42)$$

where $E_{picket,outer}$ is the laser energy in the outer beams during the early-time part of the laser pulse, A_{outer} is the surface area the outer laser beams make on the inside of the hohlraum wall, t_{laser} is the laser pulse duration, R_{hohl} is the initial inner radius of the hohlraum, and R_{cap} is the initial outer radius of the capsule. There are some regimes of hohlraum gas-fill density (e.g., < 0.3 mg/cm³) where this simple plasma-ingress picture and Eq. 42 breaks down (Izumi *et al.*, 2018), and the impact of the LEH on mode-2 symmetry is not fully understood.

In 2017, anticipating that the HDC and BigFoot designs would soon be operating at their limits of performance, the Program developed three new designs (Hybrid-B, -C, and -D), to be fielded in 2018, that incorporated the understanding and best features of existing designs and capitalized on what was learned about scaling parameters for both neutron yield and hohlraum symmetry control. The principal aspect of the HYBRID (high yield big radius implosion design) idea is to increase energy delivery to the hotspot of the implosion, since it was noted the across the database of NIF experiments that $Y \sim E_{hs}^{3.3}$ (Hurricane *et al.*, 2019b). From Eq. 27, $E_{hs} = c(\alpha_{if})m_{shell}v_{imp}^2(1 - f^2)/2$ in the limit of low levels of alpha-heating, so the essential elements needed to increase fusion yield were clearly shell mass, implosion velocity, and good low-mode symmetry. Additionally, higher adiabat directs a larger fraction of the fuel KE into the hotspot (i.e. the $c(\alpha_{if})$ factor), hence higher adiabat can help push the hotspot over the tipping point of igniting, the downside being that ultimate gain will be reduced because of reduced fuel compression.

It is more favorable to increase m_{shell} by increasing shell thickness, because of the favorable impact on hydrodynamic-stability (see Section II.C. *IFAR* discussion), but maintaining v_{imp} with a thicker shell requires a higher hohlraum T_{rad} , necessitating a more efficient hohlraum or more laser energy. Moreover, a thicker shell can complicate symmetry control due to the longer duration laser pulse required to drive it, i.e. Eq. 42. Increasing m_{shell} by increasing initial radius, without increasing thickness, absorbs more energy from a given x-ray bath due to the increased capsule surface area, so no additional laser energy or hohlraum efficiency is required. By increasing the capsule inner and DT fuel radius, controlling implosion symmetry based upon Eq. 42 and CBET, if needed, while keeping other aspects of the implosion (v_{imp} , coast time, fuel adiabat, hydro-stability, etc.) fixed as much as possible (Hurricane *et al.*, 2017a, 2019a,b), it was expected that hotspot energy could be increased without lowering hotspot pressure. By forcing

a fixed coast-time (and therefore R_{pv}) despite increased radius, the HYBRID strategy isn't equivalent to pure hydro-scaling, which has a less favorable scaling of yield with capsule absorbed energy (Clark *et al.*, 2019).

The Hybrid-B (Hohenberger *et al.*, 2020; Zylstra *et al.*, 2020a) design used a 1.000-mm-inner-radius HDC capsule in a 6.72-mm-diameter gold-lined DU hohlraum, about a 10% increase above the HDC capsule radius used in record setting shots in 2017, and retained some elements of the HDC and Bigfoot pulse shapes. The Hybrid-B design was developed in coordination with a new 2.1 MJ NIF laser capability that materialized only recently in 2022. (In 2018 a 2.1 MJ test shot without a target was performed on NIF, unexpected laser optics damage was generated from filamentation instability, so the capability could not be used for ICF experiments.) As a result, the Hybrid-B design was under driven with the standard NIF energy of 1.8 MJ, but experiments were performed anyway in order to explore how implosion properties changed with capsule scale.

The Hybrid-B experiments were also compromised by unexpected HDC capsule quality issues generated during fabrication (see Fig. 15d). Nevertheless, the Hybrid-B implosions demonstrated the highest yield for their velocity (~ 360 km/s) at the time, consistent with expectations of improvement with increased capsule scale (Hohenberger *et al.*, 2020) which results in an increase in energy delivered to the hotspot that compensates energy losses due to asymmetry and radiation. Building upon earlier work (Kritcher *et al.*, 2018b), the Hybrid-C design (Pickworth *et al.*, 2020), which used CH ablaters that previously struggled with symmetry control in a low gas-fill hohlraum, tested the efficacy of using small amounts of laser wavelength tuning (e.g., $\Delta\lambda \sim 1$ Å) to control symmetry by minimizing CBET. The Hybrid-D design, which used Be ablaters, tested the efficacy of using picketless laser pulses to minimize the growth of the outer beam plasma bubble inside the hohlraum that can occlude the inner beam propagation.

5. Experimental demonstration of burning plasmas and ignition

These initial Hybrid experiments led to the Hybrid-E experimental series that uses small amounts of wavelength tuning (i.e. CBET) to enable implosions of 1.1 mm inner radius capsules with adequate symmetry, which further increased the energy coupled to the capsule to 270 kJ (inferred). In November 2019 a Hybrid-E experiment attained a yield of 2×10^{16} and hot-spot energy record of ~ 14 kJ at a modest implosion speed of 360 km/s (Zylstra *et al.*, 2021), albeit these early versions of Hybrid-E occupied the same location in Lawson parameter space as the HDC/BF implosions (see Fig. 19b). In November 2020, using slightly smaller 1.05 mm inner radius capsules, Hybrid-E achieved yet higher yield

performance of 3.6×10^{16} (i.e., being the first to obtain ~ 100 kJ on an ICF facility) and, along with similarly performing “I-raum” (Robey *et al.*, 2018) experiments (which tests an I-beam cross-section shaped hohlraum), subsequently entered into the “burning plasma” regime (Kritcher *et al.*, 2022a, 2021; Ross *et al.*, 2021; Zylstra *et al.*, 2022a) in the Winter of 2021.

A burning plasma is a fusion plasma where alpha-particle self-heating exceeds the heating from external sources and this definition has existed in the magnetic fusion community for decades, but has not commonly been considered in ICF until recently (Betti *et al.*, 2015). For an ICF plasma, Eq. 14 implies an energy inequality statement that denotes when a burning plasma state is achieved. At stagnation, the time-rate of change of hot-spot volume, dV/dt , and therefore the heating-rate is nearly zero, so time integration of Eq. 14 is needed to define an ICF burning plasma. Mathematically, a statement of a burning plasma appropriate for ICF is (Hurricane *et al.*, 2019a,b; Zylstra *et al.*, 2022a)

$$\int_0^{t_{pf}} Q_\alpha dt > - \int_0^{V_{min}} \frac{p}{m} dV. \quad (43)$$

where t_{pf} is the time of peak fusion rate, and V_{min} is the minimum hot-spot volume. Several metrics that represent Eq. 43 exist (Zylstra *et al.*, 2022a), allowing one to test burn-average data from implosion experiments. A burning plasma is a physically identifiable threshold just shy of ignition, so knowing an experiment is in burning plasma regime is an indication of that experiment being in close proximity to ignition.

Hybrid-E and I-raum burning plasma experiments finally demonstrated burn average hotspot temperatures notably above T_{ign} , edging the implosions closer to the expected ignition boundary (see 19b). After a design modification that allowed access to even shorter coast time duration than the Winter 2021 Hybrid-E experiments (see Fig. 21), in addition to a fortuitous improvement in capsule quality (reducing the number of seeds for hydro-instability), the Lawson ignition criterion was exceeded in a Hybrid-E experiment on August 8, 2021 (see Fig. 20a) with ~ 220 kJ of capsule absorbed energy ($G_{capsule} \sim 6$). The ignition conclusion was supported by testing multiple formulations of the Lawson criterion (Abu-Shawareb *et al.*, 2022), examining other data such as the inferred hotspot power balance and observed hotspot volume (Zylstra *et al.*, 2022b), simulation analysis (Kritcher *et al.*, 2022b), qualitatively by the dramatic jump in burn average temperature and yield observed for the experiments (see Fig. 20b), and by independent analysis (Wurzel and Hsu, 2022). Further increasing the laser energy and capsule thickness of the design by 8% recently resulted in achieving $G_{target} \sim 1.5$ (LLNL, 2022). Over the past decade of indirect drive experiments on the NIF, the fusion yields have increased by a factor of

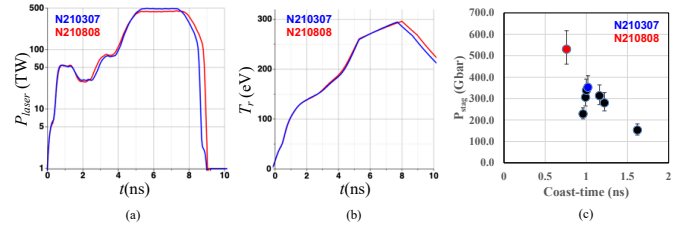


FIG. 21 (a) The as-shot laser power vs. time for a burning plasma Hybrid-E experiment (N210307) and the ignited one (N210808). The peak laser power was reduced and extended in time (conserving total laser energy) for N210808. (b) The model (Callahan *et al.*, 2020) inferred hohlraum radiation temperature vs. time for the same two experiments. The T_r for N210808 was higher than for N210307 due to the reduction in laser entrance hole size and due to the extension of the late-time laser pulse, despite the reduction in peak power. (c) As was seen in previous series of experiments (e.g. Fig. 13), the reduction in coast-time results in an increase in stagnation pressure. Black points are from other Hybrid-E experiments in the series from late 2020-mid 2021 using the same $1050 \mu\text{m}$ inner radius capsule scale as N210307 and N210808.

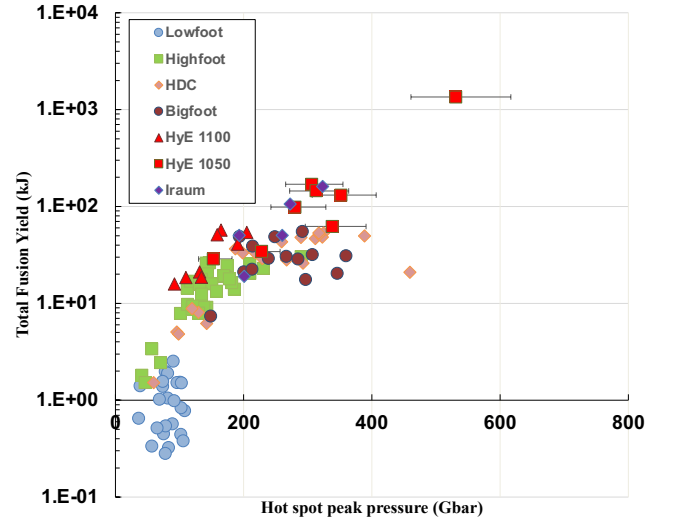


FIG. 22 NIF DT shot data of total measured fusion yield is plotted against data-inferred peak hotspot pressure with legend. The pressure at the center of the Sun is estimated to be between 200-250 Gbar.

over $1000\times$, the central pressures in the implosions have exceeded the twice the pressure at the center of the Sun (see Fig. 22), albeit high gains at high ρR (see Fig. 2b) continue to be a challenge.

6. Alternate indirect-drive concepts

Yet to be tested in potentially high-yield DT experiments are a number of novel concepts that attempt

to build upon what has been learned or explore a new part of implosion design parameter space. Namely, designs such as the ‘Hot-thick’ (Young *et al.*, 2020) drive a very thick HDC capsule (for improved hydrodynamic stability and better inertial confinement) using a very hot hohlraum (~ 320 eV as compared to the more typical ~ 300 eV) using a small case-to-capsule ratio, $CCR = R_{hohl}/R_{cap}$. ‘Hot-thick’ effectively couples more energy to the implosion via increased x-ray drive and greatly reduced coast-time. The energy absorbed by the capsule, $\sim \int T_{rad}^4 R_{cap}^2 dt$, can be increased with increased T_{rad} as well as by increasing capsule surface area, R_{cap}^2 , as is done in designs like the Hybrids, I-raum, and Frustum (Amendt *et al.*, 2019) (which tests a double frustum shaped hohlraum). Another thick-capsule design is the ‘pushed-single-shell’ design (MacLaren *et al.*, 2021), which uses a massive graded-dopant shell to lower the requirements on stagnation pressure and temperature, yet still satisfy Lawson’s ignition criteria via increased inertial confinement (longer τ , which from Newton’s law scales as the square-root of shell mass).

Resurrecting an old idea of colliding concentric shell implosions[e.g. the ‘Apollo’ capsules of the late 1970’s as mentioned in (Lindl, 1995) albeit the idea of predates the 1970’s], double shell experiments (Merritt *et al.*, 2019) explore a potential low-to-medium gain design that may be suitable for burning plasma studies. In a double shell, an outer shell of mass, m_{outer} , is accelerated inwards by ablation to a velocity, v_{outer} . This outer shell then impacts a high-Z inner shell, of mass, m_{inner} , impulsively accelerating the inner shell to velocity, v_{inner} . From conservation of energy and momentum, the velocity and kinetic energy of the inner shell can be obtained,

$$v_{inner} = \frac{2v_{outer}}{1 + \frac{m_{inner}}{m_{outer}}} \quad (44)$$

$$KE_{inner} = KE_{outer} \frac{4 \frac{m_{inner}}{m_{outer}}}{\left(1 + \frac{m_{inner}}{m_{outer}}\right)^2} \quad (45)$$

where $KE_{inner,outer} = m_{inner,outer} v_{inner,outer}^2/2$. As can be seen in Eq. 44, in the limit of $m_{inner} \ll m_{outer}$ a near doubling of implosion speed can be obtained in a double-shell implosion, but it comes at the cost of kinetic energy, Eq. 45 – by induction these relations can be applied to multiple shells. When compressed, the high-Z inner shell of a double-shell configuration is designed to be opaque to x-rays, essentially stopping bremsstrahlung losses (reducing the f_B parameter in Eq. 14), which in principal would allow access to ignition at lower temperature than a standard hot-spot ignition design. Hydrodynamic instability of the inner shell of a double-shell configuration is a major concern because of the shell-shell collision and because there is no ablative stabilization of the inner shell.



FIG. 23 Schematic of laser direct drive ICF. Multiple laser beams symmetrically illuminate a spherical shell containing DT fuel

IV. LASER DRIVEN IMPLOSIONS

A. Brief history of direct drive

Direct spherical laser illumination of a DT fuel pellet (laser direct drive or LDD, Fig. 23) was envisioned in the early days of laser fusion as a mean to achieve ignition conditions with moderate laser energy. In the seminal paper of Nuckolls (Nuckolls *et al.*, 1972), a solid DT sphere was isentropically compressed to high densities and temperatures by a shaped laser pulse with monotonically increasing power to achieve ignition and gain at laser energies as low as few kilojoules. The theory of homogeneous isentropic compression was later developed by Kidder (Kidder, 1974) assuming a time dependent pressure from laser illumination applied on the surface of a DT sphere. As shown in (Kidder, 1974; Nuckolls *et al.*, 1972), ignition of a solid DT sphere requires extremely high power levels in the late stage of the implosion (Nuckolls estimated up to 1 PW) to achieve ignition temperatures and densities in the center. This is problematic for laser driven implosions because the coupling of the laser energy to the capsule is degraded late in the implosion when the smaller capsule surface and the enhanced coronal refraction reduce the overall driving force. As shown in (Kidder, 1976), these power requirements are significantly relaxed (by a factor of $\sim 1/5$) if a hollow shell is used in place of a solid sphere. Since then, the research effort in ICF has focused on implosions of a layer of DT fuel enclosed within a spherical shell of ablator material. Even though one-dimensional hydrodynamic studies of direct drive implosions indicated the possibility of ignition and energy gain at kilojoule level of laser energy, major obstacles were found in the physics of laser plasma interaction and three dimensional effects. Initial

experiments using longer wavelength lasers (such as CO₂ with $\lambda_L = 10.6 \mu\text{m}$ (Kephart *et al.*, 1974) and Nd:glass lasers with $\lambda_L = 1.053 \mu\text{m}$ (Ceglio and Larsen, 1980)) found poor energy coupling to the target and copious amounts of energetic "hot" electrons produced by laser plasma instabilities (LPI). Hot electrons incident on the imploding shell raise the entropy of the shell material in-flight (preheat) thereby reducing the final compression. The relevant parameter for the hot electron energy is the product $I\lambda_L^2$ where I is the laser intensity on target and λ_L is the laser wavelength (Lindman, 1977). This parameter represents the oscillation energy of an electron in the electric field of the light wave. It became clear that for direct drive ICF to be a viable fusion scheme, the magnitude of $I\lambda_L^2$ had to be reduced. This led to the development of Nd:glass lasers frequency-tripled to $0.351 \mu\text{m}$ using the tripling scheme of Craxton (Craxton, 1981). There are multiple benefits from shorter wavelength lasers: (a) increased laser energy absorption from inverse bremsstrahlung due to higher critical densities (Fabre *et al.*, 1981); (b) lower hot electron energies due to reduced $I\lambda_L^2$; (c) higher thresholds for laser plasma instabilities (Kruer, 2003); (d) increased coupling of the laser energy to the target due to closer proximity of the critical surface and laser absorption region to the ablation front. All current ICF laser implosion facilities use Nd:glass lasers with UV light at $0.351 \mu\text{m}$. The Naval Research Laboratory has successfully pursued the development of even shorter wavelength drivers by exploiting the properties of excimer lasers (Obenschain *et al.*, 1996) to generate deep UV light at $0.248 \mu\text{m}$ with an electron-beam pumped KrF gas (the NIKE laser) or $0.193 \mu\text{m}$ with ArF gas (Wolford *et al.*, 2020). While hot electron generation and fuel preheat was greatly reduced with UV laser drivers, there were still crucial open issues pertaining to the interaction of UV light with the target surface and the coronal plasma. When the laser initially interacts with the target surface, short wavelength modulations (speckles) in the laser beam intensity produce local spatial variations in the ablation pressure thereby launching a rippled shock wave into the target (Gardner and Bodner, 1981; Goncharov *et al.*, 2000b; Ishizaki and Nishihara, 1998). This phenomenon is called "laser imprinting" as the rippled shock leaves behind a nonuniform target surface similar to an imprinted image of the laser speckles. Since imploding shells are hydrodynamically unstable to the Rayleigh-Taylor instability, such perturbations grow exponentially at first and then saturate producing a mixing layer advancing inside the shell at a velocity growing linearly in time (for a constant shell acceleration) (Alon *et al.*, 1995; Dimonte *et al.*, 2004). Reducing laser imprinting became of paramount importance for the viability of LDD. A straightforward reduction in on-target nonuniformities was first accomplished by maximizing the number of overlapping beams N_{beam} as the RMS standard deviation over all the speckle

spectrum decreases as $1/\sqrt{N_{beam}}$ (Pawley *et al.*, 1997). A further improvement was obtained by installing Random or Distributed Phase Plates (RPPs or DPPs) at the end of each beamline to break the spatial coherence of the beam before the focal lenses (Kato *et al.*, 1984; Kessler *et al.*, 1993). Phase plates greatly reduced the low mode intensity nonuniformities while enhancing the high frequency modulations. Those high frequency modulations were subsequently suppressed by smoothing from phase-modulated bandwidth obtained from color variations across the beam driven by two electro-optical modulators in orthogonal directions and dispersed by gratings (Smoothing by Spectral Dispersion, SSD) (Skupsky *et al.*, 1989). An alternative temporal laser smoothing scheme (Induced Spatial Incoherence, ISI) (Lehmberg and Obenschain, 1983) applied earlier in the laser chain using broad bandwidth light is currently adopted in KrF lasers and offer the additional benefit of dynamic control of the laser spot profile (zooming). Though limited to a few THz, the bandwidth introduced by SSD or ISI had also an important role in mitigating the growth of parametric instabilities from the laser plasma interaction by reducing the light coherence. In particular, low frequency laser plasma instabilities such as Stimulated Brillouin Scattering are suppressed by modest levels of bandwidth (Berger *et al.*, 1999; Maximov *et al.*, 2004; Moody *et al.*, 2001). Instead higher frequency LPIs such as SRS and Two-Plasmon Decay (TPD) instability (Kruer, 2003) remained a serious threat since their growth time is faster than beam smoothing temporal modulation. In current direct drive experiments, the SRS and TPD instabilities are controlled by limiting the laser intensity on target and by using target-based mitigation strategies such as mid-Z doping (such as Si) of the ablator material (typically CD or CH) (Craxton *et al.*, 2015). Motivated by indirect drive models of two crossing laser rays (Michel *et al.*, 2009), another degradation mechanism recently identified in direct drive ICF is CBET, a SBS-mediated transfer of laser energy from incoming light rays to outgoing ones in the coronal plasma (Igumenshchev *et al.*, 2010). CBET reduces the laser energy absorption and lowers the ablation pressure. Most of the issues arising from laser plasma instabilities can be fully overcome by rapidly and randomly varying the laser speckle patterns through the proposed STUD (Spike Trains of Uneven Duration and Delay) pulse scheme (Afeyan, Bedros and Hüller, Stefan, 2013) or with the new generation lasers currently under development (Campbell *et al.*, 2021) promising $10\times$ higher bandwidth than current lasers and expected to suppress both low and high frequencies LPIs (Bates *et al.*, 2020; Follett *et al.*, 2019).

Multi-beam UV lasers with enhanced smoothing became the standard drivers for ICF research since the 1980's with direct drive lasers such as OMEGA and NIKE featuring the highest bandwidth (few THz) to minimize laser plasma instabilities and laser imprinting. Since the

1990s, direct drive experiments were mostly carried out at the University of Rochester Laboratory for Laser Energetics, home to the OMEGA laser, at the Osaka University Institute for Laser Engineering, home of the Gekko XII laser, and at the Naval Research Laboratory, home to the NIKE laser. Recently, direct drive LPI and imprinting experiments have also been fielded on the NIF. Other smaller laser facilities have also greatly contributed to explore specific aspects of direct-drive physics. Cryogenic DT implosions on the OMEGA 60 laser have been fielded since 2008 using thin plastic shells of about 1 mm in diameter and DT ice layers of several tens of microns thickness. Such targets were imploded by shaped laser pulses of a few nanoseconds with energies of 20-30 kJ. The fusion yield of OMEGA DT layered implosions has steadily increased over the years starting from 10^{11} neutrons in the early DT implosions of $\sim 80 \mu\text{m}$ thick ice to 10^{14} neutrons of the most recent implosions from targets with thinner ice layers of $\sim 40 \mu\text{m}$ (Gopalaswamy *et al.*, 2019). When hydrodynamically scaled to NIF energies of symmetric illumination, the best performing OMEGA implosions are projected to produce over 0.5 MJ of fusion yield but they are still short of achieving ignition conditions at 2 MJ of laser energy (Gopalaswamy *et al.*, 2019).

In addition to integrated implosion experiments on OMEGA, many crucial aspects of direct drive physics are being investigated through dedicated experiments ranging from laser plasma interactions in the coronal plasma to hydrodynamic instabilities of accelerated targets, from microphysics of extreme pressure states of matter to kinetic effects and magnetic fields in the compressed core of implosions. While OMEGA experiments can address and resolve many issues related to the physics of direct drive, aspects of implosion physics related to laser plasma interactions are strongly affected by the scale length of the plasma and needs to be studied on megajoule class lasers capable of imploding larger scale capsules and generating larger scale plasmas. This is the goal of the current direct drive effort on the NIF with its main objectives being the study of the generation and transport of hot electrons and the laser-to-capsule energy coupling at ignition scale.

To adapt the existing asymmetric polar-beam configuration of the NIF laser, direct drive implosions are designed with optimized pointing to minimize the low mode non-uniformities from the NIF illumination geometry. Polar Direct Drive (PDD) (Skupsky *et al.*, 2004) requires different pulse shapes and laser intensities for beams with different angle of incidence on target. This lead to lower collisional absorption and cross-beam-energy-transfer making PDD more challenging than SDD (symmetric direct drive) (Craxton *et al.*, 2015). Wavelength detuning between different laser beams has been successfully implemented in PDD implosions on NIF to mitigate CBET and augment the ablation pressure (Marozas

et al., 2018b). Furthermore flat foil NIF experiments using Si layers demonstrated significant reduction in hot electrons produced by SRS (Rosenberg *et al.*, 2018b). Such laser and target-based solutions are just examples of the multi faceted effort to advance direct drive ICF on the NIF.

Advanced laser fusion schemes such as shock ignition (Atzeni *et al.*, 2014; Batani *et al.*, 2014; Betti *et al.*, 2007; Perkins *et al.*, 2009) and fast ignition (Atzeni, 1999; Azechi *et al.*, 2013; Kodamai *et al.*, 2002; Tabak *et al.*, 1994) also use direct laser illumination. Shock and fast ignition are considered two-step schemes because they decouple the fuel assembly from ignition by using external means such as a laser-driven strong shock or an energetic particle beam to ignite the compressed fuel. These promising schemes have two potential advantages over conventional direct drive: higher gains and lower hydrodynamic stability requirements. Since shock ignition is mostly pursued in Europe and fast ignition in Asia, these fusion schemes are not included in this review of the US ICF effort. A detailed review of laser direct drive including shock and fast ignition can be found in (Craxton *et al.*, 2015).

B. Key differences between direct and indirect drive ICF

When compared to indirect drive, direct laser illumination leads to greater coupling of the laser energy to the capsule by avoiding the inefficient intermediate step of converting laser light into blackbody radiation and reducing the laser energy absorbed by the capsule (about 10% of the laser energy for typical indirect drive targets). On the other hand, direct drive targets absorb about 70% of the laser energy. The laser energy is absorbed via inverse bremsstrahlung in the plasma below the critical density. The critical density surface is the location where the laser frequency equals the plasma frequency. Light waves can propagate in plasmas but not past the critical density in a monotonically increasing density profiles. For typical low-Z ablaters (CH, CD and DT ice) used in direct drive targets, the dominant mechanism of energy transport from the laser absorption region to the outer shell surface is electronic heat conduction. The region between the critical surface and the ablation front is called the “conduction zone” where the low-Z ablated plasma is close to full ionization and its properties are similar to an ideal plasma with heat conductivity given by Spitzer’s formula ($\kappa_{Sp} \sim T^{5/2}$). The plasma with density below the critical density expands isothermally and is referred to as the “plasma corona.” The coronal plasma is where all the laser-plasma interactions occur and where all the laser-plasma instabilities develop. Figure 24 illustrates the schematics of the density profile in a laser accelerated target with the different plasma regions.

The ablation pressure on a laser illuminated surface

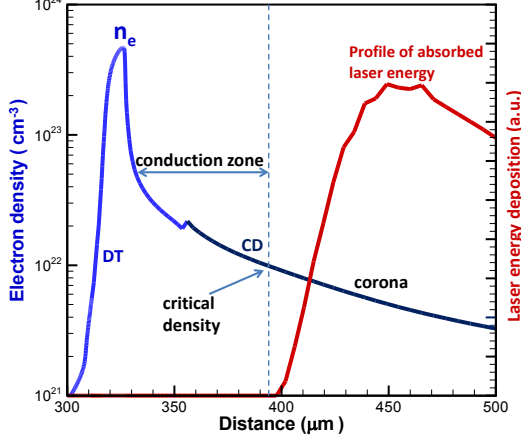


FIG. 24 Typical electron density profile (blue in DT and gray in CD) for a direct drive target. The conduction zone separates the ablation front from the critical surface. The region beyond the critical surface is the coronal plasma. Laser energy deposition (red) occurs in the CD plasma.

can be estimated using a simple planar ablative flow model assuming the laser light with intensity I is absorbed near the critical surface and the heat conducted to the ablation front through an ideal fully ionized plasma. The ablation pressure depends mostly on the laser intensity and the laser wavelength (Manheimer *et al.*, 1982) leading to

$$p_{abl,D} = \frac{m_p^{1/3}}{2^{1/3}} n_{cr}^{1/3} \left(\frac{A}{Z} \right)^{1/3} (I_L^{abs})^{2/3} \quad (46)$$

where m_p is the proton mass, A is the mass number, Z is the atomic number, $n_{cr} \simeq 1.1 \times 10^{21} / \lambda_L^2 \text{ cm}^{-3}$ is the critical density, λ_L is the laser wavelength in microns and I_{abs} is the absorbed laser intensity. Using parameter values typical of current direct drive experiments, Eq. 46 can be rewritten as

$$p_{abl,D} (\text{Mb}) \simeq 10^2 \left(\frac{0.35}{\lambda_L^{\mu\text{m}}} \right)^{2/3} \left(\frac{A}{2.5Z} \right)^{1/3} (I_L^{15})^{2/3} \quad (47)$$

where the absorbed intensity is in units of 10^{15} W/cm^2 and the ablation pressure in megabars. This shows that current UV lasers with 351 nm light can produce ~ 100 Mb pressures at intensities of 10^{15} W/cm^2 similar to the ablation pressure from x-rays in indirect drive for radiation temperatures $T_{rad} \simeq 300 \text{ eV}$.

While the ablation pressure is similar, the mass ablation rate is larger for indirect drive due to the deeper penetration of x-rays closer to the ablation front. For direct laser illumination, the specific mass ablation rate can be derived from the planar ablative flow model leading

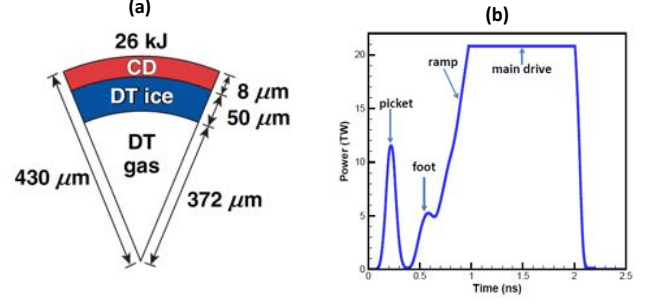


FIG. 25 Typical OMEGA DT-layered target (a) and laser pulse shape (b). The laser pulse shape is divided in four parts: the picket, the foot, the ramp and the main drive

to

$$\dot{m}_a (g/cm^2 s) \approx 2 \times 10^6 \left(\frac{0.35}{\lambda_L^{\mu\text{m}}} \right)^{4/3} \left(\frac{A}{2.5Z} \right)^{2/3} (I_L^{15})^{1/3} \quad (48)$$

which is about five times less than for indirect drive.

The fraction of ablated mass determines the conversion of the absorbed energy by the capsule to inward kinetic energy of the shell. This can be shown using a simple rocket model (Lindl, 1995; Nora *et al.*, 2014) of the imploding shell leading to

$$\eta_H \equiv \frac{E_{kin}}{E_{abs}} = \frac{1}{4} \frac{R_M (\ln R_M)^2}{1 - R_M} \quad (49)$$

where $R_M = M_f/M_0$ is the ratio of the final to the initial ablator mass. The maximum hydrodynamic efficiency $\eta_H \simeq 0.15$ occurs at $R_M \simeq 0.2$ with about 80% of the initial shell mass ablated typical of indirect drive targets. For direct drive targets, the ablated mass is about 50% of the initial mass and the hydrodynamic efficiency is only about 9%. However because of the larger laser absorption fraction (60-70% for direct drive versus 10% for indirect drive) the overall conversion of laser energy into kinetic energy for direct drive is about 6% while for indirect drive is about 1.5%.

Because of the lower ablation rate, direct drive targets use a relatively thin ablator about 6x thinner than indirect drive targets (for the same ablator material). Figure 25 shows a typical DT-layered target used on the OMEGA laser. OMEGA targets use 860-1000 μm outer diameter shells with a 7-8 μm thick CD plastic shell and DT ice layers of 40-50 μm thickness. A typical laser pulse is shown in Figure 25 (b). It consists of four parts: (a) an initial spike (or multiple spikes) in the laser power called the “picket(s)”, (b) a flat pulse called the “foot”, (c) the

power “ramp” to full power and (d) a slow varying pulse at peak power called the “main drive.” The initial picket pulse launches a decaying shock into the target. The foot launches the second shock. The ramp adiabatically compresses the capsule thickness and the main drive then accelerates shell inwards to its peak implosion velocity.

The thin plastic shell is quickly ablated off when the laser intensity reaches its peak value at the beginning of the main drive and ablation takes place in the DT fuel that becomes the main ablator material. Because of the highest value of Z/A , tritium and deuterium-tritium mixtures are the best ablator materials for direct drive with the highest ablation pressure and ablation rate as shown by Eqs. 46 and 47. The downside of DT ablators is the low laser energy absorption occurring in DT due to low Z . To optimize laser absorption, direct drive targets on OMEGA are designed with a CD layer thick enough to sustain a CD plasma in the underdense corona up to the quarter critical density through the entire laser pulse. To optimize the ablation pressure, the CD layer is thin enough to be ablated off when the main drive starts so that DT becomes the main ablator. Because of the higher Z , the presence of carbon in the absorption region of the coronal plasma enhances collisional absorption of the laser energy and heat flux towards the ablation front. Having DT as the ablator during the main drive, enhances the ablation pressure, the ablation rate, the ablation velocity and the hydrodynamic efficiency (Goncharov *et al.*, 2014). Optimized OMEGA implosions use 860-1000 μm outer diameter capsules with a CD ablator thickness of 7.5 – 8.5 μm and ice thickness of 40-50 μm .

Differences in ablation rate and laser energy coupling to the capsules between direct and indirect drive targets have profound implications on the hydrodynamic stability properties, fusion yield and energy required for ignition. In addition to the hydrodynamic properties mentioned above, direct and indirect drive targets differ with respect to LPI related issues such as hot electron preheat and cross beam energy transfer. A detailed description of the hydrodynamic stability and LPI properties of direct drive targets is presented in Section IV.B.3.

1. Direct-drive scaling advantages

Because of four fold higher energy coupling for direct drive versus indirect drive, the pressure requirements for ignition are significantly relaxed. This can be shown by starting from the Lawson ignition criterion

$$\chi \equiv P_{ign}\tau/S_f \approx 1 \quad (50)$$

where $S_f = 24T^2 / \langle \sigma v \rangle E_\alpha$ and P_{ign} is the pressure required for ignition (see Section II.A.). Assuming the onset of ignition occurs at temperatures of about 5 – 6 keV, where the fusion reactivity scales as $\langle \sigma v \rangle \sim T^3$, leads to an ignition threshold proportional to the triple

product $\chi \sim P\tau T$ (Betti *et al.*, 2010). Since the hot spot temperature scales approximately linearly with the implosion velocity (Zhou and Betti, 2007) and the pressure confinement time scales with the compressed core radius and the shell implosion velocity ($\tau \sim R_{hs}/v_{imp}$), the pressure required for ignition scales inversely proportional to the core radius ($P_{ign} \sim 1/R_{hs}$). Assuming that the hot spot internal energy $E_{hs} \simeq (3/2)P_{hs}V_{hs}$ is approximately proportional to the shell kinetic energy $E_{hs} \sim E_{kin}$, the hot spot radius can be written as $R_{hs} \sim (E_{kin}/P_{hs})^{1/3}$ and the pressure required for ignition is found to scale as (also see Section I.B.2)

$$P_{ign} \sim \sqrt{\frac{1}{E_{kin}}} \quad (51)$$

Therefore, for the same laser energy and about four times greater conversion to kinetic energy, direct drive targets require about half the core pressure of indirect drive targets to achieve the ignition conditions. Lower pressure requirements translates into lower convergence requirements for the same implosion velocity. Current direct drive ignition targets are designed for moderate convergence ratios $C_R \equiv R_0/R_{hs} \sim 23 - 25$ and relatively high in-flight adiabats $\alpha_F \simeq 3 - 4$ when compared to the more severe indirect drive requirements of $C_R \simeq 35 - 40$ and $\alpha_F \simeq 1 - 2$. As mentioned in Section IV.B.2, the higher adiabat in direct drive is also a requirement to preserve good hydrodynamic stability and its effects on the core pressure are consistent with the ignition requirements in Eq. 65. This is shown by using the familiar scaling of the pressure (Basko and Meyer-ter Vehn, 2002; Goncharov *et al.*, 2014; Kemp *et al.*, 2001; Zhou and Betti, 2007) with the Mach number leading to

$$P_{hs} \sim p_{abl,D} Mach^3 \sim p_{abl,D}^{2/5} \frac{v_{imp}^3}{\alpha_F^{9/10}} \quad (52)$$

indicating that for the same implosion velocity and ablation pressure, the hot spot pressure in direct drive implosions is about half of the indirect drive pressure due to the higher adiabat. Therefore, it is important to notice that with respect to achieving the ignition conditions, the direct drive advantage in energy coupling over indirect drive is balanced by the higher adiabats required to improve the hydrodynamic stability properties in direct drive implosions.

Unless improvements are made to the uniformity of direct drive implosions to enable carrying out lower adiabat implosions ($\alpha_F \sim 1 - 2$), the larger energy coupling in direct drive is not expected to bring direct drive implosions closer to ignition than indirect drive implosions. However, the larger direct drive targets are expected to achieve significant higher yields than indirect drive targets for the same implosion velocity and for the same proximity to ignition as determined by the Lawson parameter.

In the absence of alpha heating, the scaling of the yield with the coupled energy, adiabat and implosion velocity is $Y \sim v_{imp}^4 E_{kin}^{1.4} / \alpha_F^{0.8}$ (Betti *et al.*, 2010). Therefore for the same implosion velocity, same Lawson parameter and yield amplification from alphas heating, direct drive implosions with 4x the kinetic energy and 2x the adiabat of indirect drive lead to about 4x higher fusion yields. In ignited capsules, the fusion yields scales with the DT mass and the burnup fraction and it will again be larger by about 4x in direct drive targets due to the 4x larger fuel mass than indirect drive (for the same burnup fraction).

2. Hydrodynamic instabilities in direct-drive implosions

Hydrodynamic instabilities develop during the shock transit phase, the acceleration phase and the deceleration phase of both direct and indirect drive implosions. The Rayleigh-Taylor instability (Atzeni and Meyer-Ter-Vehn, 2004; Rayleigh, 1883) is the most deleterious of the hydrodynamic instabilities because undergoes exponential growth during the linear phase. In the classical description of two superimposed fluids in a gravitational field with the heavy fluid on top and the light fluid at the bottom (Rayleigh, 1883), a single RT Fourier mode exhibits a "bubble" of light fluid rising into the heavy one, and a "spike" of heavy fluid falling through the light one. RT bubbles are of particular interest during the acceleration phase because they penetrate into the shell while in-flight compromising the integrity of the spherical piston. Instead RT spikes are of interest during the deceleration phase when cold RT spikes from the dense shell penetrate into the hot spot thereby reducing the temperature and preventing the onset of ignition (Bose *et al.*, 2015; Kishony and Shvarts, 2001).

The ablation front in low-Z laser irradiated targets typical of direct drive exhibits a sharp density profile with a characteristic thickness L_0 set by the thermal conductivity of the cold shell (Kull, 1989; Sanz, 1994)

$$L_0 = \frac{2}{5} \frac{m_i \kappa(T_{sh})}{\dot{m}_a (1 + Z)} \quad (53)$$

where m_i is the ion mass and $\kappa(T_{sh})$ is Spitzer thermal conductivity evaluated at the temperature of the ablation front just inside the cold shell (T_{sh} is the cold shell temperature). Soft x-rays from the ablated plasma smooth out the density profiles but their effect is small especially when the ablation front is in DT. The minimum density gradient scale length is $L_m = (\nu + 1)^{\nu+1} / \nu^\nu L_0$ where $\nu = 5/2$ is the temperature power index in Spitzer heat conductivity. For typical direct drive targets, $L_0 \simeq 0.07 - 0.1 \mu\text{m}$, the ablation velocity $\sim 5 - 6 \mu\text{m/ns}$ and the acceleration $g \simeq 200 \mu\text{m/ns}^2$. Early work on the effect of mass ablation on the growth of the RT instability used heuristic closure equations to include the effect

of ablation indicating that ablation is stabilizing but incorrectly capturing the magnitude of the effect (Bodner, 1974; Kull, 1986; Mikaelian, 1992). Fitting the numerical solutions of the full set of linearized equations of motion (Takabe *et al.*, 1985) led to an approximation of the growth rate with the simple formula

$$\gamma \simeq \alpha \sqrt{kg} - \beta k v_a \quad (54)$$

with $\alpha \simeq 0.9$ and $\beta \simeq 3$ for typical direct drive conditions. Equation 54 was later generalized to different ablator materials and different plasma conditions through explicit dependencies of the coefficient α and β on the power index for thermal conductivity ν and Froude number Fr (Betti *et al.*, 1998a). The Froude number ($Fr = v_a^2 / g L_0$) determines the type of RT unstable modes. X-ray radiography experiments on OMEGA measured the growth of optical depth modulation in laser accelerated flat foils (Knauer *et al.*, 2000) and found good agreement of the RT growth rates with 2D radiation-hydrodynamic simulations and with the theoretical growth rates in Eq. 54. The first analytic self-consistent derivations of the ablative RT growth rate appeared in the literature in the mid 1990s. Since $Fr > 1$ for typical ablation fronts in DT, the unstable modes have angular and radial structure much wider than the thickness of the ablation front and the cutoff in the spectrum from ablative stabilization occurs for $k_{cut} L_0 < 1 / Fr^{\nu/(\nu-1)} < 1$ (Betti *et al.*, 1995; Bychkov *et al.*, 1994; Goncharov *et al.*, 1996; Sanz, 1994). For wave numbers above the cutoff, the instability is fully suppressed. There are several stabilizing mechanisms at play in the RT growth when $Fr > 1$. The self-consistent derivation of the ablative RT growth rate for direct drive targets (Betti *et al.*, 1998a; Goncharov *et al.*, 1996; Piriz *et al.*, 1997; Sanz, 1994) led to

$$\gamma \simeq \sqrt{A_T k g - A_T^2 k^2 v_a^2 / r_\rho} - (1 + A_T) k v_a \quad (55)$$

where $A_T = (1 - r_\rho) / (1 + r_\rho)$ is the Atwood number at the ablation front, $r_\rho \simeq (2/\nu)^{1/\nu} (k L_0)^{1/\nu} \ll 1$ is the ratio between the density in the conduction zone at a distance of $\sim 1/2k$ from the ablation front and the peak density of the shell. As shown in Eq. 55, the instability is driven by the classical RT term $A_T k g$, while the ablative stabilization enters through two terms inside and outside the square root. The stabilizing term proportional to $k^2 v_a^2 / r_\rho$ is the dynamic pressure representing the restoring force from the enhanced "rocket effect" on the RT spikes when moving towards the laser absorption region (Sanz, 1996). The term proportional to $k v_a$ represents the fire polishing effect from enhanced ablation of the spikes versus the bubbles and also the effect of vorticity convection away from the ablation front. The normalized growth rates of the ablative RT from Eq. 55 for different values of the Froude number are shown in Fig 26. Note that higher values of Fr lead to greater stabilization and smaller cutoff wave numbers.

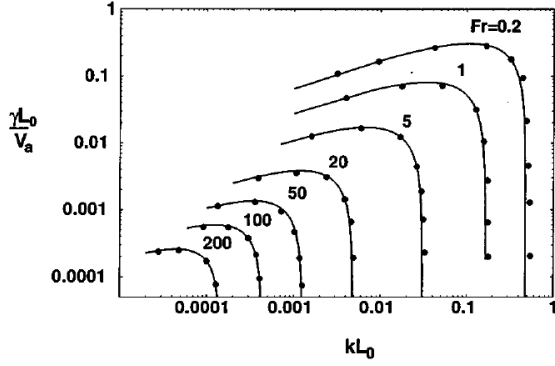


FIG. 26 Normalized linear growth rates of the ablative RTI versus normalized wave number for different Froude numbers from (Goncharov *et al.*, 1996)

With respect to indirect drive, direct drive targets exhibit lower ablation rates leading to lower ablation velocities ($v_a = \dot{m}_a / \rho_{sh}$ where ρ_{sh} is the in-flight shell density) and therefore higher growth of the Rayleigh-Taylor instability at the ablation front. Since the in-flight density of DT ρ_{sh} depends on the adiabat through the relation $p_{abl,D}(Mb) \simeq 2.2\alpha_F[\rho_{sh}(g/cc)^{5/3}]$, where $\alpha = 1$ implies a plasma in its Fermi degenerate state, the ablation velocity can be approximated using Eqs. 46 and 47

$$v_a(\mu m/ns) \simeq 5 \left(\frac{\alpha_F}{4} \right)^{\frac{3}{5}} \left(\frac{0.35}{\lambda_L^{\mu m}} \right)^{\frac{14}{15}} \left(\frac{A}{2.5Z} \right)^{\frac{7}{15}} \left(\frac{1}{I_L^{15}} \right)^{\frac{1}{15}} \quad (56)$$

Therefore direct drive targets needs to be driven on a adiabat higher than indirect drive targets to make up for the lower ablation rate and to maintain acceptable levels of ablative stabilization and RT growth mitigation.

In addition to reducing the linear growth rates, mass ablation affects the RT growth in the nonlinear phase. In the linear phase, a single Fourier mode grows exponentially until its amplitude reaches about 0.1λ . After that, the RT bubble advances at a constant velocity denoted as the "terminal bubble velocity". As shown in (Yan *et al.*, 2016), the terminal velocity depends on the buoyancy-drag balance and the fluid vorticity inside the bubble

$$U_b \approx \sqrt{\frac{g(1-r_\rho)}{C_D k} + r_\rho \frac{\Omega^2}{4k^2}} \quad (57)$$

where $C_D = 3$ in 2D and $C_D = 1$ in 3D, and Ω is the vorticity. In ideal fluids without external sources of vorticity, the potential flow description is valid and vorticity is only generated by the RT and localized at the interface between the two fluids. In this case $\Omega = 0$ everywhere

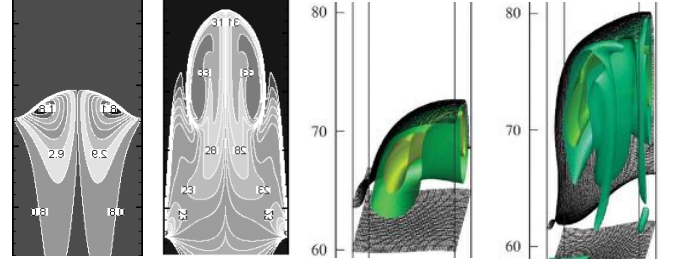


FIG. 27 Counter plots of ablation-induced vorticity inside the bubble of single mode simulations of the ablative RTI. The two figures on the left are snapshots at 1 ns and 2.25 ns from 2D simulations of a velocity perturbation with $\lambda = 10\mu m$ wavelength and $\tilde{v} = 1\mu m/ns$ in an accelerated planar target with $g \simeq 110\mu m/ns^2$ and $v_a \simeq 3.5\mu m/ns$. from (Betti and Sanz, 2006). The two figures on the right are similar snapshots from 3D simulations (Yan *et al.*, 2016). There is no color scale, but the outer (inner) surface of the top color plot corresponds to vorticity values 12.5/ns (19/ns) and the outer (inner) surface of the bottom color plot corresponds to vorticity values 25/ns (40/ns). The two on the left are from 2D simulations.

else outside the interface including inside the bubble and Eq. 57 with $\Omega = 0$ reduces to the classical terminal velocity of (Goncharov, 2002; Hecht *et al.*, 1994; Layzer, 1955). However, in the presence of mass ablation, the inter-facial vorticity is convected by the ablative flow into the ablated plasma which fills the bubble, thereby accumulating vorticity inside the bubble (Fig. 27). In the ablative RT, the vorticity scales with the ablation velocity as $\Omega \sim 2kv_a/r_\rho$ leading to a terminal velocity, Eq. 57, enhanced by ablation. The nonlinear single-mode theory of the ablative RT in (Sanz *et al.*, 2002) describes all modifications due to ablation except for the terminal bubble velocity augmentation.

Of particular importance to the acceleration phase is the penetration of the RT bubble front into the imploding shell. Unless dominated by a single mode, the seeds of the RT have a broad spectrum. In direct drive, a high-mode broad non-uniformity spectrum comes from laser imprinting (Section IV.B.3). In this case, the multimode nonlinear coupling and evolution is most relevant. Many studies of the multimode RT have been carried out, mostly for the classical case without mass ablation. It was found that when the RT is seeded by a broad multimode perturbation, small bubbles merge into bigger bubbles (bubble merger regime) or a faster growing bubble overcomes the others developing a self-similar wavelength shift to larger sizes (bubble competition regime). In both cases, the time-dependent mode spectrum exhibits a skew towards longer wavelengths as predicted by multimode theories and experiments (Casner *et al.*, 2016; Ofer *et al.*, 1996; Shvarts *et al.*, 1995). Figure 28 shows x-ray radiography snapshots of the multimode RT seeded by laser imprinting at three different times. The

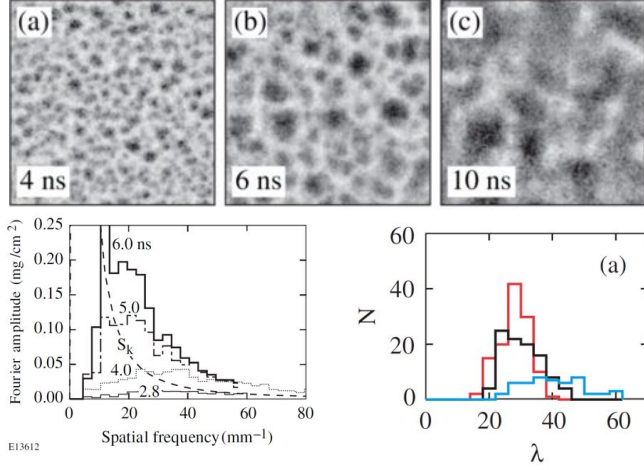


FIG. 28 (top) X-ray radiographs of the 3D broadband modulations, initially produced by imprinting of the laser beam and measured at 4, 6, and 10 ns [(a), (b), and (c), respectively] (Smalyuk *et al.*, 2005). (bottom left) Mode spectrum from the x-ray radiographs at three times (Smalyuk *et al.*, 2005). (bottom right) Simulated distributions of the bubble sizes (number of bubbles N versus bubble size λ) at three different distances traveled, 1 μm (red curve), 18 μm (black curve), and 67 μm (blue curve). from (Sadot *et al.*, 2005)

measured and simulated RT mode spectra for a broadband initial perturbation shown in Fig 28 also indicate a shift of the peak towards longer wavelengths and larger bubbles as also predicted by Haan's model (Haan, 1989).

Dimensional analysis indicates that in the deeply non-linear phase, the RT bubble front should advance proportionally to the distance traveled by the shell

$$h_b \simeq \alpha_b A_T g t^2 \quad (58)$$

where h_b is the bubble front width and $g t^2$ is twice the distance traveled for a constant acceleration. For a time dependent acceleration, $g t^2$ should be replaced by $(\int_0^t \sqrt{g} dt)^2$. The value of the constant α_b has been a subject of intense studies including dedicated experiments (Dimonte and Schneider, 2000), 2D and 3D simulations (Dimonte, 2004) indicating that α_b depends on the initial seed values of the perturbations. In (Zhang *et al.*, 2020, 2018), it was found that, for small initial seeds, ablation reduces the value of α_b with respect to the classical case

$$\alpha_b^{abl} \simeq \left(1 - \beta v_a \sqrt{k_0/g}\right) \alpha_b^{cl} \quad (59)$$

where $\beta \simeq 4.2$ and $k_0 \simeq 0.06 \mu\text{m}^{-1}$ is introduced to normalize the initial amplitudes. For values of g and v_a typical of direct drive, α_b can be significantly reduced by about 30% with respect to the classical value. However the stabilizing effect of ablation only occurs at sufficiently small initial perturbation amplitudes. Figure 29 compares values of α_b for the classical RT, ablative RT

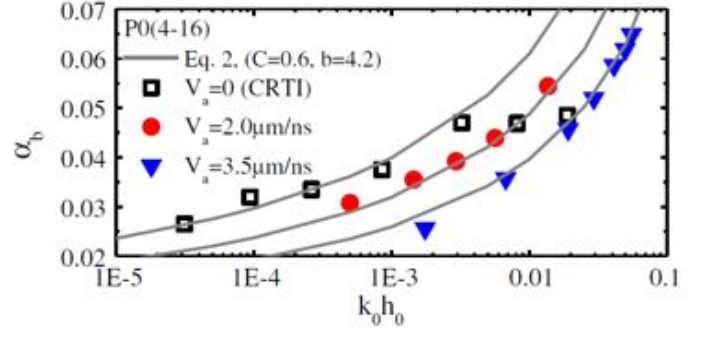


FIG. 29 Results from 2D simulations showing the dependence of α_b on the initial RMS multimode perturbation amplitude h_0 for different ablation velocities from (Zhang *et al.*, 2018).

and different initial seed amplitudes. Note that in the classical RT ($v_a = 0$ curve), the value of α_b saturates at about 0.05 when the bubble front transitions from the bubble competition to the bubble merger regime at large initial amplitudes. Instead in the ablative case, ablation induced vorticity prevents saturation and α_b exceeds the classical value at large enough initial perturbations.

If the RT bubble front penetrates through the entire shell and reaches the inner shell surface, the spherical piston is greatly decompressed and cannot efficiently compress the hot spot. An efficient energy transfer from the shell to the hot post requires a thin high-density shell whose inner and outer surfaces stagnate almost simultaneously. A turbulent shell is equivalent to a ultra high entropy plasma resulting in a very thick low-density shell with the inner surface stagnating well before the outer surface thus preventing an efficient transfer of energy. Therefore, it is crucial to control the growth of the RT bubble front to a width less than the in-flight shell thickness.

The shell thickness determines the in-flight aspect ratio $IFAR = R/\Delta_{sh}$, defined as the ratio of the shell radius R and its thickness Δ_{sh} . The IFAR is a crucial parameter determining the hydrodynamic stability to short wavelength perturbations and the penetration of the bubble front into the imploding shell. For a thin shell and in the absence of ablation, mass conservation requires that during the implosion, the shell thickness increases like C_R^2 while the IFAR decreases like C_R^{-3} .

Mass ablation reduces the shell thickness and the IFAR scaling with C_R for typical direct drive implosions is $IFAR \sim IFAR_{max}/C_R^2$ where $IFAR_{max}$ is the maximum IFAR at the beginning of the acceleration phase $IFAR_{max} = R_0/\Delta_{sh}^{min}$ (Fig 30). The figure of merit for the stability against broadband short wavelength RT is the ratio h_b/Δ_{sh} representing the relative penetration of the bubble front. If this ratio exceeds unity during the implosion, the shell breaks up. The penetration of the bubble front can be approximated with $h_b \simeq 2\alpha_b(R_0 - R)$ where R_0 is the initial radius and $R(t)$ is the dis-

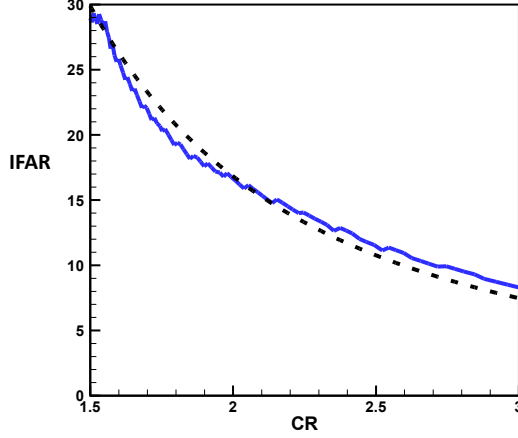


FIG. 30 Simulated evolution of the IFAR versus the convergence ratio (C_R) for a typical OMEGA DT layered target. The solid line is the simulated IFAR while the dashed line is the C_R^{-2} approximation.

tance traveled by the shell. Using the IFAR scaling leads to a relative bubble penetration

$$\frac{h_b}{\Delta_{sh}} \simeq 2\alpha_b IFAR_{max} \frac{1}{C_R} \left(1 - \frac{1}{C_R}\right) \quad (60)$$

Since the maximum value of the right hand side occurs for $C_R = 2$, the constraint on the IFAR to maintain $h_b < \Delta_{sh}$ is

$$IFAR_{max} < IFAR_{cr} = 40 \left(\frac{0.05}{\alpha_b}\right) \quad (61)$$

and the IFAR must be kept below $\simeq 40$ for $\alpha_b \simeq 0.05$. Note that by controlling the initial seeds and/or by increasing the ablation velocity, α_b can be reduced below 0.05 and the $IFAR_{max}$ can be increased. For $IFAR_{max} > IFAR_{cr}$, shell breakup occurs when the shell is at half the initial radius or earlier. The IFAR scales with the implosion Mach number ($IFAR \simeq Mach^2$, (Lindl, 1995)) and its value is determined by three variables: the in-flight adiabat, the final implosion velocity and the ablation pressure

$$IFAR_{max} \simeq 41 \left(\frac{v_{imp}^{km/s}}{400}\right)^2 \left(\frac{4}{\alpha_F}\right)^{3/5} \left(\frac{100}{p_{abl,D}^{Mb}}\right)^{2/5} \quad (62)$$

Higher adiabats and/or lower velocities lead to lower IFARs for a given ablation pressure. On OMEGA and NIF, higher velocity implosions have been successfully carried out to increase the fusion yield but at the expense of the adiabat that had to be increased in order to maintain stability by controlling the IFAR. Using the above expression for the maximum IFAR into the minimum kinetic energy required for ignition, it is found that

$IFAR_{max}$ sets the minimum ignition energy. This can be showed by using the minimum ignition energy derived by (Herrmann *et al.*, 2001; Levedahl and Lindl, 1997),

$$E_{kin}^{ign} > 126 \left(\frac{\alpha_F}{4}\right)^{1.88} \left(\frac{400}{V_{imp}^{km}}\right)^{5.89} \left(\frac{100}{p_{abl,D}^{Mb}}\right)^{0.77} \quad (63)$$

This can be rewritten using the IFAR from Eq. 62 leading to

$$E_{kin}^{ign} > 136 \left(\frac{40}{IFAR_{max}}\right)^3 \left(\frac{100}{p_{abl,D}^{Mb}}\right)^2 \quad (64)$$

Since the maximum value of the IFAR is limited by the penetration of the RT bubble set by Eq. 60, the energy required for ignition must satisfy

$$E_{kin}^{ign} > 136 \left(\frac{\alpha_b}{0.05}\right)^3 \left(\frac{100}{p_{abl,D}^{Mb}}\right)^2 \quad (65)$$

Equation 65 shows that the RT stability to short wavelength modes sets the minimum energy required for ignition. For a maximum classical RT bubble front penetration (Fig. 29), $\alpha_b \simeq 0.05$ and the minimum kinetic energy required for ignition is about $130 kJ$. The requirements on E_{kin}^{ign} are relaxed if α_b is reduced below the classical values either through higher ablation velocities or lower RT seeds (Eq. 59) or, even better, by keeping the RT growth in the linear phase.

When describing the seeds of the RT instability, it is important to recognize that the RTI occurs during the acceleration phase. Before the acceleration starts, the shell is shocked by multiple shocks (the pulse shape in Fig. 25 is a two shock design) and adiabatically compressed by the ramp in laser power. This is called the shock-transit phase of the implosion. During this phase, the acceleration of the ablation front is mostly impulsive and originates from the shocks. Shocked surfaces are unstable to the Richtmyer-Meshkov instability which grows linearly in time in a classical fluid. For a single Fourier mode with initial surface perturbation η_0 , the classical RM growth in the linear phase is $\eta_{RM}^{cl} \simeq \eta_0 (A_T k U_{ps} t + 1)$ where k is the perturbation wave number and U_{ps} is the post-shock fluid velocity. While usually referred to as an instability, the RM is simply an initial "impulse" to the perturbation that sets it in motion with a constant initial velocity causing the surface amplitude to grow linearly in time. The RM in ablation fronts (ablative RM) is different from the classical case because after the impulse, ablation determines the temporal behavior since the acceleration is negligible during the shock transit. This can be observed by setting $g = 0$ into Eq. 55 leading to damped oscillatory solutions $e^{i\omega_{RM}t - \nu_{RM}t}$ with ω_{RM} and ν_{RM} given by

$$\omega_{RM} \simeq \pm k v_a / \sqrt{r_\rho} \quad \nu_{RM} \simeq 2k v_a \quad (66)$$

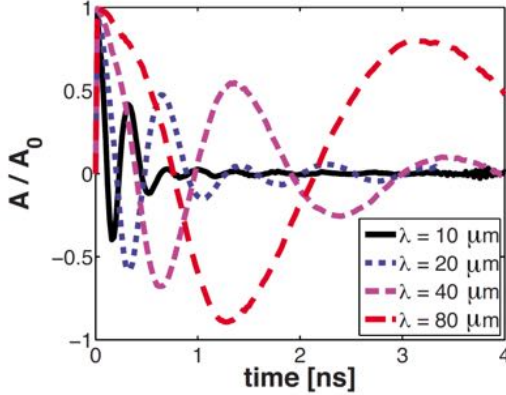


FIG. 31 RM instability at the ablation front. Perturbation evolution is shown for different wavelengths, $\lambda=10,20,40,80 \mu\text{m}$. Simulations refer to a thick D2 target irradiated by $I=4\times 10^{14} \text{W}/\text{cm}^2$, $\lambda_{\text{laser}} = 0.35 \mu\text{m}$. The target is initialized with a surface amplitude roughness $A_0 = 0.1 \mu\text{m}$. from (Marocchino *et al.*, 2010)

where $A_T \simeq 1$ has been used. Therefore, after the initial impulse, ablation turns the linear growth of the RM into damped oscillations (Goncharov, 1999) with short wavelength modes exhibiting the faster oscillations and the strongest damping (Fig. 31). In the limit of long wavelength modes, the oscillation frequency and the damping vanish and the growth is classical. The wave numbers affected by ablation are those for which the oscillation period $2\pi/\omega_{RM}$ and the damping time $1/\nu_{RM}$ is shorter than the shock transit time through the shell. Experiments on NIKE 32 and OMEGA have confirmed the theoretically predicted behavior of the ablative Richtmyer-Meshkov instability (Aglitskiy *et al.*, 2002; Gotchev *et al.*, 2006).

Another growth mechanism of surface perturbations in spherical implosions is the Bell-Plesset instability (see Section II.C). The BP is not a real instability but rather a secular growth driven by the conservation of mass in a rippled surface in convergent geometry. In spherical geometry and for large mode numbers $\ell \gg 1$, the single mode rippled surface amplitude $\xi_\ell(t)$ should be replaced by $\tilde{Q}(t) = \rho R^2 \xi_\ell$. $\tilde{Q}$ grows exponentially if the surface is unstable to the Rayleigh-Taylor instability and the surface perturbation would follow

$$\xi_{\ell \gg 1} \sim \frac{1}{\rho R^2} e^{\int^t \gamma_{RT} dt}$$

where γ_{RT} is the Rayleigh-Taylor growth rate (Epstein, 2004). During the acceleration phase, the shell density follows the ablation pressure and it is approximately constant. Therefore, in addition to the RT exponential growth, surface perturbations exhibit an additional am-

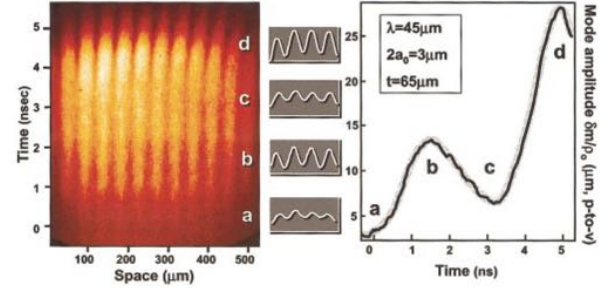


FIG. 32 Richtmyer-Meshkov instability experiments on the NIKE laser. Left: original streak record along with the amplitude lineouts taken across the ripples at four benchmarking times in the inserts. Right: the time-dependent Fourier amplitude of the dominant mode vs time for a $65 \mu\text{m}$ thick target with $\lambda = 45 \mu\text{m}$ and initial peak-to-valley amplitude $2a_0 \sim 3 \mu\text{m}$, also marked at four important times: beginning of the pulse, first peak, etc. The origin of time $t \sim 0$ here and elsewhere corresponds to the instant when the laser beam intensity reaches half maximum. The thickness of the shaded area approximately corresponds to the experimental uncertainty. (Aglitskiy *et al.*, 2002)

plification $1/R^2$ that scales with the square of the convergence ratio. This is usually referred to as the Bell-Plesset growth. Note that the BP growth persists even in the absence of acceleration (coasting phase) and it can be comparable or even exceed the exponential growth for low ℓ modes.

Several mitigation techniques have been developed to reduce the impact of hydrodynamic instabilities. Mitigation strategies can be divided into two categories: (a) reduction of the growth rates and (b) reduction of the initial perturbations (seeds). In this section, we only consider the growth rate reductions. Ablative RTI growth rates can be reduced by increasing the ablation velocity or by increasing the density gradient scale length at the ablation front. Since the ablation velocity is proportional to $\alpha_F^{3/5}$, raising the adiabat is a straightforward remedy to mitigate the ARTI (e.g. see Section III.E.2). However, simply raising the adiabat uniformly throughout the shell degrades the compressibility of the shell and increases the energy required for ignition (Eq. 65). Since only the inner portion of the shell contributes to the core confinement, it is possible to shape the adiabat profile so that the ablation front is on a high adiabat (favorable for stability) and the inner shell surface is on a low adiabat (favorable for compression and ignition). Shaping the adiabat can be done in two ways: (a) by using a strong

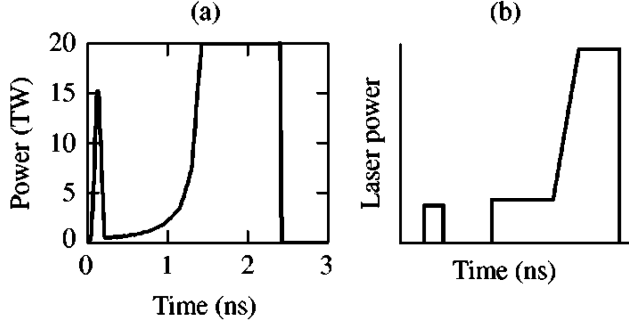


FIG. 33 Laser pulse shapes for a decaying-shock-wave picket (a) and a relaxation picket. (a) The decaying shock wave is created by the short picket at the beginning of the pulse shape. The adiabat is shaped as the shock wave decays. Figure (b) shows the pulse shape for a relaxation-picket drive. The low-intensity, narrow picket in front creates a spatial density profile that is low at the ablation region and high inside the shell. The strong shock wave resulting from the high foot intensity then propagates through this density profile shaping the shell adiabat. From (Knauer *et al.*, 2005)

decaying first shock or (b) by launching two shocks, a decaying shock with weak to moderate strength followed by a second shock. The first technique is referred to as *adiabat shaping by a decaying shock* (Goncharov *et al.*, 2003) and it requires a single-shock laser pulse shape with a strong initial picket followed by the ramp to peak power and the main drive. The strong initial picket launches a decaying shock leaving behind a monotonically decreasing adiabat profile. The second technique is referred to as *adiabat shaping by relaxation* (Anderson and Betti, 2004) and requires a two shock laser pulse shape with a weak picket followed by a foot, ramp and main drive. The weak picket launches a decay shock which moderately shapes the adiabat but also drives a rarefaction wave causing the outer shell surface to expand outward and the density profile to relax. A second shock is launched by the foot of the laser pulse. The second shock travels through the monotonically increasing density profile thereby shaping the adiabat even further. Examples of laser pulses used to perform adiabat shaping are shown in Fig. 33. Figure 34 shows the schematic of the adiabat shape (a) and compares adiabat profiles generated by different adiabat shaping techniques (b). Both adiabat shaping techniques have been shown to reduce the RTI growth rates (Fig. 35) and improve implosion performance.

A reduction of the RT growth rates can also be achieved using plastic ablaters doped with higher Z dopants such as bromine (Fujioka *et al.*, 2004). Bromi-

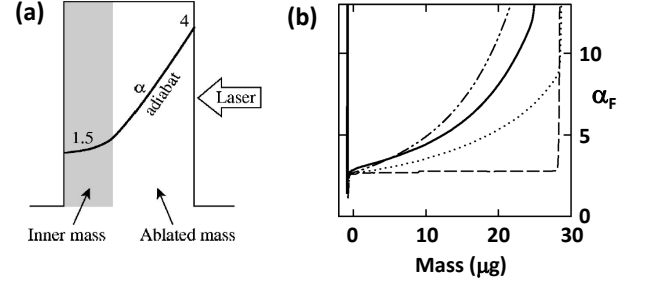


FIG. 34 (a) From (Knauer *et al.*, 2005). Schematic of a shell showing a shaped adiabat between the ablation surface and the inner surface. The shaded region is the portion of the shell that is not ablated. The adiabat is higher in the ablated material and therefore reduces the RT growth of ablation-interface perturbations. (b) From (Anderson and Betti, 2004). Simulated in-flight adiabat shapes versus mass coordinates for the three pulses (dashed is flat, dotted is the decaying shock and solid is the relaxation technique). The dashed-dotted curve is an analytic result for adiabat shaping from relaxation.

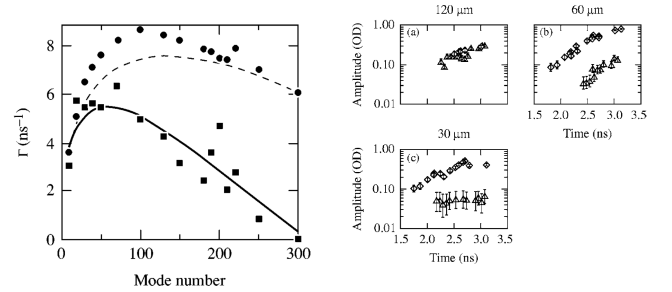


FIG. 35 (left) Comparison of simulated RT growth rates from flat adiabat (circles) and adiabat shaped targets (squares) using the relaxation technique from (Anderson and Betti, 2004). (right) Measurements of the temporal evolution of the optical depth (OD) from laser imprinting perturbations driven by a flat adiabat (diamonds) and adiabat shaped pulse (triangles) using the decaying shock technique. The different plots are for wavelengths 30, 60 and 120 μm from (Smalyuk *et al.*, 2007)

nated plastic leads to the formation of double ablation fronts (Sanz *et al.*, 2009) with an outer ablation front driven by electronic heat conduction and an inner front driven by radiation transport. The radiation-heated ablator material between the outer and inner ablation fronts exhibits lower density than the inner portion of the shell leading to higher ablation velocity and lower RT growth rates at the outer electronic ablation front (Yañez *et al.*, 2011). This technique has not yet been tested in direct drive DT layered implosions.

3. Initial nonuniformity seeding of hydrodynamic instabilities in direct drive implosions

In laser direct drive, there are multiple sources of symmetry perturbations that are amplified by hydrodynamic instabilities. The Rayleigh-Taylor is the most dangerous instability because of the fast exponential growth during the linear phase of the instability. It occurs during the main drive portion of the laser pulse when the target is accelerated inward to achieve the maximum implosion velocity at the end of the pulse. There are three types of perturbations in direct drive: low modes ($\ell < 6$), mid modes ($6 < \ell < 30$) and high modes ($\ell > 30$).

Low modes. Low modes are perturbations with wavelength longer than the shell radius. In spherical geometry, the wavelength of an angular perturbation is $\lambda = 2\pi R/\ell$ where ℓ is the mode number. Only mode numbers $\ell < 2\pi \simeq 6$ exhibit $\lambda > R$, thus the name of “low modes”. Because of their long spatial variation both angular and radial, low modes are not affected by thermal smoothing in the conduction zone or in the absorption region. Instead they are secularly driven and amplified by convergence effects. Low modes are mostly seeded by laser mis-pointing, target offset and beam power imbalance. In the case of Polar Direct Drive, the laser beam configuration exhibit a large intrinsic $\ell = 2$ mode that needs to be mitigated with appropriate beam pointing. There are also low mode asymmetries in the ice layer and in the ablator shell, but those are typically too small to affect implosion performance. Target offset away from TCC (Target Chamber Center) is caused by vibrations of the target mount. On OMEGA, a set of high speed cameras tracks the position of the target and measures the target offset at shot time. Target offsets below $10\text{--}15\mu\text{m}$ are typical on OMEGA and the resulting degradation in yield is modest ($\leq 10\%$ for a $\sim 10\mu\text{m}$ offset). Occasionally, large vibrations occur resulting in large target offset with severe consequences on the fusion yield and core compression. The $\ell = 1$ mode has also been observed as a systematic deformation in ambient plastic implosion on OMEGA (Shah *et al.*, 2017). The degradation in the fusion yield from $\ell = 1$ mode is assessed by measuring the apparent ion temperature from multiple lines of sight and by measuring the residual flow velocity of the com-

pressed core using nuclear diagnostics (Mannion *et al.*, 2020). Nuclear measurements of the ion temperature using NTOF detectors are affected by the residual isotropic and anisotropic flows (Munro, 2016; Murphy, 2014). The $\ell = 1$ mode is a shift of the core with respect to the original target center caused by a residual directional flow. This results in varying ion temperature measurements (about sinusoidal variation) depending on the LOS with a minimum and maximum value. It was shown in Ref. (Woo *et al.*, 2020) that the yield degradation from the $\ell = 1$ is a monotonic function of the ratio T_{max}/T_{min} with

$$YOC_{\ell=1} \sim \left(\frac{T_{min}}{T_{max}} \right)^{1.5} \quad (67)$$

Therefore by measuring the apparent ion temperature along different LOSs, it is possible to infer the yield degradation from the $\ell = 1$ mode.

On OMEGA, the velocity associated with the shift of the moving hot spot is inferred by measuring the second moment of the neutron spectrum among different detectors. This measurement is used (Mannion *et al.*, 2020) to determine a pre-imposed offset of the target to mitigate systematic $\ell = 1$ from laser mis-pointing or power imbalance. This mitigation technique has been successfully implemented in DT-layered implosions leading to significant yield increases.

Another low mode of interest in OMEGA implosions is the $\ell = 2$ that is routinely observed in the core self-emission images and leading to elliptical shapes of the hot spot with typical ellipticity values $\kappa = b/a \simeq 1.1\text{--}1.2$. Measurements of the soft x-ray emission from the ablated plasma in proximity of the ablation front using multiple framing cameras provide a 3D tomographic reconstruction of the ablation front capable of resolving modes $\ell = 1\text{--}3$. Using these measurements, such low modes can be suppressed by applying power corrections to the individual beams (Michel *et al.*, 2018). This was demonstrated in implosion experiments using both warm CH shells filled with D2 gas and DT layered implosions where power corrections led to rounder hot spots and modest increases in neutron yield.

Very recent 3D ray-tracing simulations of cross beam energy transfer including the effect of polarization have indicated that CBET can also be a significant source of low $\ell = 1, 2$ modes. While the amplitude of these CBET induced modes is target design dependent, their orientation appear to be systematic and can possibly explain observations of the core emission images (Edgell, 2021). Without the effect of polarization, CBET was shown to mitigate the $\ell = 1$ mode through a restoring force acting on the target (Anderson *et al.*, 2020).

Mid modes. Mid modes are moderate mode number perturbations with $6 \leq \ell \leq IFAR$ with $IFAR \simeq 30$ for typical direct drive implosions. The lower bound is

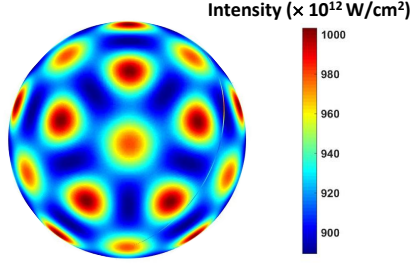


FIG. 36 Hard sphere projection of the laser illumination pattern on a typical target for 60-beam OMEGA. Courtesy of A. Shvydky (LLE).

determined by the condition for thermal smoothing during the main drive requiring that $kD_c^{main} > 1$ through the high intensity portion of the laser pulse. Here D_c^{main} is the distance D_c between the ablation front and the critical surface. Using $k \simeq \ell/R$, thermal smoothing during the main drive and decoupling of the mode from the capsule occurs when $\ell > R/D_c^{main}$. For OMEGA direct drive implosions, D_c^{main} varies between $50 - 100 \mu m$ during the main drive and the critical surface radius varies between the initial target radius and just above half its value ($\sim 300 - 500 \mu m$) leading to the condition $\ell > 6$ for mode decoupling during the main drive. This implies that mid modes are not driven secularly during the main drive but instead are seeded during the picket and foot of the laser pulse and then grow due to the Rayleigh-Taylor instability.

The upper bound in mid mode number is set by the penetration of the mode into the shell (feedthrough). Rayleigh-Taylor modes that are localized near the ablation front are called *high modes*. Mid modes differ from high modes because they radially extend across the imploding shell. The condition $k\Delta_{sh}^{in-flight}$ determines the transmission of the perturbation across the shell of thickness $\Delta_{sh}^{in-flight}$ from the ablation front to the inner surface. When $k\Delta_{sh}^{in-flight} < 1$ (mid modes) the RT modes at the ablation front decay slowly in space (like $e^{-kx} \sim 1$ with $kx < 1$ where x is the distance from the ablation front into the shell) and the ablation front distortion is transferred to the inner shell surface. Approximating $k \simeq \ell/R$ leads to the mid mode condition $\ell < R/\Delta_{sh}^{in-flight} \simeq IFAR$. The mode spectrum from a hard sphere illumination pattern of the OMEGA 60 beam configuration shows the presence of a mid mode $\ell = 10$ (Figure 36). The hard sphere illumination can be viewed as the initial laser pattern seen by the target at the beginning of the laser pulse. As soon as the plasma is formed by laser ablation around the target, the mode spectrum and amplitudes seen by the ablation front change due to the plasma buffer between the ab-

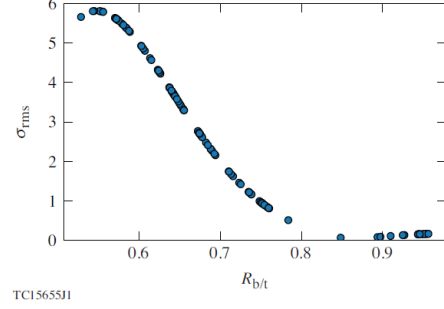


FIG. 37 RMS variation of the on-target illumination versus the ratio R_{beam}/R_{target} for OMEGA phase plates with super-gaussian profile exponent 5. Courtesy of V. Gopalaswamy (LLE).

lation front and the critical surface. During the main drive, the distance between the ablation front and the laser absorption region is large enough that mid modes from the laser illumination pattern are smoothed out before reaching the ablation front. Smoothing occurs due to the large spatial distribution of the absorbed laser energy and the diffusive effects of heat conduction [cloudy day effect, Ref (Bodner, 1981; Craxton *et al.*, 2015)]. Therefore, the source of mode $\ell = 10$ non-uniformities occur mostly during the picket of the laser pulse shape, to a minor degree during the foot and very little during the main drive. In the hard sphere illumination, the amplitude of mode $\ell = 10$ depends mostly on the ratio between the laser beam size and target size (R_b/R_t). Figure 37 shows that, on OMEGA, the amplitude of mode $\ell = 10$ is small for $R_b/R_t \geq 0.85$. For $R_b/R_t < 0.85$, the amplitude grows monotonically and implosion performance becomes dominated by mode $\ell = 10$. There are some indirect experimental evidence that significant degradation occur even at higher values of R_b/R_t but this has not been confirmed by direct probing of the mode $\ell = 10$ evolution.

High modes. The main source of short wavelength non-uniformities in direct drive is laser imprinting. Laser imprinting comes from the high frequency modulations of the laser intensity in each beam. In direct drive, high modes have mode numbers $\ell > IFAR \simeq 30$ and are driven by the Rayleigh-Taylor instability. Because of their short wavelength, they are decoupled from the source in the laser speckles during the main drive and instead they are seeded during the initial phase of the laser interaction with the target surface. Using Eq. 46 to determine the local ablation pressure from the laser speckle pattern, leads to a modulated ablation pressure launching a rippled shock in the ablator when the laser light first interacts with the target surface. Equation 46 leads to an overestimation of imprinting because it assumes that local spatial and temporal variations of the laser intensity result in local variations of the ablation

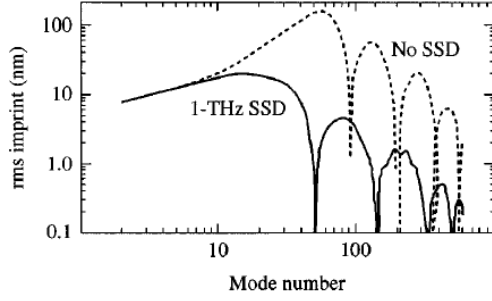


FIG. 38 Simulated ablation front mode spectrum from laser imprinting at the start of the main drive for OMEGA targets with 1 THz SSD and without SSD. From (Goncharov *et al.*, 2000b) .

pressure on the target surface (this is equivalent to 100% imprint efficiency). Instead the imprint efficiency is reduced by the initial development of the conduction zone between the critical surface and the ablation front. Initially, the conduction zone width increases with time as more laser energy is absorbed near the target surface. In the conduction zone of width D_c , the temperature rises from tens of eVs characteristic of the cold imploding shell to a few keV at the critical surface. Since the absorption region is beyond the critical surface, short wavelength perturbations with wave numbers $k > 1/D_c$ are smoothed by thermal conductivity (Bodner, 1981; Goncharov *et al.*, 2000b). Smoothing by thermal conduction is particularly effective due to the strong temperature dependence of Spitzer thermal conductivity for ideal plasmas ($\kappa_{Sp} \sim T^{5/2}$). It follows that laser imprinting occurs only during the initial picket when D_c is small, the plasma temperature relatively low and thermal smoothing is ineffective. After the initial development of the conduction zone, imprinting ceases and the growth of short wavelength modes at the ablation front is only driven by hydrodynamic instabilities. Laser smoothing, developed to mitigate laser imprinting, must therefore be effective only during the picket of the laser pulse. Two dimensional smoothing by spectral dispersion (SSD) currently implemented on OMEGA is very effective in reducing laser imprinting as shown in Fig. 38 comparing the target nonuniformity spectrum seeded by laser imprinting with and without SSD. SSD varies the interference speckle pattern of a laser beam focused by a phase plate. The smoothing requirements on SSD are set by the speckle variation being shorter than the characteristic hydrodynamic time scale of imprinting. This is achieved by adding bandwidth, typically of a few THz, to the fundamental laser frequency using an electro-optical modulator. Then, spectral dispersion is introduced causing the interference structure to vary in time and the time-averaged intensity is smoothed.

The imprinting level is also dependent on the shape

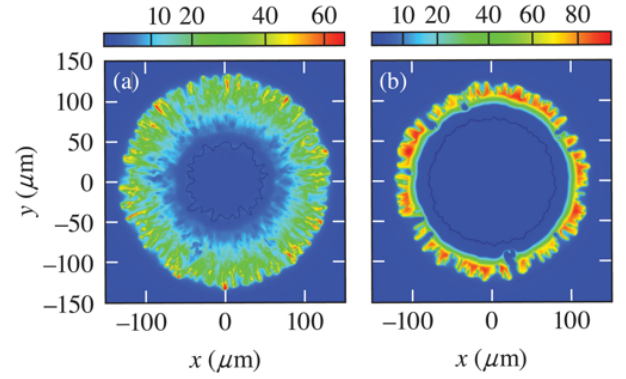


FIG. 39 Cross sections of the density (in g/cc) from 3D simulations of (left) a non-picket and (right) a single-picket pulse. Images are shown at $t = 2.45$ and 2.55 ns respectively in the moment prior to deceleration of the implosion. From (Igumenshchev *et al.*, 2019) .

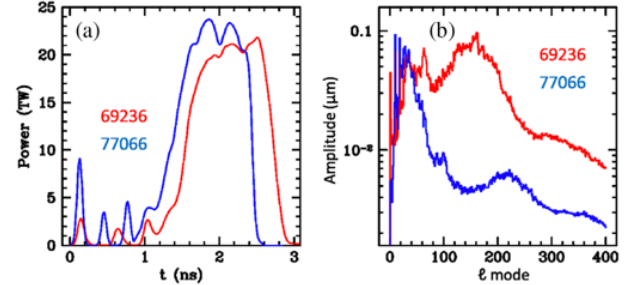


FIG. 40 (a) Laser pulses and (b) simulated spectra of surface modulation. Strong picket pulse in blue and reduced picket pulse in red. Spectra are taken at the time corresponding the shells' initial acceleration. From (Igumenshchev *et al.*, 2019) .

of the initial portion of the laser pulse. It was found in (Igumenshchev *et al.*, 2019) that laser pulse shapes with a picket develop less imprinting than laser pulses without picket and same adiabat (Fig. 39). The difference comes from the first shock which decays in space and time when launched by a picket as opposed to a supported shock launched by a foot which travels at constant velocity and strength. This occurs because the penetration depth of the ablation front perturbation into the shell is reduced by the rarefaction wave following a decaying shock. In this case, the imprinted perturbations remain localized near the ablation front where they are more effectively suppressed by mass ablation. Figure 40 shows the 3D simulated surface modulation spectra for pulse shapes with strong and weak pickets. The strong picket (and higher adiabat) pulse shape exhibits significantly lower perturbation amplitudes. The target adiabat plays a major role in governing the growth of laser imprinting after being seeded during the initial part of the laser pulse. This was observed in controlled plastic-target experiments on OMEGA where the adiabat was

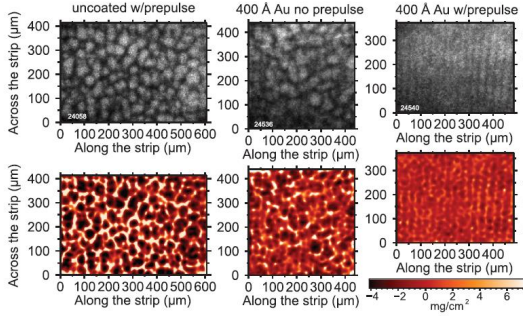


FIG. 41 Imprinting mitigation using pre-expanded Au overcoat. Framing camera images from three shots are shown (left-to-right): uncoated with a prepulse, 40 nm Au without a prepulse, 40 nm Au with a prepulse. Frame times are at 4.4 ns. The top row shows the linearized film images in grayscale. The bottom row shows the images processed using Wiener filtering with MTF correction and converted to areal mass fluctuation, displayed using the color map shown. All three shots had a preimposed 30 μm wavelength, 0.5 μm amplitude ripple (grooves oriented up-down). The uncoated shot is dominated by RT-amplified imprint from the laser speckle, while the Au coated shots with a prepulse show much lower imprint, with the pre-imposed ripple amplified by RT as the prominent feature. The shot with 40 nm Au without a prepulse shows some imprint reduction, but is still broken up by imprint. From (Karasik *et al.*, 2021).

varied between $\alpha_F = 2$ and $\alpha_F = 6$. The measured yield degradation was a strong function of the adiabat, with the lowest adiabats leading to most severe degradation (Hu *et al.*, 2016; Michel *et al.*, 2017).

Three promising approaches to reduce laser imprinting seeds have been explored in the last few years. The first was developed at NRL and uses a thin overcoat of high-Z material on the target outer surface. The high-Z layer is several tens of nanometers thick and needs to be pre-expanded before the laser pulse. Pre-expansion can be induced by heating the high-Z layer with a uniform source such as an externally-generated low-level soft x-ray prepulse of about 10 J/cm². The heated layer expands thereby generating a buffer zone between the laser deposition region and the ablation front. The effectiveness of this technique was shown in recent experiments on the OMEGA EP laser 41 using Au and Pd coating of CH targets where the imprinting level was reduced by over an order of magnitude (Karasik *et al.*, 2021).

Another mitigation strategy uses an outer thin Au-coated plastic shell surrounding the capsule. The thin shell consists of a 0.5-1 μm thick outer membrane of CH at a distance of about 300 μm from the capsule and coated with 40-60 nm of Au. A strong picket in the laser pulse heats up the gold and generates enough x-ray flux to drive the first shock into the capsule. The main drive portion of the laser pulse starts when the membrane is fully ablated and subcritical. The first shock is driven

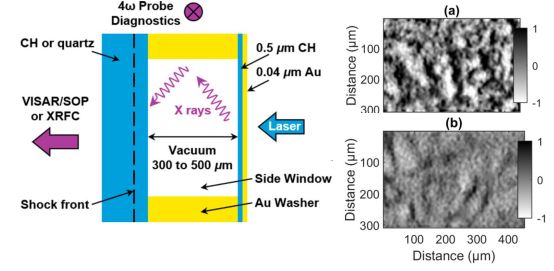


FIG. 42 Schematic of the hybrid direct-drive target design in planar geometry (left). Optical depth images of (a) a bare target and (b) a hybrid target after 4.2 ns (right). The hybrid target exhibit greatly reduced mass modulation. From (Ceurvorst *et al.*, 2020).

by x-rays and therefore highly uniform. The acceleration phase of the implosion is driven directly but a plasma buffer is left by the initial x-ray pulse and the laser is never in direct contact with the dense shell. This scheme is denoted as “hybrid”. The growth of target nonuniformities from laser imprinting was measured in hybrid targets designed for planar experiments on OMEGA EP (Fig. 42). The evolution of the imprinting spectrum in hybrid targets was compared to standard direct drive showing strong suppression of mass modulations in hybrid targets (Ceurvorst *et al.*, 2020). Another mitigation technique uses a super-critical CH foam overcoat on the capsule. The growth of imprinting is suppressed by reducing the growth rate of the Rayleigh-Taylor instability through increased ablation velocity and density scale length at the ablation surface (Hu *et al.*, 2018). Two dimensional simulations of imploding CH shells are used to study the evolution of nonuniformities seeded by laser imprinting for bare CH shells and for CH shells with a 40 μm thick layer of 40mg/cc CH foam. As shown in Fig. 43, the foam target exhibits reduced level of nonuniformities with respect to the standard CH shell. A reduction of laser imprinting was also observed in CH targets doped with higher Z elements such as silicon and germanium (Hu *et al.*, 2012). While this is an easily implementable imprint mitigation technique, it may not be sufficient to enable low adiabat direct drive implosions with $\alpha_F < 3$ requiring strict limitations on the initial imprint.

Direct measurements of the nonuniformities of the first shock front launched by direct laser irradiation in flat foils are good indicators of the seed level of laser imprinting before the amplification from hydrodynamic instabilities. These experiments were carried out using the two-dimensional velocity interferometer for any reflector

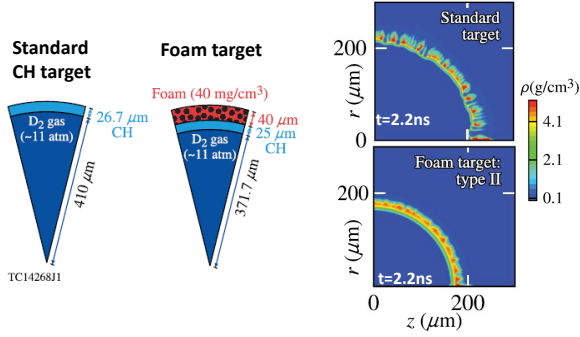


FIG. 43 The schematic diagrams for CH targets: standard bare CH shells and CH shells with a layer of foam added (left). Two dimensional simulations including laser imprinting were performed with the code DRACO showing significantly improved uniformity in foam targets with respect to standard targets (right). From (Hu *et al.*, 2018).

(2D VISAR) technique (Celliers *et al.*, 2010) which measures the fluctuations in velocity of the propagating shock front. 2D-VISAR measurements of shock front nonuniformities were carried out on OMEGA (Peebles *et al.*, 2019) and NIKE (Oh *et al.*, 2021) to assess the benefits of the aforementioned imprint mitigation techniques.

Finally, nonuniformities of the inner surface of the DT ice layer can also seed hydrodynamic instabilities occurring at the ablation front through a transfer of perturbations mediated by the rarefaction waves propagating from the inner to the outer shell surface after shock breakout. This transfer is called “feedout” and the perturbations exhibit a phase reversal before undergoing exponential RT growth (Betti *et al.*, 1998b). Feedout measurements with perturbation phase reversal followed by subsequent RT growth were carried out on the NIKE laser (Aglitskiy *et al.*, 2001).

4. Hot electron preheat in direct drive implosions

The electric field of nonlinear electron plasma waves driven by laser plasma instabilities such as Stimulated Raman Scattering (SRS) and Two Plasmon Decay (TPD) accelerate plasma electrons with velocity of the order of the phase velocity of the wave (Kruer, 2003). Such energetic “hot” ($\sim 30\text{--}100\text{keV}$) electrons intersect the imploding shell and deposit their energy in the DT fuel thereby raising the adiabat in-flight (preheat) and reducing the fuel compressibility. Understanding and quantifying preheat due to electrons from LPIs has been a long-standing effort. In OMEGA direct drive implosions, the main instability producing hot electrons is the Two-Plasmon Decay (TPD) originated from the decay of a light wave with frequency ω_0 into two plasma waves (plasmons) with half the frequency of the pump wave ($\omega_{pe} = \omega_0/2$). The condition ($\omega_{pe} = \omega_0/2$) is satisfied at

the quarter critical density $n_{cr}/4$. Strong wave-particle interactions occur through Landau damping for wave vectors $k\Lambda_{De} \simeq 0.2\text{--}0.3$ where k is the electron plasma wave number and Λ_{De} is the plasma Debye length. The electron plasma wave energy is transferred to energetic electrons through Landau damping and to the background plasma through collisional damping. In a nonuniform plasma, the TPD instability is excited above a convective threshold determined by the parameter $I\lambda_L L_n/T_e$ depending on the laser wavelength λ_L , the laser intensity I , the electron temperature T_e and the density gradient scale length L_n , all evaluated at the quarter critical density. As described in Ref. (Simon *et al.*, 1983) the TPD threshold can be approximated as $\eta_{TPD} = 1$ where

$$\eta_{TPD} = \frac{I_{14} L_n^{\mu m}}{230 T_e^{\text{keV}}} \quad (68)$$

with I_{14} is in units of 10^{14} W/cm^2 , L_n in μm and T_e in keV. In direct drive implosions, the local laser intensity driving the TPD instability depends on the overlap of several beams that share a common plasma wave vector component. This was recognized both in multi-beam experiments and simulations (Michel *et al.*, 2012a; Myatt *et al.*, 2014) and led to a general threshold parameter that depends on the quarter critical plasma properties and a polarization-adjusted sum of the intensities of each beam sharing the same angle with the plasma wave. The conversion of laser energy into hot electron energy was measured by the hard x-ray emission from hot electrons slowing down as well as from K_α emission from hot electron excitation in Mo layers embedded inside the target (Yaakobi *et al.*, 2012). Figure 44 from (Michel *et al.*, 2013a) shows the measured conversion efficiency (f_h =energy into hot electrons/ laser energy) as a function of the overlapped laser intensity (left) and as a function of the common wave gain (i.e. gain on the beam intensity) for different experimental configurations (planar and spherical) on the OMEGA laser facility

In large scale-length plasmas typical of NIF direct drive targets, SRS becomes the dominant instability and the overall conversion of laser energy into hot electron energy is expected to increase. In direct drive, SRS mostly develops at densities below $n_{cr}/4$ and its maximum gain occurs for sidescattered light waves. Like for TPD, the SRS instability threshold depends on the plasma scale length, the laser intensity and the laser wavelength, and can be approximated by $\eta_{SRS} = 1$ where

$$\eta_{SRS} = \frac{I_{14} \lambda_{L(\mu m)}^{2/3} L_{n(\mu m)}^{4/3}}{1549} \quad (69)$$

Recent direct drive experiments on NIF using planar plastic targets have shown that SRS dominates over TPD in ignition scale plasmas. The conversion efficiency of laser to hot electron energy was shown to increase with the laser intensity between 0.7 and 2.9%. Figure 45 shows

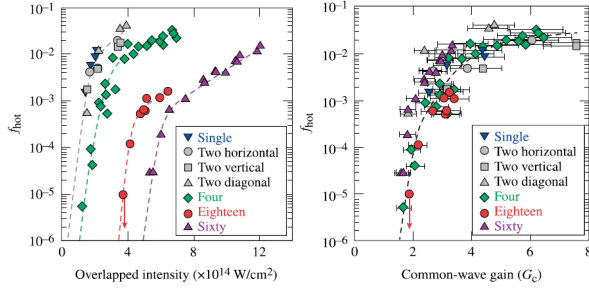


FIG. 44 (Left) Hot-electron fraction (f_{hot}) as a function of vacuum overlapped laser intensity. Single, two, and four correspond to OMEGA EP planar experiments where the beams are linearly polarized; 18 (60) corresponds to OMEGA planar (spherical) experiments where the beams have polarization smoothing. For each configuration, the overlapped intensity is given by the vacuum intensity of the laser beams on the target surface. (Right) Hot-electron fraction versus the common-wave gain for each experimental configuration tested. From (Michel *et al.*, 2013a).

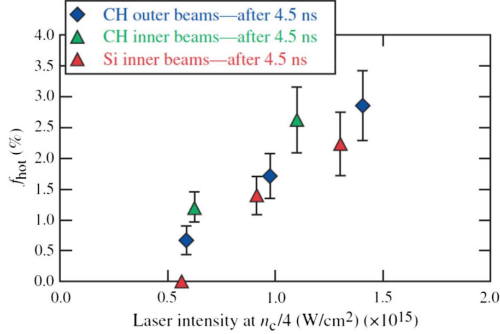


FIG. 45 Planar hot-electron measurements on the NIF. Fraction of laser energy converted to hot electrons as a function of simulated quarter-critical laser intensity, for laser drive using outer beams (diamonds) and inner beams (triangles), for CH ablators (blue, green) and Si ablators (orange). Uncertainty in f_{hot} is based on statistical uncertainty in the hard-x-ray data, propagated through EGSnrc modeling. From (Rosenberg *et al.*, 2018a).

the results of the NIF experiments including CH and Si targets, with the latter exhibiting a higher threshold for hot electron production.

Conversion factors from laser to hot electron energy above $\sim 1\%$ are of concern for the viability of direct drive as described later in this section. What is critically important is not just the total amount of energy transferred to hot electrons, but the amount of energy that such electrons deposit inside the DT fuel, and in particular, that portion of the DT fuel that produces the

areal density of the compressed core (hot electron coupling efficiency θ_h). In direct drive targets about 50% of the initial fuel mass is ablated and most hot electrons deposit their energy either in the outer ablator (typically CD or CH) or in the ablated DT plasma. Only a fraction of the hot electron energy is deposited inside the unablated DT fuel and even a smaller fraction in the inner portion of the fuel that produces most of the areal density. The minimum shell kinetic energy required for ignition is proportional to the square of the in-flight adiabat $Min[E_{kin}^{ign}] \sim \alpha_F^2/v_{imp}^6$ and a significant increase in adiabat from hot electron preheat results in more stringent requirements on the energy required for ignition. This degradation occurs because higher adiabats lead to lower hot spot pressures and lower areal densities in the compressed core. The areal density degradation from hot electron preheat can be easily quantified for a given preheat energy deposited in the fuel as shown in (Christopherson *et al.*, 2021)

$$\frac{\rho R_{hot}}{\rho R_{nohot}} \simeq \left(1 + 1.16 \frac{E_{stag}^{preheat}}{IE_{shell}}\right)^{-4/3} \quad (70)$$

where the left hand side represents the ratio of the areal density with and without preheat, $E_{stag}^{preheat}$ is the energy deposited by the hot electrons in the portion of the fuel that stagnates at peak fusion rate, and IE_{shell} is the internal energy of that same portion of the shell while in-flight. Using an ideal plasma approximations (valid for $\alpha_F \gg 1$), the energy ratio on the right-hand-side of Eq. 70 is proportional to $(E^{preheat}/Mass_{shell})/T_{shell}$ indicating that the areal density degradation depends on the hot electron energy deposited for unit mass and that lower-adiabat (e.g. colder shell) implosions are more sensitive to hot electron preheat. After a straightforward manipulation, the shell internal energy IE_{shell} can be rewritten in the following simple form

$$IE_{shell} \simeq \frac{9}{7} \frac{E_{kin}}{IFAR} \quad (71)$$

By denoting with θ_h the fraction of hot electrons depositing their energy into the unablated fuel making up the stagnating mass and assuming small areal density degradations $\delta\rho R/\rho R \ll 1$, the relation between areal density degradation and laser-to-hot-electron conversion efficiency f_h can be written as

$$\frac{\delta\rho R}{\rho R} \simeq \frac{\theta_h f_h}{\eta_k} IFAR \quad (72)$$

where $\eta_k = E_{kin}/E_L$ is the laser-to-kinetic-energy coupling efficiency. Eq. 72 sets a limit on the maximum hot electron fraction f_h based on the maximum tolerable areal density degradation

$$f_h \leq \left(\frac{\delta\rho R}{\rho R}\right)_{max} \frac{\eta_k}{\theta_h} \frac{1}{IFAR} \quad (73)$$

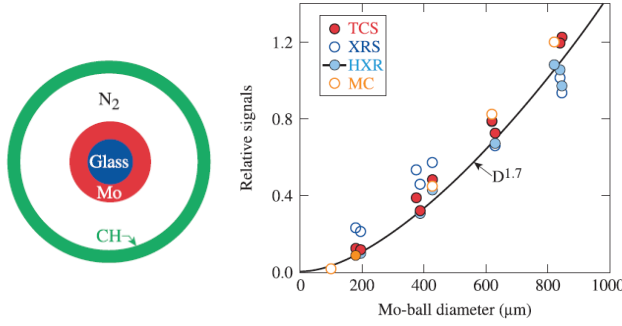


FIG. 46 (Left) Target used in divergence experiments on OMEGA. Molibdenum spheres of different sizes are enclosed within a laser driven plastic shell. (Right) Measured intensity of the Mo-K α line and hard x-ray radiation. The MC (Monte Carlo) simulation results for a divergent fast-electron beam are also shown. The curve is the best fit to the hard x-ray data. The increase of the signals with the Mo-shell diameter indicates a wide-angle divergence of fast electrons. From (Yaakobi *et al.*, 2013)

Note that the maximum f_h depends on the fraction θ_h of electrons depositing their energy in the stagnating mass. θ_h depends on the electron source divergence and the electron transport physics which is not well understood because of the complicated electron dynamics affected by the coronal electromagnetic fields, refluxing of escaping electrons and the shrinking interaction volume of the imploding shell. Measurements of the electron divergence in imploding capsules using Mo spheres of different radii enclosed within imploding shells were carried out on OMEGA (Yaakobi *et al.*, 2013). Those experiments indicated that the divergence of the electron source is wide enough that the fraction of electrons interacting with the shell at different convergence ratios decreases with the shell surface as $\sim 1/R^2$. Figure 46 shows the schematics of the targets used in the divergence experiments and the relative K α signals for different sizes of the inner Mo sphere.

Only recently, direct measurements of the hot electron energy deposited in the DT layer of cryogenic implosions and its distribution across the imploding shell mass were carried out on OMEGA (Christopherson *et al.*, 2021). Those experiments used two implosions: one of a DT layered target with a CD ablator, and one with a mass equivalent all-CD shell, both imploded by the same laser pulse. Those implosions were designed to exhibit the same coronal conditions at the quarter critical surface and therefore the same hot electron source. The differences in hard x-ray signals in the two implosions were

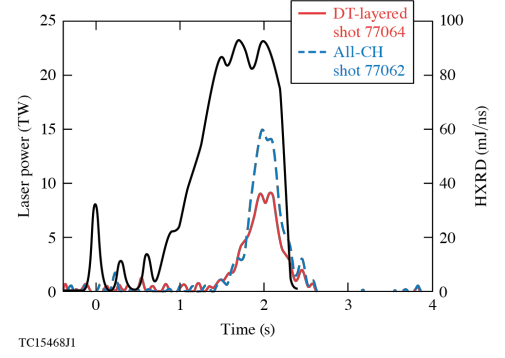


FIG. 47 The laser pulse (black) and the hard x-ray signals for DT-layered shot 77064 (red) and all-CH shot 77062 (blue) are plotted as a function of time. The difference in hard x-ray signals is proportional to the hot-electron energy deposited in the DT layer. From (Christopherson *et al.*, 2021)

caused by the slowing down of electrons in DT versus CD. Since electron slowing down in DT produces about 5x less hard x-rays than in CD, the subtraction of the two hard x-ray signals is proportional to the hot electron energy deposited in DT. Figure 47 from (Christopherson *et al.*, 2021) compares the measured hard x-ray signal of a DT layered implosion to an all-CD mass-equivalent implosion driven by the same laser pulse. The hard x-ray signal is lower in a layered target due to the hot electrons slowing down in DT rather than CD.

Other experiments using a payload of Cu-doped plastic of different thicknesses and similar preheat measurements were used to infer the distribution of the preheat energy within the unablated payload (Christopherson *et al.*, 2021). That enabled to accurately determine θ_h in OMEGA implosions. While θ_h varies for different targets, it was found that for high performance OMEGA implosions the preheat energy is uniformly distributed through the payload mass and only about 10% of the total hot electron energy is deposited in the stagnating mass ($\theta_h \simeq 0.1$). Therefore, using characteristic values typical of direct drive, the limit on the hot electron fraction f_h can be rewritten as

$$f_h(\%) \leq 0.4 \left(\frac{\delta \rho R}{0.2 \rho R} \right)_{max} \left(\frac{\eta_k}{0.06} \right) \left(\frac{0.1}{\theta_h} \right) \left(\frac{30}{IFAR} \right) \quad (74)$$

indicating that to prevent areal density degradation greater than 20%, the fraction of hot electrons should not exceed 0.4%. In high performance OMEGA implosion experiments, the hot electron temperature is $T_{hot} \simeq 50 - 60$ keV and their total energy is about 40-60 J corresponding to a conversion efficiency $f_h \sim 0.2\%$ for a 26 kJ laser pulse at peak intensity of about $9 \times 10^{14} \text{ W/cm}^2$. According to (Christopherson *et al.*, 2021), the preheat energy in the stagnating mass is about 5-7 J leading to a ρR degradation of 10 - 20% for an adiabat $\alpha \simeq 3 - 4$ and

IFAR $\simeq$ 25-30.

Because of thin ablaters used in direct drive targets, hot electrons from LPs penetrate deep into the DT fuel. Since the threshold of such instabilities is proportional to the plasma scale-length, preheat is expected to occur primarily towards the end of the pulse when the ablator material is fully ablated off and the DT fuel is exposed to hot electrons, even those with moderate energies $\sim$ 40-60 keV. Since the instability threshold for SRS and TPD is lower in long scalelength plasmas, hot electron preheat is of great concern for ignition scale direct-drive implosions. Ablator materials with higher atomic numbers have been shown to exhibit a higher threshold for LPs. For instance SiO_2 and Si ablaters (Rosenberg *et al.*, 2018a; Smalyuk *et al.*, 2010), Si-doping of CH shells (Smalyuk *et al.*, 2009) as well as buried Si-layers (Goncharov *et al.*, 2014) have significantly reduced the laser-to-hot-electron conversion efficiency. A comprehensive experimental effort is currently underway on the NIF to quantify and mitigate hot electron preheat for direct drive at ignition relevant scales (Solodov *et al.*, 2020).

5. Cross Beam Energy Transfer in direct drive implosions

As laser systems have increased their number of beams and become more powerful to drive large plasmas, the transfer of energy between laser beams in the plasma has become one of the challenges in predicting the hydrodynamic conditions of ICF experiments. The energy transferred between beams is maximized, when the frequency difference between the beams is equal to the ion-acoustic frequency in the frame of the plasma, $\Delta\omega = \omega_2 - \omega_1 = (k_2 - k_1)V + |k_2 - k_1|c_s$, where ω_1 , ω_2 and k_1 , k_2 are the frequencies and wavenumbers of the crossing laser beams; V is the plasma flow velocity; c_s is the sound speed (Fig. 48) (Randall *et al.*, 1981). For direct-drive experiments, where the laser beams typically have the same frequency ($\Delta\omega = 0$), the maximum energy transfer occurs near the Mach one surface ($V = c_s$) and this energy transfer leads to a significant reduction in the transfer of laser energy to kinetic energy of the imploding shell (6% $\rightarrow$ 4%) (Froula *et al.*, 2012; Igumenshchev *et al.*, 2010). Cross-beam energy transfer (CBET) reduces the incident energy in the central portion of the laser beams, making it possible for the incoming light to bypass the high-absorption region near the critical surface (Edgell *et al.*, 2017). Laser light in the wings of the laser beams propagates past the target and provides the seed for CBET to drive ion-acoustic waves to large amplitude. Recent work by Edgell et al (Edgell, 2021) also finds that polarization effects on CBET can significantly enhance low mode asymmetries during the main drive of OMEGA implosions.

Early direct-drive hydrodynamic simulations used classical thermal transport models (Spitzer and Härm, 1953)

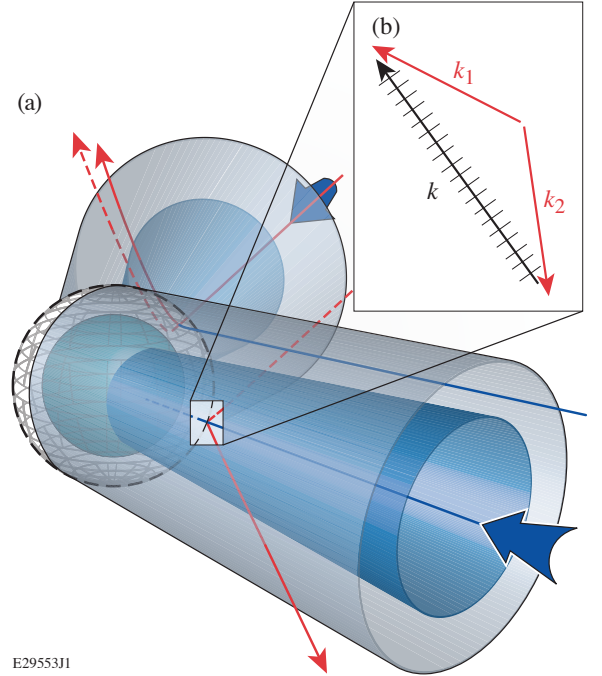


FIG. 48 (a) Light rays propagating past the target (blue) interact with light rays in the central region of another beam (red). (b) The interacting light rays seed an ion-acoustic wave near the Mach-1 surface (dashed curve). The ion-acoustic wave scatters light before it can penetrate deep into the target (dashed curves).

that required an ad hoc multiplier to match the physical observables from direct-drive implosion experiments. Much of the early discrepancies were remedied when both nonlocal thermal transport (Goncharov *et al.*, 2008) and CBET (Igumenshchev *et al.*, 2010, 2012) models were implemented in the hydrodynamic codes, but challenges in modeling the laser-beam propagation using geometric optics limited the accuracy of the CBET modeling. Improvements have been suggested to better account for energy conservation and ray caustics, which make defining intensity challenging (Colaïtis *et al.*, 2019; Follett *et al.*, 2018). Furthermore, uncertainties in plasma conditions have led to challenges in isolating errors between modeling the laser-plasma instability physics and experimental observables (Kirkwood *et al.*, 1996), which have made it difficult to understand the limitations of the linear CBET models implemented in the hydrodynamic codes.

Particle-in-cell simulations have suggested that when the ion-acoustic waves are driven to large amplitude, nonlinear effects will modify the energy transfer resulting in deviations from linear CBET theory (Cohen *et al.*, 1997; Yin *et al.*, 2019). Early experiments seemed to corroborate this picture, suggesting that nonlinear physics was required to model the interactions, but these experiments

relied primarily on hydrodynamic modeling to determine the plasma conditions (Glenzer *et al.*, 2001; Kirkwood *et al.*, 2002) and due to the uncertainties in plasma conditions, an understanding of the saturation physics was elusive.

To understand if there was significant missing laser-propagation physics within the models implemented in the hydrodynamic codes that could be masked by uncertainties in the CBET modeling, experiments were implemented that used Thomson scattering to separate the hydrodynamic uncertainties from the CBET physics (Hansen *et al.*, 2021; Turnbull *et al.*, 2020, 2017). Two important results came from these studies that are shaping the way CBET is modeled in ICF studies and the conclusions from both suggest that the current linear CBET models used in hydrodynamic codes are sufficient to accurately predict energy transfer provided the underlying hydrodynamic conditions are properly determined.

When the product of the intensities of the crossing beams was sufficiently small to limit the total energy transferred between beams, the linear CBET models implemented in the hydrodynamic codes accurately reproduced the measured results, but a sensitivity in the CBET resonance to the shape of the electron distribution function (Afeyan *et al.*, 1998) was identified (Turnbull *et al.*, 2020). The experiments showed that the non-Maxwellian electron distribution functions measured by Thomson scattering (Milder *et al.*, 2020), and inherent to laser produced plasmas (Langdon, 1980; Matte *et al.*, 1988; Milder *et al.*, 2021), modified the CBET resonance [Fig. 49(a)]. When accounting for the modified electron distribution functions in the CBET calculations, the simulations accurately reproduced the measured energy transfer.

The second significant result was obtained by increasing the probe intensity to where the ion-acoustic wave amplitudes were expected to significantly exceed $dn/n > 1\%$. These experiments demonstrated the robust nature of the linear CBET models implemented in the hydrodynamics codes, even when nonlinear laser-plasma instability physics was occurring [Fig. 49(b)]. During the 500 ps high-intensity CBET experiment, the ion temperature was measured to increase by a factor of 7 (120 eV to 900 eV), resulting in increased damping of the ion-acoustic waves, which limited the energy transfer to a regime where the linear CBET model accurately reproduced the energy transfer (Hansen *et al.*, 2021). Large Vector Particle-In-Cell (VPIC) (Bowers *et al.*, 2008) simulations demonstrated the physical mechanisms that limited the CBET to the linear regime. On the very fast time scales, the CBET driven ion-acoustic waves were driven to large amplitudes where they accelerated ions. The fast ions then equilibrated with the bulk ion distribution function resulting in an increased ion temperature. Fundamentally, this increase in ion temperature was expected through conservation of energy (Manley-

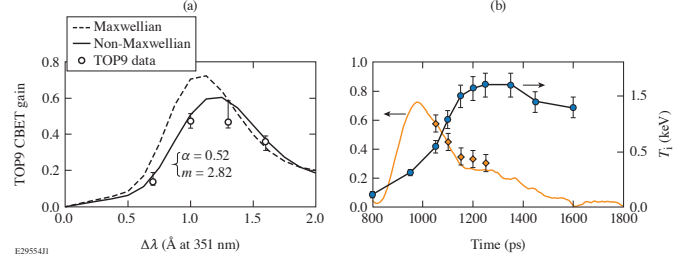


FIG. 49 (a) CBET data (points) measured when energy transfer was relatively low due to the low seed intensities ($0.4 \times 10^{15} \text{ W/cm}^2$) are shown for conditions where the many overlapping laser beams used to heat the plasma drove non-Maxwellian electron distribution functions. Calculations that (dashed curve) assume a Maxwellian electron distribution function and (solid curve) used the measured non-Maxwellian electron distribution function determined from Thomson-scattering (Turnbull *et al.*, 2020). (b) For CBET experiments at higher TOP9 (probe) beam intensities ($4 \times 10^{15} \text{ W/cm}^2$) where significantly more energy was transferred, the CBET gain was measured (solid curve) to decrease as a function of time. Over this same time period where CBET was active, Thomson-scattering measurements showed a rapid increase in the ion temperature (circles); the final ion temperature was nearly a factor of two larger than the electron temperature. When accounting for this increased ion temperature, the linear CBET calculations (diamonds) were in excellent agreement with the measurements (Hansen *et al.*, 2021).

Rowe relations (Manley and Rowe, 1956)) and was explored theoretically in the context of indirect drive ICF in Ref. (Michel *et al.*, 2012b). Although these results show the robustness of linear CBET, it is important to recognize that CBET inherently transfers energy to the ions modifying the hydrodynamic conditions and therefore this physics must be accounted for in hydrodynamic codes to accurately determine the plasma conditions and subsequently laser-plasma instabilities (Michel *et al.*, 2013b; Strozzi *et al.*, 2017).

Due to the ability of CBET to significantly reduce the hydrodynamic efficiency for direct-drive implosions, concepts for mitigating CBET have been explored (Bates *et al.*, 2018; Edgell *et al.*, 2017; Follett *et al.*, 2019; Igumenshchev *et al.*, 2012). Figure 50 shows the implosion trajectories from direct-drive experiments on the NIF where the relative frequency of the laser beams was used to reduce CBET and increase the hydrodynamic efficiency by moving the CBET resonances to regions of plasma where energy transfer was less efficient (Marozas *et al.*, 2018b). Although these wavelength detuning concepts are encouraging, they are ultimately complicated by the fact that a reduction in CBET increases the laser intensity near the quarter critical density, making any design using this tactic more susceptible to hot electron generation by two-plasmon decay or stimulated Raman scattering. To account for both hot electron genera-

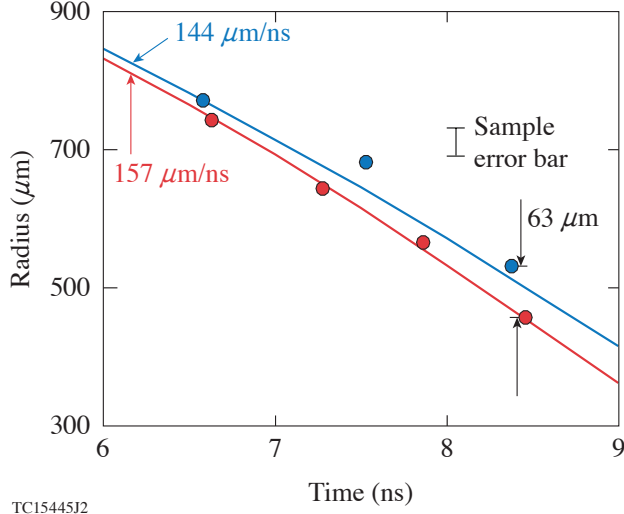


FIG. 50 Equatorial shell trajectories from postprocessed simulated (solid lines) and experimental (symbols) backlit radiographs show improved coupling when the relative frequency between crossing beams was changed. The lower lines and symbols represent the baseline zero frequency detuning experiment. The upper lines and symbols represent the average of the two detuning experiments. Courtesy of (Marozas *et al.*, 2018b).

tion and CBET mitigation, large bandwidth ultraviolet (UV) lasers (Dorrer *et al.*, 2020; Obenschain *et al.*, 2020; Weaver *et al.*, 2017) are a leading concept (Follett *et al.*, 2019). Significant breakthroughs in both gas laser ($\Delta\omega/\omega \sim 0.6\%$ at 193 nm) (Obenschain *et al.*, 2020) and solid state laser ($\Delta\omega/\omega \sim 5\%$ at 1053 nm) (Dorrer *et al.*, 2020) technologies along with novel broadband frequency conversion concepts ($\Delta\omega/\omega \sim 1.5\%$ at 351 nm) (Dorrer *et al.*, 2021) are under development. Simulations suggest increasing the UV bandwidth of the laser beyond $\Delta\omega/\omega \sim 0.5\%$ (an order of magnitude beyond current UV solid state laser bandwidths) will mitigate CBET (Bates *et al.*, 2020, 2018), while increasing the bandwidth beyond $\Delta\omega/\omega > 1\%$ will increase the intensity thresholds for hot electron generation by factors of several (Follett *et al.*, 2021), allowing current direct-drive designs to be implemented without significant laser-plasma instabilities.

C. Present status of direct-drive implosion performance and proximity to ignition

Recent progress in direct-drive inertial confinement fusion has considerably improved the prospects for achieving thermonuclear ignition with megajoule-class lasers. When hydrodynamically scaled to laser energies typical of the National Ignition Facility, recent OMEGA implosions are expected to produce about 600 kJ of fusion

yield and 74% of the Lawson triple product required for ignition at 1.9MJ of symmetric illumination. Those implosions have benefited from a significant increase in implosion velocity obtained through larger-diameter targets. A new statistical approach (Gopalaswamy *et al.*, 2019) used in designing OMEGA targets and laser pulse shapes has demonstrated a considerable predictive capability, thereby enabling the design of targets with improved performance, leading to tripling the fusion yield and increasing the areal density. Systematic experiments and statistical modeling were used to identify and quantify the different mechanisms of performance degradation and implosion optimization (Lees *et al.*, 2021) leading to a recent record neutron yield of 1.74×10^{14} .

1. Hydrodynamic Scaling from OMEGA to megajoule laser energies

Hydrodynamic scaling is a simple way to extrapolate the performance of OMEGA implosions to higher laser energies. This scaling assumes that the same core conditions of hot spot pressure and shell density achieved on OMEGA can be reproduced at higher energies on a larger spatial scale (Nora *et al.*, 2014).

To keep the hydrodynamics the same at different scales, the target mass and volume must scale with laser energy, the fraction of laser energy coupled to the capsule must be the same and the shell adiabat must be the same. This leads to scale invariant implosion velocity, stagnation pressure and shell density. The hydro-scaling relations can be summarized as follows

$$Mass \sim E_L; R \sim E_L^{1/3}; \frac{E_{kin}}{E_L} = const; \alpha = const \quad (75)$$

$$v_{imp} = const; P_{hs}^{stag} = const; \rho_{sh}^{stag} = const \quad (76)$$

This simple scaling does not include non-ideal effects such as alpha heating, thermal transport, radiation transport and laser plasma instabilities. Experimental tests of hydrodynamic scaling from Omega to NIF have been limited and, so far, limited to cylindrical geometry (for instability growth diagnosability) (Sauppe *et al.*, 2020). In spherical geometry, thermal transport is not scale invariant because the heat losses depend on the volume to surface ratio. At larger scale the ratio $Volume/Surface \sim R$ increases with size R leading to lower thermal losses from the hot spot and higher hot spot temperatures.

This can be shown by using the mass conservation equation for the hot spot (Betti *et al.*, 2001)

$$\dot{M}_{hs} = 4\pi R_{hs}^2 \dot{m}_a \quad (77)$$

where M_{hs} is the hot spot mass and $\dot{m}_a$ is the ablation rate off the shell inner surface into the hot spot. The ablation rate scales as $\dot{m}_a \sim \kappa_{Sp}(T_0^{hs})/R_{hs}$ where T_0 is the hot spot central temperature, $\kappa_{Sp} \sim T^{5/2}$ is Spitzer

thermal conductivity and the hot spot mass is $\rho_{hs}V_{hs} \sim P_{hs}V_{hs}/T_{hs}$. Using the above scaling relations and the deceleration time scaling $t_{dec} \sim R_{hs}/v_{imp}$ into Eq. 77 leads to

$$T_{hs} \sim (P_{hs}R_{hs}V_{imp})^{2/7} \quad (78)$$

For hydro-equivalent implosions with constant P_{hs} and v_{imp} , the hot spot temperature increases with size and laser energy as

$$T_{hs} \sim R_{hs}^{2/7} \sim E_L^{2/21} \quad (79)$$

This enables the derivation of the scaling of the fusion yield in the absence of alpha heating ($Y_{no\alpha}$). For a 50-50 DT plasma, the neutron yield can be written as

$$Y \sim \frac{1}{4}n_i^2 < \sigma v > V_{hs}\tau = \frac{1}{16}P_{hs}^2 \frac{< \sigma v >}{T_{hs}^2} V_{hs}\tau \quad (80)$$

where τ is the burn duration or confinement time. For ignition relevant hydro-equivalent implosions, the $no\alpha$ hot spot temperature is in the 4 – 6 keV range where $< \sigma v > \sim T^3$, the $no\alpha$ pressure and velocity are invariant, the volume scales as $V_{hs} \sim R_{hs}^3$ and the confinement time scales as $\tau \sim R_{hs}/v_{imp}$ leading to

$$Y_{no\alpha} \sim P_{hs}^2 T_{hs} R_{hs}^4 / v_{imp} \sim T_{hs} R_{hs}^4 \sim R_{hs}^{30/7} \sim E_L^{10/7} \quad (81)$$

While the size scaling of the hot spot thermal transport can be easily taken into account in the one-dimensional hydrodynamic scaling, the lower heat losses lead to lower ablative stabilization of the deceleration phase Rayleigh-Taylor instability (Betti *et al.*, 2002) thereby affecting the three-dimensional evolution. At larger scales, more bremsstrahlung radiation is trapped by the hot spot and shell. This improves the shell stability during the deceleration phase by increasing the density gradient scale length at larger scales. This stabilizing effects counteracts the destabilization from lower ablation velocity at larger scales so that the overall Rayleigh-Taylor growth factor during the deceleration phase is approximately scale invariant (Bose *et al.*, 2015).

With regard to the thermal transport from the coronal plasma to the ablation front, the diffusive component is approximately scale invariant because planar in nature as long as the distance between the critical surface and the ablation front is much smaller than the target radius. Instead the nonlocal effects vary with scale because the ratio of the electron mean free path to target size is not scale invariant but it decreases at larger scales. Changes in coronal thermal transport, laser energy absorption, cross beam energy transfer occurring at different scales are combined into a single parameter; the laser-to-capsule energy coupling η_k =shell kinetic energy/laser energy= E_{kin}/E_L . Because of the large uncertainties in the scaling of energy coupling for direct drive and the

lack of relevant experimental data at megajoule energies, dedicated experiments are underway on the NIF to assess the value of η_k . Within the frame of the hydrodynamic scaling, the value of η_k is assumed to be scale invariant.

The next step is to assess the effect of alpha heating which becomes important at megajoule scales. For this, we apply the analytic theory of Ref. (Betti *et al.*, 2015; Gopalaswamy *et al.*, 2019) that approximates the yield amplification from alpha heating using the simple expression

$$\hat{Y}_{amp} \equiv \frac{Y_\alpha}{Y_{no\alpha}} \simeq \frac{1}{(1 - 0.96\chi_{no\alpha})^{0.75}} \quad (82)$$

where $\chi_{no\alpha}$ is the no-alpha normalized Lawson parameter $P_{hs}\tau/[P\tau]_{ign}$ rewritten in terms of neutron yield and areal density

$$\chi_{no\alpha} = \rho R^{0.61} \left(\frac{0.12Y_{no\alpha}^{16}}{M_{stag}} \right)^{0.34} \quad (83)$$

where ρR is the total neutron-averaged areal density in g/cm², $Y_{no\alpha}^{16}$ is the no-alpha neutron yield in units of 10¹⁶ and M_{stag} is the stagnating DT mass at bang time in mg. For ignition relevant direct-drive targets the stagnating mass is about half of the unablated DT mass ($M_{stag} \simeq M_{unabl}/2$).

As shown in Ref. (Christopherson *et al.*, 2019, 2020; Lindl *et al.*, 2018) the normalized Lawson parameter (sometimes rewritten in different forms) is suitable for describing all the different thermonuclear regimes of ICF implosions, from weak alpha heating to burning plasmas and up to the onset of ignition. The scaling of the Lawson parameter with size and energy follows the scaling of the confinement time $\chi \sim \tau \sim R \sim E_L^{1/3}$ leading to the following hydrosaling formulas for the fusion yield from OMEGA to NIF energies of symmetric illumination

$$Y_{no\alpha}^{NIF} = Y_{no\alpha}^{OMEGA} E_{ratio}^{10/7} \quad (84)$$

$$\chi_{no\alpha}^{NIF} = \chi_{no\alpha}^{OMEGA} E_{ratio}^{1/3} \quad (85)$$

$$\hat{Y}_{amp} = (1 - 0.96\chi_{no\alpha}^{NIF})^{-0.75} \quad (86)$$

$$Y_\alpha^{NIF} = Y_{no\alpha}^{NIF} \hat{Y}_{amp} \quad (87)$$

where $E_{ratio} = E_L^{NIF}/E_L^{OMEGA}$ is the laser energy ratio between NIF and OMEGA. It is important to emphasize that OMEGA laser beam configuration is spherically symmetric and therefore the above hydrosaling relations only apply to a spherical symmetric illumination similar to 60-beam OMEGA. This scaling does not account for the polar laser configuration of NIF which is expected to significantly degrade the implosion performance.

2. Current OMEGA implosion performance and proximity to hydroscaled ignition

The highest fusion performance of OMEGA DT layered implosions was achieved in 2018-2021 using larger size

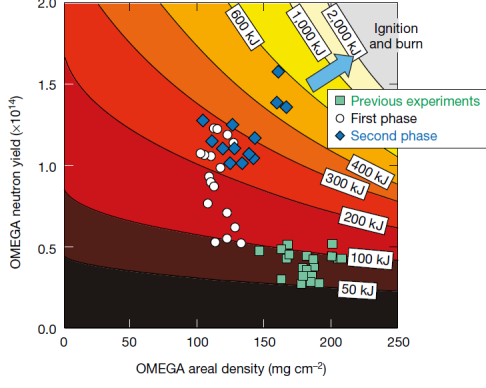


FIG. 51 Results from three sequential implosion campaigns on OMEGA extrapolated to 1.9MJ of symmetric illumination using the hydroscaling relations. The contours represent curves of constant extrapolated fusion yield (in kJ) (squares, circles, diamonds are in a temporal sequence starting from 2015). The arrow points in the direction of future experiments. The increase in yield since 2015 resulted in implosions that would produce about 500-600 kJ of fusion energy at NIF laser energies of 1.9 MJ with about a third of the yield from hydrodynamic compression and two-thirds from alpha-particle self-heating. From (Gopalaswamy *et al.*, 2019).

capsules with outer radius 12-23% larger than the laser beam radius. Statistical modeling of experimental and simulation databases has enabled the development of an accurate predictive capability (Gopalaswamy *et al.*, 2019) used to design higher performance implosions.

Figure 51 shows contours of constant fusion yield of high performance OMEGA implosions extrapolated to 1.9 MJ of symmetric illumination using the above hydroscaling relations. Three sequential experimental campaigns (“phases”) are shown to indicate advances in fusion yield since 2015 on OMEGA when the milestone core pressure of 50 Gbar was achieved (Bose *et al.*, 2016; Regan *et al.*, 2016). The neutron yield increased from $4\text{--}5 \times 10^{13}$ in 2015 to $1.5\text{--}1.7 \times 10^{14}$ in 2018-21. This increase was accomplished by driving the implosions on high adiabats of about $\alpha_F \simeq 5$ and by using larger size shells with outer radius larger than the laser beam radius so to improve the laser energy coupling to the capsule, partially due to CBET mitigation (Igumenshchev *et al.*, 2013).

As shown in Fig. 51, if the same hydrodynamic ($\text{no}\alpha$) core conditions of best OMEGA implosions are achieved at a larger spatial scale (hydroscaling) to match the higher laser energy of NIF (about 4x in spatial scale from OMEGA to NIF), the expected fusion yield at NIF scale is about 600 kJ. While such a yield would represent significant progress in performance, it is still well below the requirements for ignition.

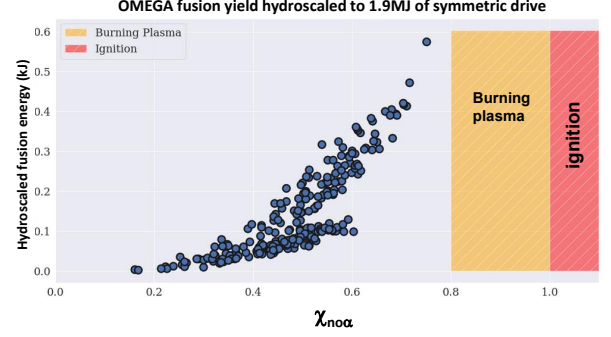


FIG. 52 Fusion yield of OMEGA DT layered implosions hydrodynamically scaled to 1.9 MJ of symmetric illumination versus the no-alpha Lawson parameter scaled to 1.9 MJ. Courtesy of V. Gopalaswamy (LLE).

The proximity to ignition is better identified by the value of the no-alpha Lawson parameter requiring $\chi_{\text{no}\alpha} \simeq 1$ for ignition. Figure 52 shows the fusion yield of OMEGA implosions scaled to 1.9 MJ of laser energy versus the no-alpha Lawson parameter. The best performing OMEGA implosions achieved an areal density of about 0.16 g/cm^2 and yield of $1.5\text{--}1.7 \times 10^{14}$ with a stagnated mass of $\sim 0.011 \text{ mg}$ and laser energy of $27\text{--}28 \text{ kJ}$. The corresponding Lawson parameter is $\chi_{\text{no}\alpha} \simeq 0.18$. After scaling $\chi_{\text{no}\alpha}$ to 1.9MJ of laser energy using $E_{\text{ratio}}^{1/3} \simeq 4.1$, the scaled Lawson parameter is $\chi_{\text{no}\alpha}^{1.9\text{MJ}} \simeq 0.74$. This indicates that about a 50% increase in the Lawson parameter is required to achieve the ignition conditions at NIF scale for symmetric illumination and hydroequivalent implosions.

Statistical modeling is also used to quantify the different degradation mechanisms affecting OMEGA implosions (Lees *et al.*, 2021). The fusion yield is found to be reduced by the ratio of laser beam to target radius (a proxy for the illumination nonuniformities from the OMEGA beam geometry), the asymmetry in inferred ion temperatures (a measure of the magnitude of the $\ell = 1$ mode from target offset, laser mispointing and LPI), the time span over which tritium fuel has decayed (a measure of the He^3 accumulation from tritium decay) and parameters related to the implosion hydrodynamic stability such as in-flight adiabat and aspect ratio. The recent record fusion yield of 1.74×10^{14} on OMEGA was achieved using a 1.02 mm outer diameter capsule and implementing active mitigation of the $\ell = 1$ mode using a pre-imposed target offset (Mannion *et al.*, 2021). When adjusted for tritium decay and $\ell = 1$ mode, the highest yield in OMEGA cryogenic implosions is predicted to

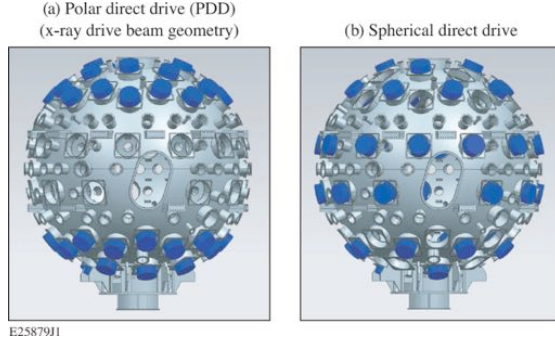


FIG. 53 (a) mechanical drawing of the target chamber for the National Ignition Facility with the blue cylinders indicating the beam port positions used for laser indirect drive (i.e., x-ray drive). Polar direct drive uses this beam geometry. (b) same as (a) except for spherical direct drive where the beams are uniformly around the target chamber. This is the ideal beam arrangement for laser direct drive. Figure from (Hohenberger *et al.*, 2015).

exceed 2×10^{14} fusion reactions (Lees *et al.*, 2021).

3. Polar Direct Drive

An alternative laser-direct-drive approach (Craxton *et al.*, 2005; Skupsky *et al.*, 2004) called polar direct drive is being pursued as a means to perform laser-direct-drive experiments on the National Ignition Facility (Campbell *et al.*, 2017; Goncharov *et al.*, 2016; Regan *et al.*, 2018) without extensive facility modifications. Since a primary goal of the National Ignition Facility is to achieve ignition and modest gain first with laser indirect drive and then with laser direct drive, the facility is only optimized for laser indirect drive (Spaeth *et al.*, 2016). The two approaches have different requirements for the arrangement of laser beams around the target chamber and phase plates controlling the shape of the laser spot on target. The laser beams on the National Ignition Facility are directed through polar beam ports in each hemisphere of the spherical target chamber (see Fig. 53a) to optimize the laser energy coupling to the cylindrical hohlraum for laser indirect drive (Campbell and Hogan, 1999). The hohlraum is positioned at the center of the chamber with its symmetry axis aligned with the vertical axis, and the polar beam ports have cone angles with respect to the vertical axis of 23.5° , 30.0° , 44.5° , and 50.0° , which are referred to as ring 1, ring 2, ring 3, and ring 4, respectively. The phase plates on the National Ignition Facility are designed to form an elliptical focal spot that passes through the laser entrance hole of the hohlraum. For laser direct drive, the optimal beam configuration to maximize on-target laser irradiation uniformity and the energy coupling from the laser to the target is to

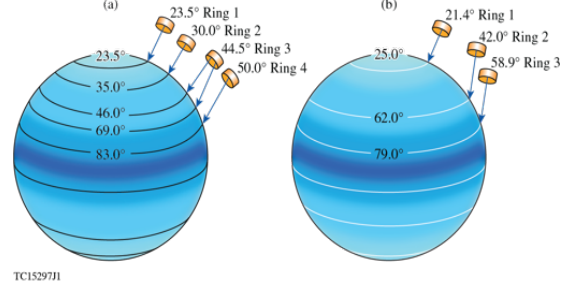


FIG. 54 The beam repointing schemes for Polar Direct Drive on the National Ignition Facility and the OMEGA laser are shown in (a) and (b), respectively. The cylinders to the upper right of each image indicate the beam port locations and ring identifications and the blue arrows indicate the beam repointing to the polar angles shown on target to achieve optimized laser-drive symmetry in the Polar Direct Drive geometry. Figure from (Marshall *et al.*, 2016).

have the beams uniformly distributed around the target with a normal incidence on the target (i.e., spherical direct drive, see Fig. 53b), and to have the laser spot size comparable to the target diameter (Craxton *et al.*, 2015). Polar-direct-drive target designs absorb a smaller fraction of the incident laser irradiation compared to spherical direct drive designs (Craxton *et al.*, 2005; Skupsky *et al.*, 2004). Although the target chamber of the National Ignition Facility has beam ports for laser direct drive as well as laser indirect drive, switching from the current laser-indirect-drive configuration to the laser-direct-drive one requires a major, resource- and time-intensive engineering effort.

The polar-direct-drive concept had been considered during the planning stages of the National Ignition Facility (Eimerl, 1995a,b; Eimerl *et al.*, 1995). In contrast to spherical direct drive, each ring of the NIF polar beams are uniquely repointed toward the equator of the target and defocused for polar direct drive to minimize the laser-drive asymmetry on target (i.e., the elliptical laser spots formed with the laser indirect drive phase plates are defocused to match the size of the target diameter). Proof-of-principle polar direct drive implosions were performed on the OMEGA laser system (Boehly *et al.*, 1997) using forty of the sixty beams (Craxton and Jacobs-Perkins, 2005; Marozas *et al.*, 2006; Marshall *et al.*, 2006). The beam repointing schemes for Polar Direct Drive on the National Ignition Facility and the OMEGA laser are shown in Fig. 54. Low-convergence ($CR \leq 10$), polar-direct-drive implosions that are less susceptible to laser imprint have been conducted on the National Ignition Facility by repointing and defocusing the laser beams (Cobble *et al.*, 2012; Hohenberger *et al.*, 2015; Radha *et al.*, 2016, 2012; Yeaman *et al.*, 2021; Zylstra *et al.*,

2020b)(Marozas *et al.*, 2021). Defocusing a laser beam smoothed with a phase plate is a compromise, since it could lead to undesirable modulations in the far-field intensity spatial profile. Assessing implosion performance degradation due to the polar-direct-drive geometry compared to spherical direct drive is a key issue for the laser-direct-drive ignition approach (Hohenberger *et al.*, 2015; Radha *et al.*, 2016, 2012, 2013). In addition, target solutions are being investigated to improve the symmetry of a polar-direct-drive implosion by contouring the shell to alter the mass distribution in the ablator shown in Fig. 55a (Marshall *et al.*, 2016) and by using a Saturn target shown in Fig. 55b (Craxton and Jacobs-Perkins, 2005; Marshall *et al.*, 2006) (i.e., the addition of a low-Z ring around the capsule equator, where refraction in the plasma formed around this ring permits time-dependent tuning of the capsule drive uniformity). Implosion symmetry, energy coupling (Marozas *et al.*, 2018a,b; Zylstra *et al.*, 2020a), preheat (Michel *et al.*, 2019; Rosenberg *et al.*, 2018b, 2020; Solodov *et al.*, 2020), and laser imprint (Hohenberger *et al.*, 2016) have been investigated on the National Ignition Facility for polar direct drive (Hohenberger *et al.*, 2015; Marozas, 2021; Radha *et al.*, 2016, 2012, 2013). As shown in Fig. 50, proof-of-principle experiments using a modest level of wavelength detuning (i.e., $\pm 2.3\text{\AA}$ in the ultraviolet) demonstrated an increase in the energy coupling of polar direct drive implosions on the National Ignition Facility. Polar direct drive implosions of DT-filled capsules on the National Ignition Facility for neutron effects experiments have produced up to 33.6 kJ of fusion output with 1.26 MJ of ultraviolet laser energy (Yeaman *et al.*, 2021).

The higher convergence ratio ($CR \geq 20$) polar-direct-drive implosions needed for ignition designs on the National Ignition Facility require facility modifications (Campbell *et al.*, 2021; Marozas, 2021). Facility enhancements to the on-target laser drive uniformity include polar direct drive phase plates; high-contrast, multiple-picket pulse shaping; beam smoothing with multi-FM (frequency modulation) smoothing by spectral dispersion (Hohenberger *et al.*, 2016) or 2-D smoothing by spectral dispersion (Regan *et al.*, 2005; Skupsky *et al.*, 1989) and possibly polarization smoothing (Boehly *et al.*, 1999); and $6\text{\AA}$ UV wavelength detuning with beam remapping (Collins and Marozas, 2018; Collins *et al.*, 2012; Marozas, 2021; Marozas *et al.*, 2018a,b). Facility enhancements to the DT cryogenic target handling system are required (i.e., a scaled-up version of the OMEGA fill-tube DT cryogenic target-handling system on the National Ignition Facility (Harding *et al.*, 2018)). Target solutions to mitigate preheat are needed (Solodov *et al.*, 2020). The use of silicon ablaters was found to mitigate the hot-electron preheat by increasing the threshold laser intensity for hot-electron generation from $3.5 \times 10^{14} \text{ W/cm}^2$ in plastic to $6 \times 10^{14} \text{ W/cm}^2$ in silicon. The overall hot-electron production is also reduced in silicon ablaters

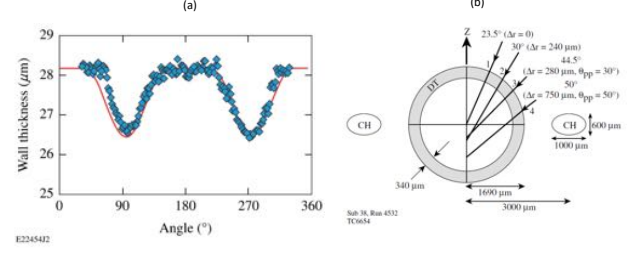


FIG. 55 Target solutions are being investigated to improve the implosion symmetry of Polar Direct Drive implosions. (a) The measured contoured-shell thickness (diamonds) as a function of polar angle is compared to the ideal contoured profile. Reducing the mass around the equator of the target compensates for the reduced energy coupling of Polar Direct Drive in this region. Figure from (Marshall *et al.*, 2016). (b) The addition of a low-Z ring around the capsule equator, where refraction in the plasma formed around this ring permits time-dependent tuning of the capsule drive uniformity. Figure from (Craxton and Jacobs-Perkins, 2005).

when the intensity threshold is exceeded. The laser beam smoothing, laser pulse shaping, wavelength detuning, and DT cryogenic target handling system (Gibson *et al.*, 2009; Johal *et al.*, 2009) on the National Ignition Facility (Spaeth *et al.*, 2016) are all optimized for laser indirect drive and do not meet the requirements of laser direct drive (Hansen, 2012; Marozas, 2021). Each of the proposed changes is required for laser direct drive, regardless of whether the National Ignition Facility is configured for polar direct drive or spherical direct drive. The only differences between polar direct drive and spherical direct drive would be the population of optics in the equatorial ports for spherical direct drive and differing phase-plate details between the two approaches.

The strategy for polar direct drive research over the next decade on the National Ignition Facility is defined by four physics goals (Campbell *et al.*, 2021; Marozas, 2021). The first one is to optimize symmetry for low-convergence ($CR \leq 10$), warm polar-direct-drive implosion (i.e., DT-filled capsules using plastic or an alternative ablator) using the current National Ignition Facility configuration, pulse shaping, beam pointing, and target solutions (i.e., contouring or shimming and a Saturn ring) with a goal of achieving a 50-kJ fusion yield. The second one is to improve the energy coupling and symmetry for $CR \leq 10$, warm polar direct drive implosions using wavelength detuning with beam remapping and polar-direct-drive phase plates. Beam remapping is needed to satisfy the polar-direct-drive requirement that the crossing beams at the target equator have different wavelengths with wavelength detuning of $6.0\text{\AA}$. The

wavelength detuning is needed to mitigate cross beam energy transfer and increase the energy coupling (Marozas *et al.*, 2018a,b). The goals for this stage of the research are to achieve a laser absorption exceeding 85% and a fusion yield of 100 kJ with these modifications. The third and fourth goals require the modifications of a laser-direct-drive cryogenic target-handling system and laser beam smoothing on the National Ignition Facility to increase the convergence ratio to 15 and then to 20 with a polar-direct-drive DT cryogenic implosion for the alpha burner and burning-plasma regime (i.e., 1-MJ fusion yield) designs, respectively. A ten-year plan is needed to implement the polar-direct-drive upgrades to the National Ignition Facility to optimize the performance of polar-direct-drive DT cryogenic implosion targets in the burning-plasma regime, and to understand the laser facility requirements (i.e., energy, power, size, laser beam smoothing, etc.) needed to achieve ignition and multi-MJ yield. The physics program underway on OMEGA and the NIF will inform and validate these modifications (Campbell *et al.*, 2017; Goncharov *et al.*, 2016; Regan *et al.*, 2018).

V. MAGNETIC DRIVE IMPLOSIONS

A. History of Pulsed Power Fusion

1. Development of Pulsed Power

In this section the developments of pulsed power leading to the Z pulsed power facility (Z) as a driver of fusion experiments are briefly described. This narrative is based on a review article by Smith (Smith, 2006), a Sandia National Laboratories report (Van Arsdaal, 2007), and a book by Martin (Martin, 1996). In addition, Martin (1996) and Van Arsdaal (2007) include numerous references to works on the early pulsed power devices described in this section. Another recent review of work on the Z pulsed power accelerator can be found in (Sinars *et al.*, 2020).

In 1960 a group at the Atomic Weapons Research Establishment (AWRE) headed by J. C. “Charlie” Martin began development of x-ray sources to radiograph explosive events. The first source used a radio frequency accelerator to generate a 35 MeV electron beam that was focused onto a high atomic number target to produce x-ray photons by bremsstrahlung. Although the system worked, much higher intensities were desired. Charlie was convinced that a high voltage (~ 6 MV) delivered to a pair of electrodes separated by a vacuum gap would not simply breakdown if the pulse length was short (tens of ns) but would generate an electron beam with ~ 100 kA of current. For this purpose, the AWRE group developed stacked strip-line pulse generators. Transmission lines formed by two planar electrodes separated by di-

electric insulating material (usually polyethylene) were charged by a simple capacitor bank. These transmission lines stacked in a Blumlein configuration delivered the charge voltage to an electron accelerating gap. The two-way transit time along each of the transmission lines determined the pulse length. An early machine built in 1961, “Dagwood,” used 10 ft long transmission lines to produce a 1.5 MV voltage pulse of 20 ns duration. In this section we will refer to pulsed devices as “machines,” although they are often referred to as “accelerators.” Dagwood was used to develop electron tubes. The accelerating gap of an electron tube must operate in vacuum to avoid electric breakdown. The interface between the insulator and the vacuum is subject to surface breakdown. AWRE found that avalanche breakdown could be minimized by orienting this surface at an angle to the electric field, so the electrons were driven away from the surface to avoid secondary ionization. This work led to the vacuum insulator stack configuration used in modern machines. The AWRE group then built a larger machine named SMOG (Six Megavolt Or Goodbye), using 20 solid dielectric Blumlein transmission lines charged in parallel to 200 kV and then switched in series to generate a 4 MV pulse to drive electron tubes to radiograph explosive events.

The need to test the effects of prompt gamma-rays and soft x-rays on weapons components grew in the early 1960’s. Devices existed that could test small volumes, but larger machines were required to test military components. The U.S. government agencies contracted the development of nuclear weapons effects simulators (NWES) to industry. Sandia National Laboratories was given the responsibility for the simulation of warhead packages. In 1964, AWRE delivered to Sandia a machine similar to SMOG, which produced a pulse of 4.5 MV and a radiation dose of about 15 rads at 1 m. Prompt gamma-ray simulation work was done with this machine until 1967, but the machine was difficult to operate. The solid dielectric switches had to be replaced after every shot.

At that time, Physics International (PI) was designing an operationally convenient and more powerful machine using Marx banks and Blumleins both insulated by transformer oil. At the end of 1964, the Pulserad 730 was operational, producing a 50 kA, 4 MV, 40 ns pulse that could deliver 50 rads at 1 m. PI named this new technology “pulsed power.”

In 1967 the Defense Atomic Support Agency (DASA) asked for proposals to build a machine capable of delivering 50 krad. This led to the development of the “Aurora” machine, which had 4 Marx-oil Blumlein modules delivering power to 4 electron tubes. Switches with adequate time jitter to synchronize these modules were developed, which led to the modern approach of modular accelerators. Moreover, electrical power was delivered 3 meters to the electron tubes via vacuum coaxial transmission lines with electrons not crossing the gap due to

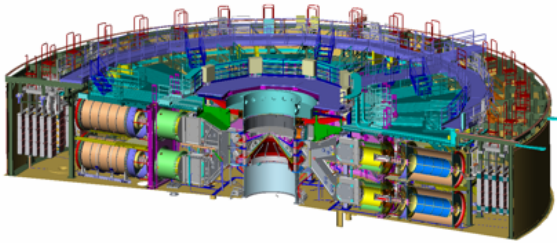


FIG. 56 Z at Sandia National Laboratories.

the magnetic field generated by the current. This was the first demonstration of self-magnetically insulated transmission lines (MITL's), which are still used in modern pulsed power machines such as Z.

The pulsed power machines described so far were high impedance, i.e. high voltage and low current. These machines were suitable for simulating the prompt gamma-ray flash from a weapon with many MeV photons, but there was a need for lower energy x-rays (< 1 MeV). In 1967, the Naval Research Laboratory (NRL) built the low impedance Gamble I machine designed to deliver < 1 MV 1 MA 50 ns pulses. Gamble I used water rather than oil for a dielectric in the pulse forming lines, which increased the permittivity by a factor of about 6 and thus reduced their length. The current rise-time of a switch at a voltage, V , is limited by the inductive voltage of the switch, i.e., $V = LdI/dt$. Simply increasing the voltage doesn't help because the switch gap, and thus L , will need to be increased. This was not a problem for previous low current machines. Gamble I used switches with multiple channels to overcome this problem. Marx generators charged intermediate storage capacitors that were then discharged into the pulse forming lines. These were switched in 200 ns at 500 kA to 2 MV to a transmission line transformer, which increased the current and lowered the voltage. This scheme further reduced dI/dt in the switches. The 1 TW Gamble II based on similar technology, which was built in the mid 70's, generated electron beams at 1 MA/cm². NRL's 1 TW Gamble II, built in the mid 70s to generate electron beams at 1 MA/cm², was based on similar technology.

Dagwood, Aurora, Gamble-I, and Gamble-II demonstrated several important technologies used in modern modular pulsed power machines. Further developments such as low jitter triggered gas switches developed at Sandia allowed the construction of Particle Beam Fusion Accelerator-I (PBFA-I now Saturn) and PBFA-II, which was reconfigured to the present Z Machine, which we simply refer to as Z. Z has 36 parallel modules delivering power radially inward, see Fig. 56. Each module starts with a Marx generator, which charges an interme-

diate storage capacitor. Triggered gas switches transfer this energy into pulse forming lines. Multi-channel water switches then transfer the energy to transformer transmission lines. The Marx generators are immersed in a tank containing oil and the rest of the components are immersed in deionized water. At a radius of 1 meter, an insulator stack separates the water section from the inner vacuum section. The stack is very similar to those designed at AWRE for electron tubes. Four levels of disk-shaped transmission lines propagate the power inward toward the load. These transmission lines use the principle of self-magnetic insulation that was demonstrated on the Aurora machine. These four levels are connected in parallel to increase the current by a post-hole convolute at a radius near the load. Power is then delivered to the load by a single MITL referred to as the final feed. The Z machine delivers 18-28 MA, depending on the impedance of the loads.

2. Electron beam driven fusion

In the 1960s electron beam driven bremsstrahlung sources were providing multi-megavolt photons, which could be used to simulate the effects of the prompt gamma-rays. However, intense x-ray sources at lower photon energies were still needed. Lower voltage high-current machines were being built to provide this need. Winterberg (Winterberg, 1968) proposed that such electron beams could be used to drive ICF implosions in a manner similar to that of lasers. Al Narath and Everet Beckner started an electron beam driven fusion program at Sandia in 1973, with help from NRL using Gamble-I and Gamble-II, motivated by NWES and possible generation of commercial electricity. Such fusion explosions would provide the needed soft x-rays (Van Arsdall, 2007). Numerical simulations of capsules driven by 1 MeV electron beams indicated that extremely high electron-beam current (360 MA), current density on target (1400 MA/cm²), and beam intensity on target (~ 1400 TW/cm²) would be needed (Sweeney and Clauser, 1975), but the much higher energy efficiency of pulsed power (5-10 %) as compared to Nd lasers (< 1 %) was a strong motivation. It was demonstrated that electron beams could be focused to current densities of 5 MA/cm² (Yonas *et al.*, 1973), but the stopping distance was too large to effectively couple to an ablatively driven capsule. The low energy deposition per unit mass resulted in low implosion velocities even when the beam was allowed to pass through the shell. Nevertheless, a magnetized target driven by an electron beam did produce measurable deuterium-deuterium (DD) fusion neutrons of $\sim 10^6$ (Chang *et al.*, 1977; Widner *et al.*, 1977a). That result may have been the first successful demonstration of Magneto-Inertial Fusion (MIF) (Lindemuth and Kirkpatrick, 1983). However, scaling to higher yields

with electron beams did not look promising.

3. Ion beam driven fusion

The stopping range of ions is much smaller for electrons at a given voltage, which in principle alleviates this problem. Magnetically insulated diodes demonstrated the feasibility of generating intense ion beams (Dreike *et al.*, 1976). Numerical simulations indicated that a 10 MA, 10 MV proton beam could drive a breakeven capsule (Clauser, 1975) Slutz proposed and provided analytic estimates of performance of a target that converted the ion beam energy into x-rays to drive the capsule (Slutz, 1995). The target consists of a spherical gold shell filled with CH foam that contains a radiation-driven fusion capsule similar to those now being shot on NIF, see Fig. 25. In this design the ion beam, which is focused radially inward, penetrates the gold shell and stops in the foam where it deposits most of its energy. The hot foam emits x-rays that are trapped within the gold shell, forming a radiation filled hohlraum. The radiation is symmetrized in the same manner as laser indirect drive hohlraums discussed in section III. Furthermore, the radiation intensity is higher than the beam intensity due to a “greenhouse effect,” The ions have a relatively long range in gold because most of the electrons are bound. Thus, they penetrate easily as does visible light through glass. The ion range is shorter in the low atomic number CH foam because there are more free electrons to interact with the moving ion. Consequently, the ion energy is absorbed in the foam as is the visible light absorbed by the floor of a greenhouse. The gold shell has a high opacity to the x-rays emitted by the foam and so the radiation is trapped just as glass is opaque to the infrared emitted by the heated floor of a greenhouse. Thus, both the beam intensity and symmetry requirements are reduced. With these promising developments, Sandia switched from an electron beam to a light ion fusion program. During this period Sandia built PBFA-I and PBFA-II to test ion diodes. The most studied was the applied magnetic-field diode (Johnson *et al.*, 1985), which used external coils that generated a magnetic field of 2-3 Tesla to allow ions, but not electrons, to cross the accelerating gap. Electrons emitted by the cathode formed a cloud of negative space-charge that acted as a virtual cathode. The anode surface was designed to breakdown electrically to form a source of ions. Since the breakdown involved hydrocarbons the dominant ions were protons. The anode surface had a shape similar to a barrel to focus the ion beam to the cylindrical axis. Unfortunately, the beams did not form a tight focus due to high emittance and the diode impedance decreased with time which reduced the coupling of energy to the beam. The beam intensity at the focus was 1.5 TW/cm², which was much too low to drive ICF capsules which require more than 30 TW/cm².

Protons with an energy of 4 MeV at a nearly optimal range in an indirect drive ion target design (Slutz, 1995), and lithium ions with 30 MeV have nearly the same range. A new approach then pursued based on inductive energy storage and plasma opening switches (Cook *et al.*, 1986; Mendel *et al.*, 1987) to accelerate lithium ions to 30 MeV. The idea was to build up magnetic energy over about 100 ns by driving current through the MITLs into a plasma opening switch initially in a conducting mode. The switch would then open quickly to generate high voltage, much like the ignition system in older cars. In principle this would multiply the power. Unfortunately, the switches did not open quickly into anything but a low impedance load, i.e., not an ion diode.

There were other problems as well. The beams were not pure lithium, but due to the relatively poor vacuum ($\sim 10^{-5}$ Torr) were contaminated with hydrocarbons (Quintenz *et al.*, 1996). Cuneo *et al.* (Cuneo *et al.*, 1997) developed cleaning techniques that removed most of these impurities (Cuneo, 1999). Perhaps the most damaging problem was that an instability in the ion accelerating gap led to unacceptably large emittance despite attempts to control that instability (Desjarlais *et al.*, 1991). The instability induced fluctuations allowed the electrons to diffuse toward the anode, which lowered the diode impedance over time. Solutions were proposed (Slutz, 1987), but these were not successful.

A probable explanation is that charge exchange of the beam ions with neutrals in the anode plasma also contributed to a falling impedance of the diode (Litwin and Maron, 1989; Pointon, 1989). In the end, the beam intensity needed to drive fusion capsules was not achieved.

4. Z pinch-driven indirect drive

During the period that light ions were the main approach to pulsed power fusion, a small group at Sandia was studying the generation of x-rays with fast z-pinches. The motivation of this research was the need to test the radiation effects of intense soft x-rays (< 5 keV) on weapon components. Electron-beam-driven Bremsstrahlung sources were not adequate to test these effects, so alternative approaches were being pursued. Imploding wire arrays, which formed hollow cylindrical z-pinch plasmas, were found to be effective radiators (Matzen, 1997; Sanford *et al.*, 1996). Research was started to determine if these radiation sources could be used to drive ICF implosions and PBFA-II was converted to the Z Machine, see Fig. 56. Hammer *et al.* (Hammer *et al.*, 1999) proposed a concept to generate highly symmetric radiation fields from these z-pinches to drive ICF capsules. In this approach a capsule is contained in a cylindrical secondary hohlraum that has radiation entrance holes in the two end walls, Fig. 57. The geometry is essentially the same as for the NIF indirect drive

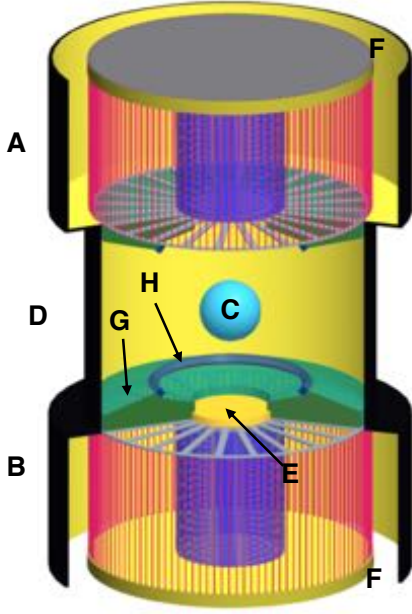


FIG. 57 Double-ended z-pinch hohlraum concept, with (A) and (B) top and bottom primary hohlraums, each containing a notional wire array z-pinch with an internal pulse-shaping target, (C) high yield fusion capsule, (D) secondary hohlraum containing the capsule, (E) on-axis permanent shine shield and radial spoke electrode structures, (F) upper and lower electrical power feeds, (G) secondary entrance tamping foam, and (H) P_4 symmetry shield.

approach described in section III of this paper. However, instead of laser beams entering the entrance hole, radiation flows in from two primary hohlraums that each contain a wire array z pinch implosion. At stagnation these z pinches radiate x-rays, which fill the primaries. A shine shield protected the capsule from direct radiation from these z pinches, while the radiation driving the capsule is symmetrized by radiation smoothing within the secondary hohlraum. Simulations by Hammer *et al.* (Hammer *et al.*, 1999) indicated that this *double-ended pinch* approach could produce fusion yields of 400 MJ if both z pinches were driven with 60 MA of current. Experiments based on this configuration were performed on Z with 19 MA driving the two pinches in series (Bennett *et al.*, 2002), which drove a capsule implosion to a convergence of 20 and indicated a highly symmetric radiation drive. Numerical simulations and view factor calculations also indicated that high radiation symmetry could be obtained within the secondary hohlraum (Vesey *et al.*, 2003).

The simulations of pinches at 60 MA by Hammer *et al.* (Hammer *et al.*, 1999) indicated that the primary hohlraums would reach a temperature of about 280 eV, while the secondary hohlraum containing the capsule would reach a temperature of 210 eV. Z experiments at 20 MA produced radiation temperatures of 145 eV in a

primary hohlraum and 95 eV in a secondary hohlraum (Cuneo *et al.*, 2005). We can show by simple scaling that the measured primary hohlraum temperature is consistent with simulations at 60 MA. The energy in a z pinch implosion is determined by the integral of $\int PdV$, which is proportional to the square of the drive current, I . The radiation energy lost to the hohlraum walls is given by the time integral of Eq. 33, with Eq. 34. Using these relations and the Z experimental results we can then estimate the drive current needed to reach a primary hohlraum temperature of 280 eV. The result is $I = 19(280/145)^{5/3} = 57$ MA, which is close to the detailed calculation. As with NIF capsules the radiation needs to be pulse shaped. Pulse shaping was demonstrated experimentally using nested wire arrays (Cuneo *et al.*, 2005). These results indicate that the double-ended pinch hohlraum could be a viable approach to fusion with pulsed power machines delivering about 60 MA to two pinches. This approach should be revisited when a pulsed power machine is built that can deliver such current.

Although the radiation temperatures achieved on Z were not high enough to get a significant yield from an ICF implosion, they were high enough to study capsule ablation at conditions similar to the foot pulse of a NIF target. Radiographs at 6.151 keV showed a spikey structure to the ablator surface of a CH capsule at a convergence of 4.1 (Cuneo *et al.*, 2012) and showed the effect of a fill tube on the capsule uniformity (Bennett *et al.*, 2007).

Another indirect drive approach was studied that could produce much higher radiation temperatures, which was referred to as the *dynamic hohlraum* at Sandia (Matzen, 1997) and the *flying radiation case* at Los Alamos (Brownell *et al.*, 1998). In this approach tungsten wire arrays implode onto a foam target driving a radiative shock into the foam, see Fig. 58. The wire array plasma would remain relatively cool and trap radiation, generating high radiation temperatures (~ 300 eV) within the foam (Nash *et al.*, 1999). Simulations (Lash *et al.*, 2000) indicated that this indirect drive approach might produce high yields on future pulsed power machines with larger drive currents. The collision of the tungsten onto the foam generated a radiative shock (Bailey *et al.*, 2002), while the high opacity tungsten acted as a radiation case. Simulations (Slutz *et al.*, 2003) were used to optimize the dynamic hohlraum configuration for driving small plastic capsules filled with deuterium. Experiments with this configuration (Ruiz *et al.*, 2004) generated $\sim 5 \times 10^{10}$ DD fusion neutrons. Subsequent numerical simulations were in good agreement with these results (Slutz *et al.*, 2006). Generating radiation fields with the high symmetry required for fusion was the greatest challenge for the dynamic hohlraum approach and led to the design of opacity tailored capsules that according to numerical simulations could accept non-uniformities as large as

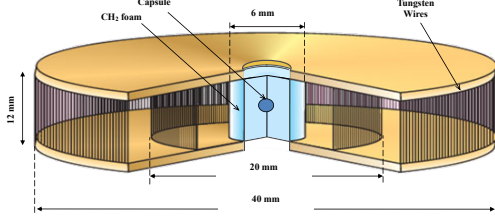


FIG. 58 Schematic of the dynamic hohlraum.

20% and still implode symmetrically (Slutz *et al.*, 2007). Two dynamic hohlraums were also used in the double-ended pinch configuration (Sanford *et al.*, 1999), which produced a radiation temperature of 130 eV in the secondary hohlraum. The dynamic hohlraum approach is still being used for opacity measurements (Rochau *et al.*, 2014).

Although both of these indirect drive approaches still have promise, research at Sandia is now focused on magnetic direct drive concepts which are unique to pulsed power and can deliver large energies to the implosion.

B. Direct magnetic drive using the MagLIF concept

1. Basic Principles

Wire arrays driven by Z, which stores 22 MJ, have produced nearly 2 MJ of x-rays. This demonstrates the high efficiency of pulsed power, but can fusion implosions be directly driven by magnetic pressure from a pulsed power machine? Since the fuel must be contained, the z pinch should be a metal tube rather than a wire array as illustrated in Fig. 59. The axial current within the tube produces an azimuthal B-field given by $B = \frac{\mu_0 I}{2\pi r}$, where in Section V units are meters-kilogram-seconds unless stated otherwise. The B-field exerts a pressure $P_{mag} = \frac{B^2}{2\mu_0} = \frac{\mu_0}{8} \left(\frac{I}{\pi r}\right)^2$, which can be quite high, e.g., 100 Mbars for $I = 25$ MA and $r = 1$ mm. This pressure is comparable to the ablation pressure of a NIF indirect-drive capsule implosion. The wall thickness, Δr , of the tube must be thick enough to avoid disruption by the Magneto-Rayleigh-Taylor (MRT) instability (Harris, 1962). We define the aspect ratio $A_R = r_o/\Delta r$, where r_o is the outer radius of the tube, which by convention is called a liner. Experiments and simulations (Dimonte, 2000) have shown that bubbles from the Rayleigh-Taylor instability penetrate a distance $h = \alpha g t^2$, where g is the acceleration and t is time. The factor α is about 0.05 so $A_R < 10$ is needed to avoid bubble penetration through the liner.

We can estimate the liner implosion velocity using the

assumption of a thin liner. The total force driving the liner inward is $F = P_{mag} 2\pi r L$, r and L are the radius and length of the liner. Equating this to the liner mass, M , times acceleration and putting the resulting equation in dimensionless form we obtain

$$\frac{d^2 x}{d\tau^2} = \frac{-8 (I/I_x)^2}{\sqrt{3} x} \quad (88)$$

where $\tilde{\tau} = t/tp$,

$$t_p = \sqrt{\frac{8}{\sqrt{3}} \left(\frac{4\pi}{\mu_0}\right) \left(\frac{r_0}{I_x}\right) \frac{M}{L}}, \quad (89)$$

$x = r/r_0$, r_0 is the initial radius of the liner, and I_x is the peak current. This equation has the simple solution $x = 1 - \tilde{\tau}^4$ for a current profile given by $\left(\frac{I}{I_x}\right)^2 = \frac{3\sqrt{3}}{2} \tilde{\tau}^2 (1 - \tilde{\tau}^4)$. The implosion velocity is then

$$v_{imp} = -4 \left(\frac{C_R - 1}{C_R}\right)^{3/4} \left(\frac{I_x}{t_p}\right)^{1/2} \left[\frac{\sqrt{3}\mu_0 A_R^2}{32\pi^2 \rho_L (2A_R - 1)} \right]^{1/4}, \quad (90)$$

where A_R is only used to calculate the mass of the liner, the convergence ratio is defined to be $C_R = r_0/r_s$ with r_0 and r_s the initial and stagnated radius of the liner (similar to Section II.C). The implosion velocity decreases with liner density so a low density is desirable. Some of the liner material could be mixed into the fuel during the implosion, which would enhance radiation losses. Since radiation losses increase with atomic number [e.g. Eq. 4], a low-atomic number material is advantageous. Lithium would be an optimal choice but is difficult to machine and handle. Beryllium is a practical alternative.

Assuming a beryllium liner with $A_R = 10$, $I_x = 20$ MA, $t_p = 120$ ns and $C_R = 30$, the implosion velocity is 105 km/s, which is too low for conventional ICF, e.g., NIF capsules implode at ~ 400 km/s (see Section II). The high implosion velocity of conventional ICF serves two purposes. First, a strong shock wave heats gaseous fuel, which lowers the amount of compression that is needed. Second, the high velocity produces compressive heating that overwhelms loss mechanisms. The temperature of a shocked DT gas is approximately $\theta_s = 0.0047 v_{shock}^2$ eV, where v_{shock} is the shock velocity km/s, so $\theta_s = 750$ eV for a 400 km/s implosion and $\theta_s = 47$ eV for a 100 km/s implosion. The fuel density will increase by roughly a factor of 4 (strong planar shock limit); hence, during the shock heating phase, the liner or capsule will converge by $C_s = 4^{1/d}$, where the dimensionality is $d = 2$ for cylindrical and $d = 3$ for spherical implosions. Subsequent compression further heats this preheated gas quasi-adiabatically to fusion temperatures and forms a hot spot. Ignoring thermal losses, the temperature after compression will be $\theta_c = \theta_s C_c^{d(\gamma-1)}$,

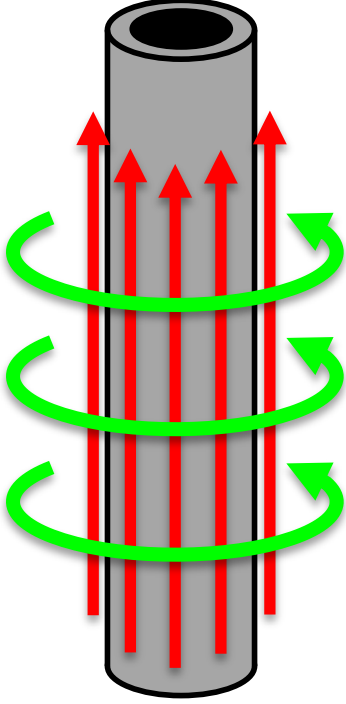


FIG. 59 A schematic of the magnetic implosion of a liner.

where $\gamma = 5/3$ for a monatomic gas and C_c is the convergence after shock heating. The overall convergence is then $C_R = 4^{1/d} \left(\frac{\theta_f}{\theta_s} \right)^{\frac{1}{d(\gamma-1)}}$. Assuming a final temperature $\theta_f = 6$ keV, which is comfortably above the ignition temperature of 4 keV (see Section I.B.1), we find a convergence greater than 76 is required for the 100 km/s cylindrical implosion, but only 4.5 for a 400 km/s spherical implosion. Clearly shock heating is not sufficient for the relatively slow magnetically driven cylindrical implosions and some kind of fuel preheat is required. This is accomplished by laser heating in the MagLIF (Magnetized Liner Inertial Fusion) concept (Slut *et al.*, 2010), which is presently being studied on Z at the Sandia National Laboratories, with a frequency doubled (0.53 μ m) Neodymium glass laser (Z Beamlet) providing the preheat.

So far, we have only considered adiabatic compression, but radiative and conductive cooling are important during the implosion (as discussed in Section I.B). If we assume a cylindrical implosion with uniform fuel density and use the ideal gas equation of state, the rate of compressive heating is given by the expression $P \frac{dV}{dt} = 5.0 \times 10^{11} \rho \theta r v_{imp}$ W/m. Thermal conduction

is dominated by electrons due to their low mass and subsequent high velocity. The thermal flux is $F_{ce} = \frac{9.6 \times 10^{14} \theta^{2.5} \nabla \theta F(\omega \tau)}{\ln \Lambda}$. The temperature profile can be found by assuming the divergence of the thermal flux is independent of the radius. This leads to the expression $L_{ce} = 5.7 \times 10^{14} \theta^{3.5} F(\omega_{ce} \tau_{ei})$ W/m for the rate of conductive cooling, where we have assumed $\ln \Lambda = 6$, θ is the central temperature and $F(\omega_{ce} \tau_{ei})$ accounts for the effect of a strong magnetic field. The degree of magnetothermal-insulation is a function of the Hall parameter, which is the product $\omega_{ce} \tau_{ei}$ where ω_{ce} is the cyclotron frequency of the electrons and τ_{ei} is the average time between electron-ion collisions. The magnetization term is (Braginskii, 1965)

$$F(\omega_{ce} \tau_{ei}) = \frac{1 + 0.39 (\omega_{ce} \tau_{ei})^2}{1 + 3.9 (\omega_{ce} \tau_{ei})^2 + 0.26 (\omega_{ce} \tau_{ei})^4} \quad (91)$$

$$\approx \frac{1}{1 + (\omega_{ce} \tau_{ei})^2} \quad (92)$$

for $\omega_{ce} \tau_{ei} > 3$ where $\omega_{ce} \tau_{ei} = \frac{1.28 \theta^{1.5} B}{\rho}$.

During the liner implosion the compressive power done on the fuel minus the loss rate equals the rate of change in the fuel energy, thus

$$P \frac{dV}{dt} (1 - f_L) = \frac{3}{2} \frac{d}{dt} (PV) = \frac{3}{2} \left[P \frac{dV}{dt} + V \frac{dV}{dt} \right], \quad (93)$$

where f_L is the ratio of the thermal and radiative losses over $P \frac{dV}{dt}$. In this discussion we assume f_L is constant to illustrate the physics. The basic ideas remain true when more detailed numerical calculations are done. This can be solved to yield $PV^\gamma = \text{const}$, where $\tilde{\gamma} = \frac{5-2f_L}{3}$ [note the similarity to Eq. 23]. The fuel temperature from compressive heating including losses is then $\theta = \theta_0 \left(\frac{r_0}{r} \right)^{2(\tilde{\gamma}-1)}$, where θ_0 is the temperature after preheat and before compression. At stagnation the ratio of the temperature after compression over the preheated temperature is thus $\frac{\theta_s}{\theta_0} = C_R^{2(\tilde{\gamma}-1)}$. This ratio is one half of the adiabatic case for $f_L = 0.15$ assuming that $C_R = 30$. Clearly the loss fraction

$$f_L = \frac{1}{v_{imp}} \left[\frac{1.9 \times 10^5 \Gamma}{\theta^{0.5}} + \frac{1.1 \times 10^3 F(\omega_{ce} \tau_{ei})}{\Gamma} \right], \quad (94)$$

where $\Gamma = \rho r$ needs to be kept small. This expression is minimized by setting $\Gamma = 0.007 \theta^{1.5} F(\omega_{ce} \tau_{ei})^{1/2}$ to find the minimum loss fraction $f_{Ln} = \frac{2.9 \times 10^4 \theta F^{1/2}(\omega_{ce} \tau_{ei})}{v_{imp}} = 1.7 F^{1/2}(\omega_{ce} \tau_{ei})$ at stagnation assuming $\theta = 6$ keV and $v_{imp} = 100$ km/s. This indicates that magnetization is required for such implosions. A simple axial field is effective since the heat flow in the axial direction is not large due to the small area of the ends and the much lower axial temperature gradient. We estimate the initial

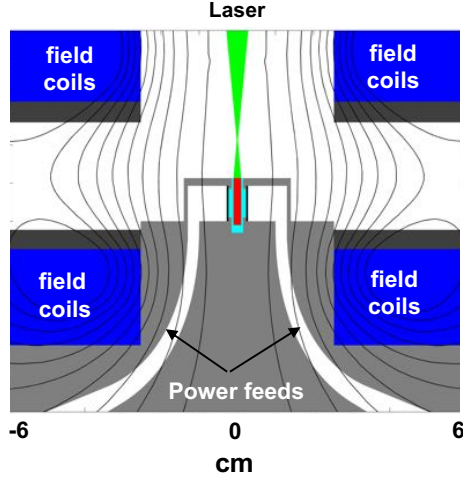


FIG. 60 A schematic of the MagLIF configuration within Z.

fuel density of 1.4 kg/m^3 from the expression for f_L by assuming that $F(\omega_{ce}\tau_{ei})$ is small, $f_L = 0.1$ and $C_R = 30$.

External magnetic field coils are used to provide an initial magnetic field of 10-20 Tesla for MagLIF experiments. A schematic of the MagLIF configuration within Z is shown in Fig. 60. A more detailed schematic of just the liner is shown in Fig. 61 and the three steps of MagLIF are illustrated in Fig 62. First, the fuel and liner are magnetized by the external coils over a relatively long timescale of 3 ms. Second the laser heats the fuel to an average temperature of 100-200 eV. This is followed by the magnetic compression of the liner. On the time scale of the implosion (50-100 ns) the axial magnetic field is nearly frozen into both the fuel and the liner due to high conductivity of both the fuel and the liner. The magnetic Reynolds number is greater than 1000 in the fuel. Simulations of MagLIF using the code LASNEX (Zimmerman and Kruer, 1975) indicate that about 50% of the initial magnetic flux in the fuel inside a beryllium liner is retained up until stagnation. Thus, an initial field of 15 T results in a magnetic field strength of 6000 Tesla and a α -particle cyclotron radius just 0.4 of the fuel radius at stagnation for $C_R = 30$. Clearly the field will strongly affect α -particle heating. The simulations assuming a laser preheat energy of 1 kJ, which Z Beamlet can provide, indicate a stagnation temperature of about 3 keV and that 50% of the fuel mass is lost out the open ends during the implosion so the final fuel density is about 630 kg/m^3 ; hence $\omega_{ce}\tau_{ei} = 63$ and $F(\omega_{ce}\tau_{ei}) = 0.00037$ so radial thermal transport is strongly suppressed. The range of an α -particle (Atzeni and Meyer-Ter-Vehn, 2004) is $\Gamma_\alpha = 0.15\theta^{1.5} = 0.8 \text{ kg/m}^2$, which is much smaller than $\rho L = 6.4 \text{ kg/m}^2$ for a 1 cm long liner. Therefore, most α -particles will be absorbed in the fuel.

MagLIF is a specific example of MIF. When ICF was proposed (Nuckolls *et al.*, 1972) it was already known

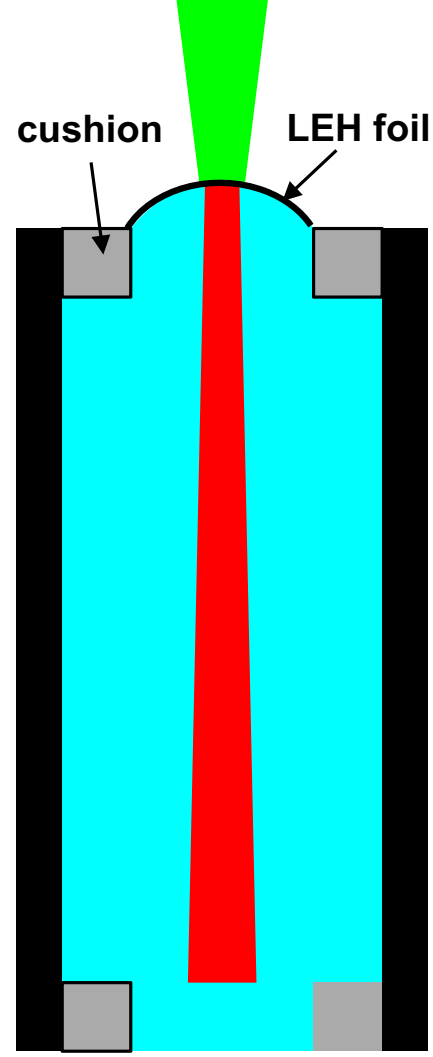


FIG. 61 A schematic of MagLIF. The laser beam is green, the heated fuel is red and the liner is black.

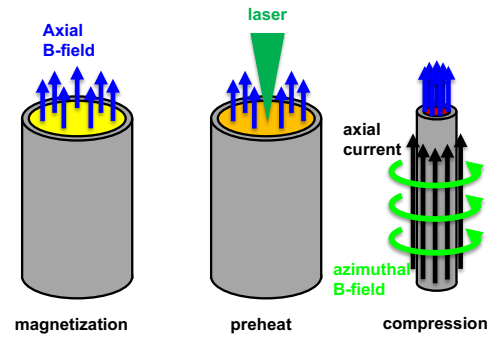


FIG. 62 The three steps of MagLIF.

that a magnetic field could significantly reduce electron thermal conductivity (Landshoff, 1949). However, it was not until several years later that experiments demonstrated that a magnetic field could improve ICF yields (Chang *et al.*, 1977; Widner *et al.*, 1977b) as described above. Other methods to magnetize and pre-heat fusion fuel have been proposed. A Soviet proposal, MAGO (MAGnitnoye Obzhatiye, or magnetic compression), would generate an inverse z pinch to inject high velocity plasma into a chamber where it would be compressed by a liner implosion (Khariton *et al.*, 1976). Subsequently, a joint US-Soviet research effort studied magnetically driven liners for that purpose (Lindemuth *et al.*, 1995). These liners were large compared to MagLIF and imploded in several μs . Liner driven field-reversed plasma configurations (FRC) have also been proposed (Schoenberg and Siemon, 1998). FRCs were originally developed as magnetic confinement fusion systems (Tuszewski, 1988). It was believed that they are self-sustaining long enough (tens of μs) to move them to within a metal liner for subsequent compression (Intrator *et al.*, 2002). FRCs have plasma densities of $\sim 1 \times 10^{-4} \text{ kg/m}^3$ and plasma temperatures of 250 eV, which make them suitable for liner compression with implosion times of $\sim 10 \mu\text{s}$ (Intrator *et al.*, 2004). The Air Force Research Laboratory and Los Alamos National Laboratory launched an experimental effort (Intrator *et al.*, 2008) to generate, translate, and the compress FRCs using the Shiva Star pulsed power driver. Degnan *et al.* (Degnan *et al.*, 2013) concluded that the FRCs produced in this experiment did not have a sufficient lifetime for translation and compression. The parameter space for MIF has been explored numerically (Lindemuth and Kirkpatrick, 1983; Lindemuth, 2015). These studies have shown that significant gain could be obtained even for very slow implosion velocities ($< 10 \text{ km/s}$) when the initial fuel densities are very low ($\sim 10^{-3} \text{ kg/m}^3$). This could allow simple highly efficient pulsed power drivers and enable a low-cost approach to fusion (Siemon *et al.*, 1999)

2. Experimental demonstration of MagLIF concepts

The first experiments integrating all three phases of the MagLIF concept validated the basic underlying physics. These experiments, which were first designed numerically (Sefkow *et al.*, 2014) and then tested on Z (Gomez *et al.*, 2015, 2014), produced DD neutron yields of 2×10^{11} to 2×10^{12} as measured by activation and time of flight techniques (Hahn *et al.*, 2016, 2014). Deuterium plasma temperatures of 1.1-3.0 keV were measured both by neutron time of flight and x-ray spectroscopy (Hansen *et al.*, 2015). In addition, secondary DT neutrons produced in the aneutronic branch of the DD reaction, were measured with yields up to 5×10^{10} , indicating a high degree of magnetization (Knapp *et al.*, 2015; Schmit *et al.*, 2014).

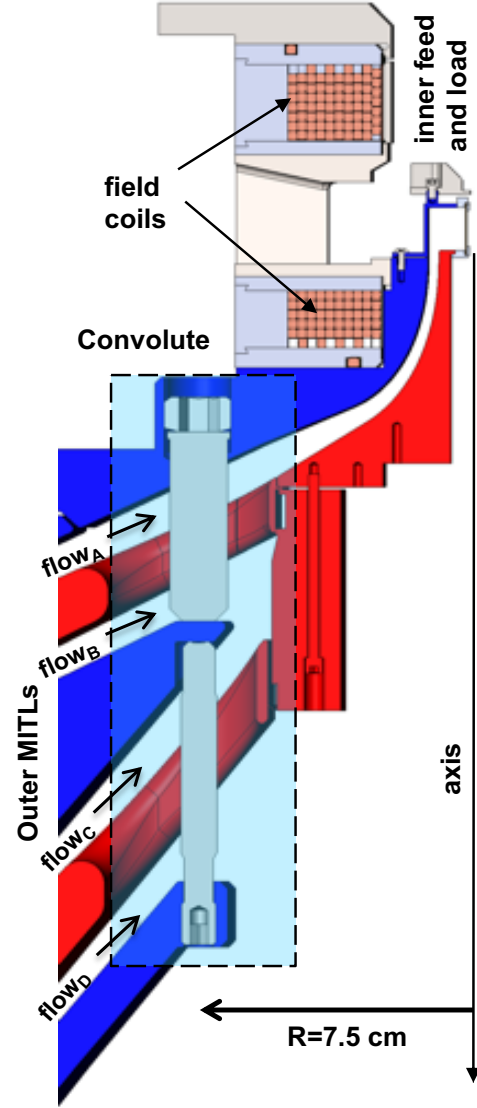


FIG. 63 A schematic of the Z convolute. The convolute adds the four inflow currents into one feed current.

The yields in these early experiments were modest for several reasons. First, a large fraction of the laser energy was absorbed in the $1.5\text{-}3.5 \mu\text{m}$ thick polyimide foil covering the laser entrance hole (LEH), see Fig. 61. The Z Beamlet laser was obtained from LLNL to provide x-rays for backlighting (Rambo *et al.*, 2005). Beam smoothing was not required for this application. Furthermore, Z Beamlet is a Neodymium glass laser with only frequency doubling ($0.53 \mu\text{m}$) as compared to NIF and Omega, which are frequency tripled ($0.35 \mu\text{m}$). Laser-only experiments using Z Beamlet (Geissel *et al.*, 2014) indicate that due to the large variations in laser intensity only 10% of the 2 kJ of laser energy penetrated a $2.0 \mu\text{m}$ thick foil in the early MagLIF experiments. Intensity variations drove laser plasma instabilities (LPI) that pre-

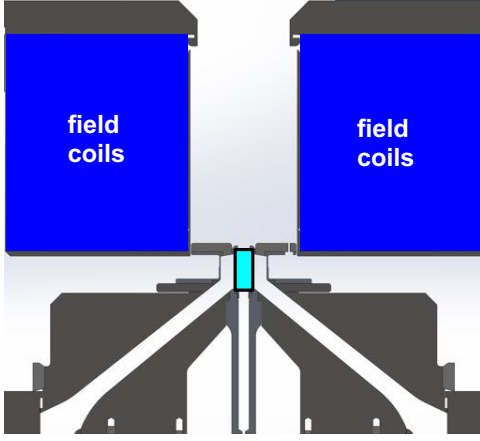


FIG. 64 A schematic of MagLIF with the new lower inductance feed and new magnetic field coils. This feed geometry approaches the minimal inductance configuration obtained from a variational calculus method (Hurricane, 2004; Waisman and Cuneo, 2009).

vented rapid burn-through of the foil. Later experiments using the University of Rochester’s Omega-EP laser and the Z Beamlet laser at Sandia have shown that the energy coupling could be significantly improved by using distributed phase plates to smooth the beam (Geissel *et al.*, 2018; Harvey-Thompson *et al.*, 2015, 2016). Experiments have since been conducted that delivered 1.2 kJ of energy into the fuel. Second, the drive current in these experiments was only about 16 MA because of the relatively high feed inductance (7.2 nH) of these early MagLIF loads as compared to other loads such as the dynamic hohlraum, which has a feed inductance of only 3.0 nH and peak currents up to 28 MA.

Increased load inductance raises the voltage on the convolute and increases current loss. The convolute (see Fig. 63) adds the currents from four MITLs thus delivering 4 times the current of each MITL. A model developed by (Hutsel *et al.*, 2018) has been incorporated into LASNEX, which can now predict peak drive current as a function of load inductance. New final feed and magnetic coil designs have lowered this inductance (Fig. 64) and increased the current from 16 MA to 19.5 MA (Fig. 65). Third, 2D simulations (Slutz *et al.*, 2017) indicated the yield increases with the applied magnetic field strength. The new field coils have allowed experiments to be conducted with 16.9 T fields as compared to the early experiments with 7-10 T fields.

All of these improvements have resulted in an order of magnitude increase in yield to 1.1×10^{13} DD neutrons and nearly twice the burn-averaged ion temperature to 3.1 keV, both trends predicted by LASNEX simulations (Gomez *et al.*, 2020). Simulations by Slutz *et al.* (Slutz *et al.*, 2017) indicate that further increases in the drive

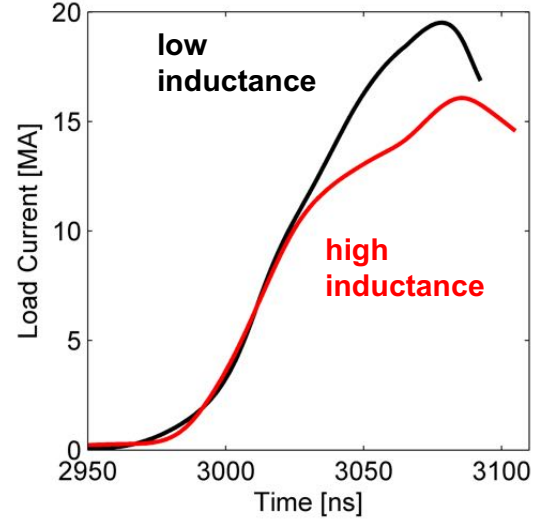


FIG. 65 The measured currents are plotted as a function of time for the old and new MagLIF feeds.

parameters to a 30 Tesla B-field, 6 kJ preheat energy absorbed, and 22 MA peak load current could enable another order of magnitude increase in fusion yield on Z.

The applied magnetic field can be increased to 20-30 Tesla when a set of coils just above the final feed and below the MagLIF liner is implemented, see Fig. 64. Much higher fields could be attained using the auto-magnetizing liner (AutoMag) concept (Slutz *et al.*, 2017). An AutoMag liner consists of helical conduction paths separated by insulating material, see Fig. 66. The electric field across the insulators is proportional to rate of current rise (dI/dt) of the drive pulse. The drive current is profiled to have a foot that rises to ~ 1 MA in 100 ns, which by design generates an electric field across the insulators that is below the breakdown threshold. Consequently the drive current flows along the helical conducting paths to magnetize the fuel. The foot is followed by the main pulse, which rises to full current (~ 20 MA for Z experiments) also in about 100 ns. Since dI/dt increases by a factor of 20 the insulators break down so the current flows axially as in a normal liner implosion. Generation of fields as large as 100 Tesla has been demonstrated experimentally with AutoMag (Shipley *et al.*, 2018, 2019).

Although 2D simulations have predicted the performance trends, the simulated yields are roughly 3 times the experimental yields. These 2D simulations did not include the effects of mix and cannot accurately calculate the 3D nature of the stagnation. Progress is being made on performing 3D simulations of MagLIF.

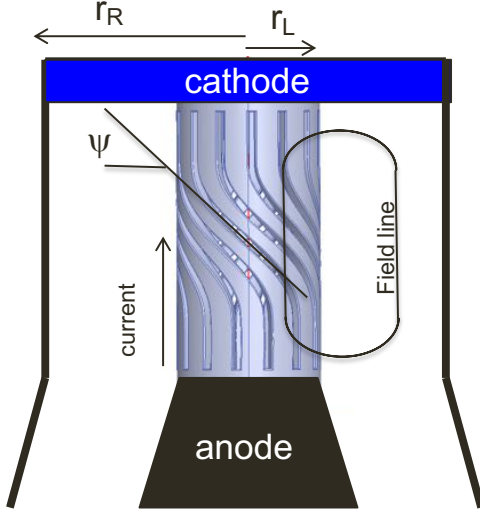


FIG. 66 A schematic of the AutoMag concept. The gaps in the metal would be filled with an insulator such as epoxy.

3. The effect of mix

Although 2D simulations have predicted the performance trends, the simulated yields are roughly 2-3 times the experimental yields. These 2D simulations cannot accurately model the development of the MRT instability (McBride *et al.*, 2012), which could lead to the mixing of liner material into the fuel. Progress is being made on performing 3D simulations of MagLIF, which could in principle model such a mixing process.

Materials with an atomic number greater than hydrogen in the fuel will lower performance by enhancing radiation losses. A series of LASNEX simulations were performed by (Slutz *et al.*, 2018) with various dopants to determine the yield dependence with dopant fraction. The opacities were calculated using the Detailed Configuration Accounting (DCA) method without assuming Local Thermodynamic Equilibrium (LTE) (Scott and Hansen, 2010). LASNEX test simulations with this opacity model compared well to the detailed radiation code SCRAM (Hansen *et al.*, 2007). The results indicated that a 0.6% mixture of beryllium at the start of the simulation decreased the yield by a factor of 2 and a 2.5% mixture would lower the yield by a factor of ten. The effect of other dopant materials scales as the cube of the atomic number, e.g., 1% Be has the same effect as $(4/13)^3 \times 1\% = 0.03\%$ Al.

The LEH window absorbs a significant fraction of the laser energy and can be injected into the fuel during expansion. This is consistent with observed cobalt spectral lines in experiments performed by (Harvey-Thompson *et al.*, 2018) with thin (1 nm) cobalt layers on the inside of the window. The preheated fuel forms a blast

wave that could also mix material from either the liner or the cushion, see Fig. 61. Experiments with coatings on the cushion near the LEH confirm that cushion material is mixed into the fuel. Indeed, experiments with aluminum cushions have had lower yields than experiments with beryllium cushions. As with all ICF approaches, the interface between the metal liner (pusher) and the fuel is unstable to the Rayleigh-Taylor (RT) instability when the pressure of the fuel decelerates the liner. Spectral features of iron (a liner impurity of known concentration ~ 110 ppm) have been observed (Hansen *et al.*, 2015), which indicate beryllium is mixed into the fuel.

A two-pulse laser profile has been used to improve the laser penetration of the LEH foil. Nevertheless, a significant fraction ($\sim 28\%$) (Harvey-Thompson *et al.*, 2018) of the laser energy is still absorbed by the window material, but no mix is observed. An alternative approach (Slutz *et al.*, 2018) is to weaken the foil several μ s before the main laser pulse so that the fuel pressure of several atmospheres breaks the foil. The window then opens in several μ s like a gate in response to the pressure of the gaseous fuel. The foil could be weakened by a laser pulse focused to a star pattern on the top of the window or by current-heated wire in contact with the foil around its periphery. Both approaches have demonstrated the desired foil opening (Galloway, 2020; Miller *et al.*, 2020), but have not yet been integrated into MagLIF experiments on Z.

Iron spectral line data implies a beryllium atomic fraction of about 5% at stagnation (Hansen *et al.*, 2015) assuming uniform mix into all of the fuel. Simulations predict that such a high uniform mix fraction would essentially dud the yield if it occurred early in the implosion. Analysis of iron helium-like resonance to the lithium-like satellite line ratios indicate a temperature of 50-70% of the burn temperature suggesting that Be is not mixed into the central portion of the fuel. A series of LASNEX simulations were performed that introduced Be into a layer of the fuel next to the liner at peak implosion velocity to approximate the effect of RT mixing during the deceleration of the liner (Slutz *et al.*, 2018). The layer thickness and the amount of Be were varied. It was found that both the difference between the two measured temperatures and a 50% reduced yield were indicated with a layer thickness roughly $1/3$ of the fuel radius at stagnation. These simulations also indicated that the mixing of cold fuel (DD or DT) instead of Be would have a much smaller effect on the yield. Thus, a cryogenic fuel layer on the inner surface of MagLIF liners should reduce the effect of mix. DT ice layers can be grown by the process of “beta layering” (Hoffer and Foreman, 1988; Martin *et al.*, 1989), a process that could be used for MagLIF liners as well. Ice layers of just deuterium can be made as well and will be used for near term MagLIF experiments. However, at the low temperatures needed (< 18 K) to maintain solid DT, the vapor pressure (gas den-

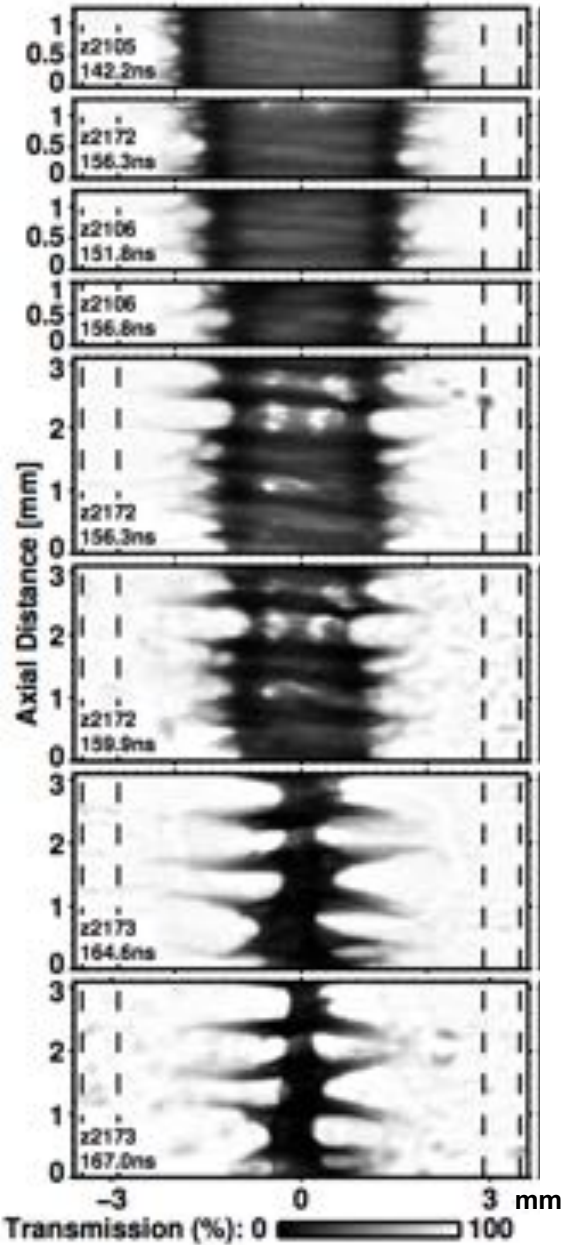


FIG. 67 Radiograph of a beryllium liner implosion at different times without an axial magnetic field.

sity $\sim 0.3 \text{ kg/m}^3$) is substantially below the gas density needed for MagLIF liners ($> 1.0 \text{ kg/m}^3$). One solution is to form the ice layer and then puff gas into the liner. Another is to provide a foam layer on the inside surfaces of the liner that can wick liquid fuel. We will come back to this in subsection 5 below.

4. Implosions Instabilities

The RT instability plays an important role in all ICF approaches (see Section II.C). The acceleration phase of a MagLIF implosion is driven by the azimuthal magnetic field produced by the axial current flow. Such implosions are susceptible to the MRT instability. Unlike radiation or laser direct drive, there is no ablative stabilization mechanism, thus the MRT instability poses a serious threat to liner implosions. A z pinch with no applied axial field is most susceptible to the $m = 0$ mode, because that mode does not bend the driving azimuthal field lines, i.e., the k vector is in the axial direction, which is perpendicular to the magnetic field. Therefore, the MRT instability produces bubbles and spikes that form bands around the cylindrical z pinch. Liner implosion experiments without an axial field exhibited this behavior (McBride *et al.*, 2012), as illustrated in Fig. 67. When an axial field is applied the k vector of the fastest growing mode will have an azimuthal component and the bubble and spike bands will be helical with a pitch angle $\Phi = \frac{B_z}{B_\theta}$. The drive current is about 1/2 of the maximum when the liner starts to move (Sultz *et al.*, 2010) so $B_\theta > 600 \text{ T}$ while $B_z = 10 \text{ T}$ when the MRT starts to develop. This implies a very small pitch angle, which is not consistent with the experimental results (Awe *et al.*, 2014, 2013), see Fig. 68. The leading hypothesis (Ryutov and Dorf, 2014) for the large pitch angles observed is that low density plasma is swept from the MITL feed to the liner region. As the current rises this plasma entrains the axial magnetic field and compresses against the liner. Such a mechanism could make the axial field on the outside of the liner comparable to the azimuthal field. Extended magnetohydrodynamic (MHD) simulations (Seyler *et al.*, 2018) of the power flow have demonstrated this mechanism. The need to include extended MHD terms increases the difficulty of performing 3D simulations that can accurately simulate the effect of MRT on liner implosions. Simulations (Jennings, 2020) using the 3D Eulerian Gorgon code, that does not have extended MHD, (Chittenden *et al.*, 2004) have qualitatively reproduced the stagnated images MagLIF experiments at stagnation, see Fig. 69. However, the helical structure is only produced when an initial helical perturbation is applied. Similar results are produced (Weis, 2020) with the 3D ALE code HYDRA (Farmer *et al.*, 2017). 3D simulations with sufficient physics to exhibit the helical instability without imposing initial helical perturbations are required to determine if the helical MRT instability is better or worse than the $m = 0$ mode.

The MRT instability can be seeded by both surface roughness and density nonuniformities. Another possible source (Peterson *et al.*, 2012) is an Electro-Thermal Instability (ETI) that is caused by positive feedback when current is driven through a material that has a temperature-sensitive resistivity. ETI can be seeded by

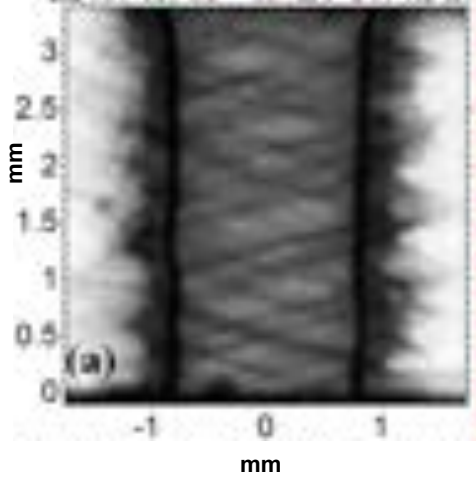


FIG. 68 Radiograph of a beryllium liner implosion at a convergence of 2.7 with an applied 7 Tesla axial magnetic field.

temperature, current density, or resistance variations. Experiments performed with aluminum and copper rods machined to be as smooth as possible have exhibited initial growth due to ETI that was larger than expected from the MRT instability. Later, when the ETI has produced significant density variations the MRT instability becomes important. These rods had rms surface roughness of 11 nm for Cu and 27 nm for Al and also had some pits and bumps on the surface. Simulations show that development of the ETI is not strongly affected by the surface roughness over a range of 37-296 nm (Peterson *et al.*, 2013).

A thin coating of dielectric material can reduce the ETI and hence the MRT instability significantly (Awe *et al.*, 2016; Peterson *et al.*, 2014). This technique has been used to improve the stagnation stability of MagLIF. Figure 70 shows x-ray images of the stagnated fuel from an $A_R = 6$ liner implosion a) without (Knapp *et al.*, 2019) and b) with a dielectric coating (Gomez *et al.*, 2020) on the outside of the liner. Clearly the coating has improved stability, but the DD neutron yield decreased from 3.2×10^{12} to 7.6×10^{11} . Liners with $A_R = 9$ have produced higher yields with coatings as compared to without coatings (Ampleford, 2020). There is not presently a good explanation for the decreased yield on the $A_R = 6$ shots.

There are other options for reducing the effect of the MRT instability on MagLIF that have not yet been tried. The growth rate of the MRT instability would be reduced in a liner with density profile that is low at the outer surface and increases toward the inner surface (Velikovich *et al.*, 1998). The magnetic pressure will then drive a snowplow-like implosion with nearly constant velocity. Another possibility is the dynamic screw pinch

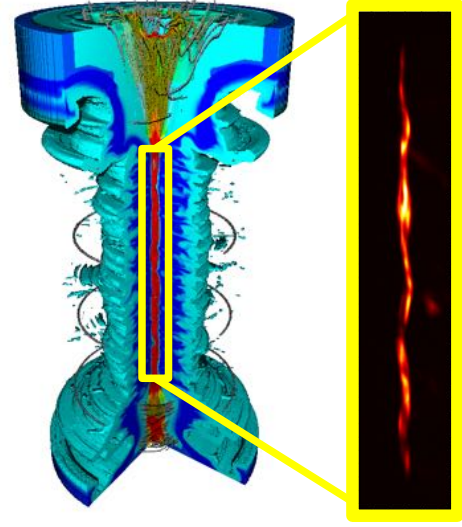


FIG. 69 A 3D Gorgon simulation compared to a self-emission x-ray image of the stagnated fuel in a MagLIF experiment on Z.

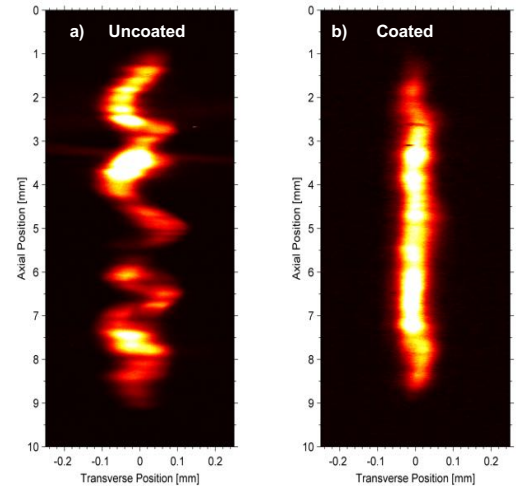


FIG. 70 X-ray images of stagnated fuel for MagLIF experiments with and without a dielectric coating on the liner.

(Schmit *et al.*, 2016; Shipley *et al.*, 2019), which produces a time-dependent axial magnetic field. We have already mentioned that a dense cryogenic layer of fuel would help mitigate the effects of mix.

5. High yield and gain from MagLIF on future pulsed power machines

When MIF was first being studied it was thought that, although a magnetic field would improve volume burn, it would inhibit the propagation of a deflagration burn

wave into any surrounding fuel (Jones and Mead, 1986). This would limit MIF to modest gains. Later Slutz and Vesey (Slutz and Vesey, 2012) found numerically that propagating burn and gains approaching 1000 were possible for MagLIF with drive currents exceeding 60 MA. A layer of cryogenic (frozen) DT on the inside surface of the liner was included in these simulations. The optimum thickness of this layer increases with drive current. Several conditions are required for burn propagation. First, modest initial fields (~ 10 T) are optimum. Larger fields inhibit propagation, while smaller fields result in poor hot spot performance. Second, the liner areal density must be large ($\Gamma_L \sim 50$ kg/m²) to provide sufficient confinement time. Finally, the fuel areal density, Γ_F , also needs to be sufficiently large. This requirement depends on the Γ_L , for $\Gamma_L = 50$ kg/m², which is attained in simulations of MagLIF liners driven by 60 MA, $\Gamma_F > 0.7$ kg/m². Note that the fusion rate is proportional to P^2V which is proportional to Γ_F^2 . Simulations indicate optimal initial gas densities are 3 kg/m³ at 60 MA and rise to ~ 5 kg/m³ at 70 MA (Slutz *et al.*, 2018). We discuss later how to get such gas densities that are not in equilibrium with an ice layer.

Z has enabled research on dynamic material equations of state, opacities and inertial fusion and has also provided radiation and neutron sources for weapon components testing. The maximum current delivered by Z ranges from 20 MA for high impedance loads such as MagLIF to 28 MA for low impedance loads such as the dynamic hohlraum. All of this research would benefit from higher currents. Work is presently underway to design a new pulsed power machine capable of delivering 40-60 MA. One possible approach is to use Linear Transformer Driver (LTD) technology (Douglass *et al.*, 2018). Two LTD based pulsed power machines have been designed to deliver 48 MA and 65 MA to high impedance loads such as MagLIF (Stygar *et al.*, 2015). An equivalent circuit of these two designs has been used in 2D LASNEX simulations to find optimal MagLIF designs as a function of the driver current (Slutz *et al.*, 2016). These 2D simulations indicate that propagating burn could be possible for currents greater than 55 MA, with a yield of 1 GJ at 60 MA. The simulations indicate that the optimum preheat energy increases with current from several kJ at 20 MA to 30 kJ at 50 MA. This level of preheat has been demonstrated by experiments (Pollock *et al.*, 2020) performed at the NIF. Although the effects of the MRT instability and mix were not included, the simulation scaling is encouraging.

The energy delivered by a 60 MA machine is nine times the energy delivered by the 20 MA Z machine, while the simulated yields increase by a factor of 1,000 without propagating burn. The physical parameters of the fuel and implosion change significantly for this predicted scaling. Recently, a more conservative scaling has been presented (Schmit and Ruiz, 2020), which holds most of the

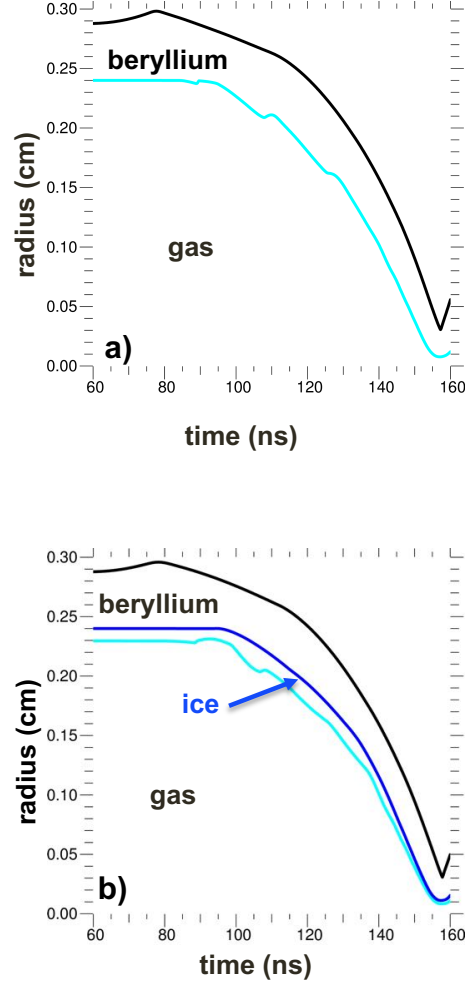


FIG. 71 Interface positions as a function of time from LASNEX simulations with and without a layer of cryogenic fuel on the inside of the liner.

dimensionless physical quantities constant. This scaling predicts the yield increases with drive current as $(I^3 - I^4)$ if alpha particle heating is ignored. This corresponds to yield increase of about 80, but according to simulations the yield would be increased another factor of 10 by including alpha particle heating. Thus, both scalings indicate that 50-100 MJ yields may be possible with a 60 MA driver even without propagating burn into a dense fuel layer.

1D LASNEX simulations of MagLIF liners without and with a cryogenic fuel layer are shown in Fig. 71. The trajectory of the inner surface of the liner is strongly affected by the blast wave induced by laser heating when no cryogenic layer is present, while the trajectory of the inner surface is smooth when a layer is included. A new set of simulations has been performed with a cryogenic DT wetted foam (CH) layer on the inside of the liner. The re-

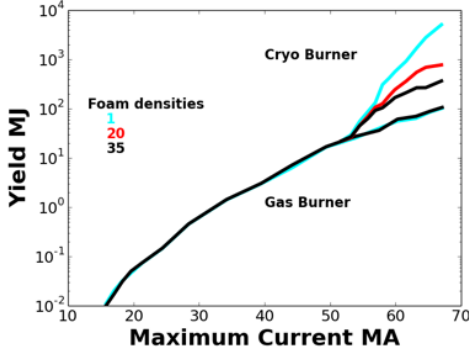


FIG. 72 LASNEX simulated yields of optimized MagLIF with a cryogenic wetted foam layer on the inside of the liner.

sults for several foam densities are shown in Fig. 72. The layer serves as a mix mitigation layer for “Gas burners” and as extra fuel for high gain “Cryo Burners.” The layer was 100 μm thick for the “gas burner” and increases from this value with current for the “cryo burners.” Note that the optimum liner radius is smaller for cryo burners than gas burners. Wetted foam has the advantage that the gas density is determined by the temperature, e.g., the vapor density of liquid DT is 4 kg/m^3 at 23 K. The disadvantage is that the layer is mixed with carbon, which will make it harder to burn. Foam densities of 35 kg/m^3 have been used in NIF capsules. Lower foam densities may be possible, e.g., a density of 14 kg/m^3 is used for dynamic hohlraum experiments. The wetted foam has essentially no effect on the gas burners but reduces the yield of cryo burners as the foam density increases. Yet even for the highest foam density (35 kg/m^3) the simulated yields are significantly greater than gas burners at high current drive.

VI. SUMMARY AND OUTLOOK

Across the US ICF Program understanding of what physics and engineering principles are key is improving at a rapid pace. There are many commonalities to the problems indirect-drive, direct-drive, and magnetic-drive face and, of course, the underlying physics principles of ICF are the same regardless of approach.

While the initial predictions for indirect-drive on NIF appear to have been optimistic, a strategy of identifying and addressing the physics and engineering obstacles in steps has paid off in terms of understanding the physics and demonstrating increased fusion performance. Using a strategy of leveraging increased capsule scale with what has been learned about implosion symmetry and instability control, a key turning point was achieved on the NIF, increasing the record fusion yields discussed in Sec-

tion III.D to 170 kJ, exceeding $G_{\text{fuel}} > 6$ (e.g. Fig. 2), and achieving a major fusion physics milestone of a burning plasma (Kritcher *et al.*, 2022a; Ross *et al.*, 2021; Zylstra *et al.*, 2022a), then shortly thereafter ignition by Lawson’s criterion (Abu-Shawareb *et al.*, 2022; Kritcher *et al.*, 2022b; Zylstra *et al.*, 2022b) with 1.37 MJ total fusion yield, $G_{\text{fuel}} > 70$, and $G_{\text{capsule}} \sim 6$. Then, most recently $G_{\text{target}} \sim 1.5$ followed (LLNL, 2022) with a modest and incremental increase in laser drive (the development of which was set in motion in 2015) and a corresponding increase in capsule thickness. Since the early indirect drive experiments on NIF a decade ago, fusion yields have increased by over a factor of 1000 $\times$. What remains is to build upon these recent results by increasing energy coupling to the indirect-drive targets and then understanding and correcting the limitations presently observed with respect to compression.

In conjunction with the mainline indirect-drive approaches on NIF that have led to tangible gains, a number of higher-risk concepts are also being explored in the hopes of accelerating both learning and progress towards yet higher gains.

Machine learning has been used to great effect to improve the fusion performance of direct-drive implosions at LLE and teased-out key sensitivities that were not previously recognized. When scaled to NIF energies with an assumption that NIF is reconfigured to a spherical direct-drive configuration, the best performing OMEGA implosions are *projected* to produce over 0.5 MJ of fusion yield.

Pulsed power has evolved since the 1960s and is still evolving. Z can deliver megajoules to a load with a relatively high efficiency of 5%. Future pulsed power machines are now being designed to deliver 10 MJ to a load. Pulsed power is used to efficiently generate x-rays with wire array and gas puff implosions. These x-rays are used for weapon component testing and for studying the properties of materials, e.g., the dynamic hohlraum approach is being presently used to measure the opacity of materials relevant to understanding our Sun.

Concepts to produce fusion in the laboratory using pulsed power are also continuing to evolve. That research started with electron beams. Although, the stopping distance of electron beams was too long, the principles of Magneto-Inertial Fusion were first demonstrated with electron beam driven capsules. The program then switched to ion beams, which have a more favorable deposition, but were found difficult to focus. Pulsed power research then focused on generating x-rays to indirectly drive capsules. Two approaches were developed using wire array implosions. These are the double-ended pinch and the dynamic hohlraum, which could be viable on a future pulsed power machine. The research was then redirected toward the possibility of direct magnetic drive to deliver much more energy to the implosion. This change motivated the development of MagLIF. The prin-

ciples of MagLIF have been demonstrated experimentally and fusion yields have improved by increasing the B-field, the preheat energy, and the drive current. Simulations indicate that high fusion gain and yields could be possible with MagLIF on a future pulsed power machine that delivers more current.

Given that we now have an existence proof of fusion ignition and scientific breakeven in the laboratory, the future of ICF looks bright.

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